Research and Evaluation of the Schools Excellence Fund Digital and STEM Programme (2018-2022)

Final Evaluation Report

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Table of Contents

Executive summary 2
Introduction 2
Key findings 3
Recommendations for future projects 5
Final word 6
Section 1: Introduction 7
Section 2: SEF Digital & STEM – Programme overview 8
Programme objectives 8
Programme funding 10
Programme timeline 11
Section 3: Evaluation approach 12
Section 4: Interim evaluation report – Key findings 15
Clusters and schools 15
Numbers of teachers and learners 17
Most frequent terms used to describe projects 18
Most innovative elements of the project 19
Project achievements and issues 20
Curriculum and goals 25
Pedagogy 28
Schools, teachers and wider partnerships 31
Technology 34
Oide TiE Advisor feedback 36
Section 5: Post-Project Survey – Experience of the project 40
Overview of the survey 40
Experience of the project 40
Section 6: Post-Project Survey - Lessons learned 52
Section 7: Post-Project Survey - Possible future digital education initiatives 64
Section 8: Programme recommendations 73
Establish clear programme goals 74
Initial organisation of the programme 75
Programme flexibility 75
Project administrative requirements 75
Funding of digital technologies 77
Ongoing support structures 78
Programme dissemination 79
Final word 80
Appendix 1: List of SEF Digital & STEM clusters 81
Appendix 2: Department of Education SEF Digital & STEM documentation 83
Appendix 3: SEF Digital & STEM case studies 86
Appendix 4: SEF Digital & STEM teacher stories: 107
Appendix 5: Post-project survey 132
List of abbreviations 135

Executive summary

Introduction

The *Digital Strategy for Schools (2015-2020): Enhancing Teaching, Learning and Assessment* contained an action to encourage a culture of innovation in schools in relation to digital learning. The Department of Education and Skills (now the Department of Education and Youth)[1], subsequently launched the Schools Excellence Fund Digital Programme (SEF Digital) which was later expanded to the SEF Digital & STEM Programme. In April 2018, Minister Richard Bruton announced funding for 265 schools through the Programme. The purpose of the Programme was to demonstrate the innovative use of digital technologies in teaching, learning and assessment through clusters of schools collaborating on specific projects.

The schools applied for inclusion in the Programme via an open call to all schools which was launched in December 2017 by the Department[2]. The Programme sought to create collaborative environments in school clusters where teachers could work jointly on innovative projects that integrated digital tools into their teaching practices. The participating primary and post-primary schools were encouraged to design projects that addressed specific curricular needs, including STEM education and curricular reform in primary languages and maths.

The projects were expected to demonstrate how digital technologies could improve learning outcomes, strengthen collaboration among teachers, and trial innovative approaches to assessment. The Programme aimed to build links between schools, higher education institutions, and industry partners, to broaden the focus of cluster activities and provide students meaningful, real-world contexts for their digital learning.

Evaluation approach

The evaluation process commenced in early 2019 and was finalised in 2024. The work of the Programme, like many other school activities, was severely disrupted by the COVID-19 pandemic. The schools commenced their projects in 2018 and were very active up to March 2020 when schools were closed on a rolling basis until June 2021. Schools gradually reopened in September 2021, and the Programme held a key event in April 2022 where the clusters reconvened with the Department and other key stakeholders in Croke Park. As a result of the disruptions, the evaluation was carried out in two phases: Phase 1: September 2018 to June 2020 and Phase 2: April 2022 to June 2023. In 2020 the evaluation team, anticipating further disruption to school activities decided to produce an Interim Evaluation Report based on a range of data including documentation provided by the clusters, and on two surveys, one for Cluster Coordinators and one for Oide TiE Advisors[3]. This report was shared with the Department in September 2020. This document was subsequently revised and updated in May 2022, and key findings are summarised in Section 4.

[1] The Department was known as the Department of Education and Skills (to October 2020) and subsequently the Department of Education (to May 2025). It is currently the Department of Education and Youth. Throughout this report, the Department of Education and Youth is referred to as "the Department".

[2] https://www.dcu.ie/stemeducationinnovationglobalstudies/news/2017/12/minister-bruton-launches-school-excellence-fund

[3] Clusters were supported by Oide Technology in Education (TiE).

Phase 2 of the evaluation commenced with the Croke Park event held in April 2022 where the clusters shared their reflections on the Programme and lessons learned. This was an extremely successful event, and it also provided useful insights to inform the development of a Post-Project Survey for Cluster Coordinators, teachers and Oide TiE Advisors that was sent out during the 2023 school year, when schooling was returning to normal. The final survey captured data on three areas:

1. Experience of the project
2. Views on lessons learned (related to good classroom practice, infrastructural models, support models and embedding digital education), and
3. Views in relation to possible future digital initiatives.

The findings are presented in Section 5, 6 and 7. The evaluation also captured a series of case studies and teacher stories that illustrate how clusters engaged with the Programme and document how they used a range of digital technologies with their learners. These appear in Appendices 3 and 4.

Key findings

The Programme had a significant positive influence on both teaching and learning practices in the participating schools. Teachers reported increased confidence in using digital tools, greater collaboration with colleagues, and more effective engagement with students. Learners exhibited increased levels of motivation and more developed key skills, particularly in STEM-related subjects, while they also benefited from the interactive and creative use of digital tools for activities such as coding, robotics, and multimedia storytelling.

The collaborative structure of the SEF Digital & STEM Programme proved to be one of its strongest elements. Schools worked together within their clusters to share resources, strategies, and experiences. This collaboration allowed educators to exchange knowledge and expertise, fostering a culture of support and innovation. Schools that had previously been isolated, or lacked access to resources, found value in connecting with their peers, strengthening their overall capacity to engage in digital education.

The Interim Evaluation Report at the end of Phase 1 found that:

- Most of the teachers involved in the project found the experience very positive, fostered good links between teachers across different schools, facilitating knowledge and expertise sharing, and advanced students' digital skills.
- The SEF Digital and STEM Programme experience was helpful in the move to online learning during lockdown, as learners and staff had developed their skills in the use of online learning tools through project activities.
- The clusters involved in the Programme had been successful in disseminating their activities to the wider community, through the use of websites (60%) and social media (mainly using Twitter/X). Some clusters had also participated in (or initiated) large-scale public events.

This evaluation captured data from 35 of the 41 clusters and found a wide range of innovative practices supported by digital technologies across six categories:

1. Use of sophisticated technology (e.g., the use of equipment/infrastructure allotted for the project e.g., use of iPads, Lego WeDo kits, VR headsets etc.)
2. Inter-cluster collaboration
3. Teacher collaboration
4. Upskilling of learners (e.g., increase in learners' skills e.g. project management, problem-solving, 21st-century skills)
5. Curriculum planning
6. Engagement of young women in STEM careers.

These findings are discussed further in Section 4, with Phase 2 data presented in Sections 5, 6 and 7. The responses to closed questions in the Post-Project Survey in Phase 2 are captured in Section 5, showing a high-level of satisfaction with the projects and that learners responded positively to being involved in the Programme. Some issues were raised in relation to Programme supports (funding, advice, guidelines etc.) and in terms of support for Cluster Coordinators.

An analysis of the open questions in the Post-Project Survey reinforced these findings and it highlighted a range of successes and challenges faced by the clusters, which are summarised below.

Table 1: Successes and challenges identified in the Post-Project Survey.

Overall Experience	Most participants had positive experiences of the project, noting high student engagement, collaboration between schools, and the development of innovative teaching approaches. Coordinators and teachers identified challenges that came up in some clusters, which included high staff turnover, problems around project communications, and some technical and infrastructure issues.
Impact of COVID-19	All clusters were badly disrupted by the pandemic and school closures, though some were more able to adapt than others.
Collaboration	Collaboration between schools was a major element of the success of the clusters. It is notable that a third of the school clusters continued to collaborate after the end of the project, this was most often in STEM or computing related areas.
Long-Term Impact	The Advisors noted the positive impact of the projects on digital skills, on the integration of technology into teaching and learning processes, and on the collaborative working that had developed with other schools. They also noted that where staffing was consistent and project goals were clear, digital practices were sustained after the project.
Funding	Some clusters reported issues over funding, and in particular the impact lack of funding had on the sustainability of their projects, especially in terms of ongoing procurement and repair of equipment.
Regulation and Data Privacy	The survey did not ask specifically about regulation, ethics or data privacy, but it was perhaps surprising that only one respondent raised issues here, this was in relation to "support for managing data and compliance with data protection regulations".

Section 6 of the Final Evaluation Report captures the responses to open questions related to lessons learned in the Programme and these are summarised below.

Table 2: Lessons learned as identified in the evaluation.

Good Classroom Practice	Effective strategies included student-led projects, collaborative learning and personalised learning. In such cases teachers often moved from structured to more open approaches over time.
Infrastructural Models	Need for: • Strong broadband. • Physical space for housing and securing technology. • Hardware and devices, for both technologies for general use, but also specialist equipment for specific areas. • Software and platforms (i.e. Google, Microsoft and learning management systems). • Standardisation and clear policy.
Support Models	• Importance of continuing professional development and training and for peer-to-peer collaboration. • Participants wanted practical, up-to-date training tailored to their needs. In-school and peer-led models were preferred. Some questioned the value of private providers (i.e. those who developed certain technologies or software products providing training). • Technical support. • Administrative and institutional support. • The sustained Oide TiE support model was widely praised.
Embedding Digital Technologies	• Embedding required long-term funding, training, time, collaboration, curriculum alignment. • Relation to assessment needs consideration. • Evaluation of effectiveness of technology integration is required.

Recommendations for future projects

The evaluation suggests that there is a need for similar research projects in the future and recommends that if such projects are undertaken, then the Department might make some adjustments to further enhance the experience for the participating schools, and to further inform wider policy and practice in digital education.

The report recommends that any future programme or projects should:

- Establish clear goals for the use of digital technologies in schools. While this Programme sought to “demonstrate the innovative use of digital technologies”, it was not always clear what this might look like.
- Revise the application process and provide feedback to clusters on their applications, so that proposals are more realistic in light of the timeframe and the resources available to schools.
- Build in greater flexibility at the design phase to enable projects to change direction if their activities are not progressing according to their original plans. In particular, such an approach might adopt a phased funding release model, as is sometimes the case in other jurisdictions.
- Revisit future monitoring and evaluation requirements. While a project of this nature will require participating schools to complete regular reports, particularly in relation to their

project expenditure, it is recommended that this element is revised in the future. The clusters did receive guidance and advice from the Department, but this was deemed to be insufficient by some Cluster Coordinators who expressed a need for additional supports. This element of reporting also resulted in an increased workload for the Department, and they too need adequate resourcing to deal with such intensive projects going forward.

- Reconsider how funding is distributed to clusters of schools. While schools were readily able to draw down funding to procure digital technologies at the outset of the Programme, there were issues as time went on. The evaluation suggests revisiting the kinds of financial supports that might be made available to schools and how these could be drawn down over a prolonged period of time.
- Further strengthen the supports made available to clusters. While many of the supports, such as the role played by the Oide TiE Advisors, were successful the report identifies the need for a range of additional supports that might include technical support and critical feedback on project activities over time.
- Provide greater clarity on how projects are expected to disseminate the activities they engage with using digital technologies, so that the wider school community can learn from their activities.

Final word

The SEF Digital and STEM Programme was deemed an overwhelming success by those who remained active through to the end. The Programme enabled schools to work together to use digital technologies in a supported way across the curriculum. The majority of teachers who participated continue to use these technologies beyond the Programme and believe this will have a lasting impact in embedding digital practices in their schools. The Programme has gathered a range of valuable practices, which are captured on project websites and in the case-studies and teacher stories. These could be further developed by the Oide TiE team to inform all schools in the future.

New and emerging technologies, such as Artificial Intelligence, are now impacting on a range of school practices and there is an ongoing need to explore what impact, good and not so good, they can have on educational practices now and in the future. Thus, there is a need to engage in further such innovative projects and to continually evolve the model so that it works for all stakeholders, the Department of Education and Youth, support services and, most importantly, for schools.

Section 1: Introduction

H2 Learning was contracted by the Department of Education to carry out an evaluation of the Schools Excellence Fund Digital and STEM Programme (SEF Digital & STEM) and began work on the evaluation in March 2019. H2 Learning shared the Interim Evaluation Report in September 2020, with the expectation at that time of producing a final evaluation report by late 2021. The purpose of the Interim Evaluation Report was to test the evaluation approach, to identify emerging trends, to flag any issues that might need additional attention during the final phase of the project, and to give the Department an opportunity to comment on the emerging report. However, the rolling school closures that occurred between March 2020 and Spring 2022, a result of the COVID-19 pandemic, severely disrupted the project and hence the production of the Final Evaluation Report.

This Final Evaluation Report builds on the Interim Evaluation Report, providing further insights into the effectiveness of the SEF Digital & STEM Programme. It describes some of the issues that arose during the Programme and provides recommendations for consideration in developing possible future initiatives in this area. The Final Evaluation Report includes elements of the Interim Evaluation Report, particularly those elements which illustrate the richness of the activities that took place in schools, and data gathered from Cluster Coordinators and PDST Advisors[4] (later renamed Oide TiE Advisors), at the end of the project in June 2022.

The Final Evaluation Report also includes the three case studies created for the Interim Evaluation Report, as these illustrate the innovative nature of the selected projects. These case studies were selected to capture the range of innovative practice across the Programme under the criteria outlined in the European Commission report, entitled *Study on supporting school innovation across Europe (2018)*: Students, Teachers, Schools Leaders, Shared Vision and Wider Partnerships[5].

The remainder of this report is laid out as follows:

- **Section 2:** SEF Digital & STEM – Programme overview
- **Section 3:** Evaluation approach
- **Section 4:** Interim Evaluation Report- Key findings
- **Section 5:** Post-Project Survey - Experience of the project
- **Section 6:** Post-Project Survey - Lessons learned
- **Section 7:** Post-Project Survey - Possible future digital education initiatives
- **Section 8:** Recommendations
- **Appendix 1:** List of SEF Digital & STEM clusters
- **Appendix 2:** Department of Education SEF Digital & STEM documentation
- **Appendix 3:** SEF Digital & STEM case studies
- **Appendix 4:** SEF Digital & STEM teacher stories
- **Appendix 5:** Post-project Survey

[4] In the initial phase of the project the schools were supported by the Professional Development Service for Schools (PDST) and in the final phases of the project this organisation was restructured into a new entity Oide Supporting the Professional Learning of Teachers and School Leaders. Within SEF Digital the clusters were supported by Oide Technology in Education (TiE). For more, visit https://oide.ie/digital-technology/home/

[5] PPMI. (2018). *Supporting School Innovation Across Europe. Final Report to DG Education and Culture of the European Commission*. Publications Office of the European Union. https://doi.org/10.2766/466312

Section 2: SEF Digital & STEM – Programme overview

The *Digital Strategy for Schools (2015-2020): Enhancing Teaching, Learning and Assessment* contained an action, 3.4, to encourage a culture of innovation in schools in relation to digital learning. The Department launched the Schools Excellence Fund Digital (SEF Digital) in December 2017 and subsequently announced in April 2018 that 265 schools were to be funded through the programme. Subsequently the Department took the decision to expand the programme to the SEF Digital and STEM programme.

The Department document, *Description of Scheme – Schools Excellence Fund – Digital*, set out the purpose of the scheme as follows[6]:

> *"The purpose of the scheme is to demonstrate the innovative use of digital technologies in teaching, learning and assessment, through clusters of schools collaborating on specific projects".*

The Programme also included 10 STEM clusters, who used a range of digital technologies to support STEM education practices. The document then set out more detailed objectives organised in several categories (i.e. scope, purpose, outcomes and expectations), and these are set out below. The evidence related to how each of these specific objectives were met is embedded within the report. In the Interim Evaluation Report an explicit mapping of evidence to the specific objectives was provided, but this has not been reproduced here.

Programme objectives

Scope

Schools were initially invited to apply for inclusion in the Programme, where they could collaborate on the design and delivery of specific projects that utilised digital technologies in an innovative way in the context of:

a) Post-primary – curricular reform and innovation at Junior and Senior Cycles.
b) Primary – curricular reform of Primary Languages Curriculum, Primary Maths Curriculum and innovative delivery of existing subject areas.
c) Projects involving cross-sectoral topics.
d) Projects relating to key policy initiatives (e.g. STEM report).
e) Projects that reflect the diversity of the educational environment.

Purpose

The projects were to be of three-year duration (i.e. commencing September 2018 and concluding in June 2021) and this included the evaluation and dissemination phases, and they were to demonstrate how digital technologies can be used to enhance teaching, learning and assessment practices and, particularly, how the outcomes from the projects can inform:

[6] https://www.oidetechnologyineducation.ie/projects-initiatives/digital-clusters/

1. Classroom practice in relation to the embedding of digital technologies and constructivist approaches to teaching, learning and assessment at primary and post-primary levels.
2. The infrastructural models that could best support innovation in relation to teaching, learning and assessment, communication, and collaborative planning.
3. The most appropriate support models that could be put in place to facilitate innovation in schools.
4. How the embedding of digital technologies in teaching, learning and assessment within and across all schools could be achieved.

It was considered that a three-year period would be the minimum time period necessary to assess local implementation and Programme impact at a national level and to inform longer-term decision making by the Department.

Expected outcomes

It was expected that the work of the clusters of schools would provide outcomes including some or all of the following:

a) Evidence of use of digital technologies to enable sharing of information and collaborative practice between teachers in the cluster/professional network, such as development of resources, blog posts, articles, and best practice videos.
b) Evidence of enhanced learning by students through teachers' engagement in action research, collaborative and independent learning, including discovery and problem-based learning by students/pupils in each school in the cluster, mediated through the use of a range of digital technologies. Outcomes of the various projects will include evidence of the students' own work.
c) Evidence of shared learning/teaching experiences between the schools in each cluster. For example, this may include joint videos produced, collaboration on how particular aspects of new curriculum might best be delivered especially through the effective use of technology.
d) Evidence of use of technologies (hardware, software, or mechanisms to share the use of technology between the schools in the cluster) to demonstrate how such technologies can be used effectively. Report output from each cluster (which may take the form of an evaluation of their chosen technology, or video demonstrations on the effective use of those technologies).
e) Evidence of how linkages with third level institutions enhance the design, development, delivery and the evaluation of the projects.
f) Evidence of best practice in relation to forming effective links with business and industry.

Programme expectations

In rolling out the Programme, the Department provided a range of advice and support to schools, through the Professional Development Service for Teachers (PDST) in the area of Technology in Education (PDST TiE), which is now Oide Technology in Education (Oide TiE)[7]. For

[7] https://www.oidetechnologyineducation.ie/

consistency, this report uses the term Oide TiE when referring to this team of advisors and their role in the Programme.

The clusters were expected to:

a) Develop clear project implementation plans, with timelines, identified targets, and estimates of associated costs.
b) Establish links with local businesses, industries, and third-level institutions where possible.
c) Maintain a focus on how the use of digital technologies can enhance teaching, learning and assessment.
d) Develop and disseminate a range of project outcomes.
e) Disseminate the project outcomes.
f) Engage with ongoing evaluation.
g) Report on progress to an agreed schedule.

Programme funding

In 2018, €1m was allocated for the SEF Digital & STEM Programme, comprising €750k to issue to schools via grants for relevant infrastructure and associated training required, support for events etc. Schools were required to submit budget and project plans, outlining how the funding would be utilised. The budgets were reviewed and approved, with the approval and agreement by schools to the terms of the programme, including progress reporting etc, confirmed via a letter signed by a department official and a representative from the lead school in each cluster. The funding varied by project, based on the number of schools involved and the nature of the infrastructure needed, with the average grant being €31k per cluster. As the allocation for the programme fell into 2018, with no guarantee of ring-fenced funding for years 2 and 3, the bulk of funding issued in year 1, and this was advised to schools during the budget submission process.

The balance of €250k was allocated within the Department to provide substitution cover so that participating teachers could be released for related activities. Cluster Coordinators could take up to nine days annually, with up to five days available for one teacher in each of the other schools in the cluster.

Programme timeline

The actual timeline of the project is presented below with the key milestones in terms of reporting and project data gathering that took place between December 2017 and September 2024. The original planned timeline was severely disrupted by the number of rolling school closures due to COVID-19.

Year 0: 2017-18

- Call for proposals released by Department of Education - Dec 2017
- Schools selected to commence their projects in September - April 2018

Year 1: 2018-19

- Project commencement - Sept 2018
- H2 Learning appointed as evaluator - March 2019
- SEF Digital & STEM Event Athlone - May 2019

Year 2: 2019-20

- School closures due to COVID-19 - March 2020
- Survey (Cluster Coordinators and Oide TiE) - June 2020
- Collection of online data use – Summer 2020
- Internal Interim Evaluation Report - Sept 2020

Year 3: 2020-21

- Further school disruption due to COVID-19

Year 4: 2021-22

- Gradual reopening of schools during the school year - Sept 2021
- Croke Park SEF Digital & STEM Event - April 2022

Year 5: 2022-23

- Post-Project Survey administered to Cluster Coordinators and Oide TiE staff - June 2023
- Initial draft of Final Evaluation Report shared with the DE – Sept 2024

Figure 1: SEF Digital & STEM Project Timeline.

Section 3: Evaluation approach

COVID-19 and the associated school closures had a major impact on the formulation of this report, resulting in the production of an extensive Interim Report in September 2020. The Interim Report recognised that there could be additional school closures and that this might impact negatively on further data collection and on some clusters completing the project. Thus, the extensive data collected during this initial phase was very important and captures where clusters were at this key juncture in their projects.

As expected, COVID-19 continued to disrupt the Programme, and a further round of data collection took place in 2022 and 2023. Evaluators had not initially intended to conduct two distinct phases of data collection; this was necessitated by pandemic related disruptions. While many clusters continued to remain active and completed many aspects of their projects, some clusters did not remain active. The approaches adopted in collecting data from the clusters during these two phases is outlined below.

Phase 1

The first stage of the evaluation collected and reviewed data from a wide range of sources as outlined below. The DE shared the original cluster proposals and project plans[8] and all the completed progress reports and expenditure reports that had been submitted. This was supplemented by reviewing school dissemination material, interviews, focus groups and the collection of survey data. The dataset therefore consisted of:

- The proposals (40 out of the 41 were received and reviewed), and three-year plans (40/41).
- Twice-yearly Progress Reports (57/82) and Expenditure reports (32/82).
- Websites, Twitter accounts and other social media accounts associated with the clusters (as identified in documentation, surveys, or online searches).
- Discussions with Department officials.
- Discussions and focus group with Oide TiE Advisors (6-11-2019, 13-12-2019).
- Attendance and data collection at SEF Digital & STEM Event in Athlone Institute of Technology (now Technological University of the Shannon (TUS) Athlone) 29-5-2019.
- Cluster Coordinator Survey April 2020 (35 respondents from 41 clusters - 85% response rate).
- Oide TiE Advisor Survey June 2020 (9 respondents).

The steps in our analysis of this data were as follows:

- Initial systematisation and preliminary review of the proposals, reports and online data.
- Identification of important themes through discussion with the Department and the Oide TiE Advisors, and discussions and observations at the Athlone event.
- Use of identified themes to inform the development of a survey for Cluster Coordinators.

[8] Note the evaluation reviewed 40 of the 41 proposals and project plans.

- Selection of three clusters judged particularly active and with diverse data, to develop case studies. This enabled testing and refinement of the themes, examining progress in terms of potential for supporting innovation.
- Detailed analysis of the Cluster Coordinator survey, with coding developed both top-down (thematic) and bottom-up (textual) and amalgamated into larger categories. Frequencies quoted in this report refer to numbers of clusters (one coordinator per cluster).
- Identification of data gaps relating to support received, leading to an additional survey with Oide TiE Advisors.

The Key findings identified in the Interim Evaluation Report are summarised in Section 4 of this report.

Phase 2

The second stage of the evaluation took place after the period of school closures because of the COVID-19 pandemic, and the subsequent impacts on schooling post-pandemic. The period of the Programme was extended for a year because of the challenges schools had faced, and most clusters resumed work on their projects once disruption passed. Few further progress reports were received in this period, and projects were not able to follow up fully on the main Interim Report recommendations (notably those related to dissemination and project evaluation). Two further data collection exercises were undertaken, firstly at a Croke Park event in 2022, and a Post-Project Survey in 2023.

Croke Park event

In April 2022 an event was held at Croke Park bringing the clusters together with a focus on supporting projects in dissemination, and a beginning was also made in sharing reflections on the progress of the projects and lessons learned. These discussions informed the thinking of the evaluation team in constructing a post-project survey.

The event provided an opportunity to bring projects together in a face-to-face setting, and for them to share their experiences over the course of the Programme. The event was extremely successful, and it showcased the progress that clusters made in relation to using digital technologies, despite the challenges they encountered during the COVID-19 pandemic. The event also provided a forum for the DE and H2 Learning to outline their plans for the final phase of the project, where they planned to survey all project clusters and to gather a selection of teacher stories from the clusters which were designed to showcase the work schools engaged in over the lifetime of the Programme. The event also provided reassurance to the participants in relation to what they achieved during the lifetime of the project, as many felt that COVID-19 had disrupted their plans and their activities. Despite this disruption, the majority of clusters adapted their plans and continued to engage with the support services to complete their projects.

Following the event, H2 Learning developed the post-project survey to capture sample teacher stories. This work then informed the final data collection phase which was carried out during the 2022-2023 school year.

Post-project survey

The post-project survey (see Appendix 4) was sent to Cluster Coordinators, teachers and Oide TiE Advisors, with tailored questions for each group. The survey consisted of both closed (multiple choice) questions and open questions. The open questions explored three themes:

1. Experience of the project
2. Views on lessons learned (classroom practice, infrastructural models, support models and embedding digital education)
3. Views on possible future digital initiatives.

In total 61 responses were received, 36 from Cluster Coordinators, 16 from other teachers and 9 from Oide TiE Advisors. The responses related to 34 clusters from the original 41 clusters.

The results of the quantitative data from the closed questions are summarised as bar charts and are presented in Section 5. Responses to the open questions were grouped under the three main themes and examined using thematic analysis to identify sub-themes. Representative responses are included in this report to illustrate key findings. The results related to experience of the project are presented in Section 5, those related to lessons learned are presented in Section 6, and those related to possible future digital initiatives in Section 7. Frequencies quoted usually refer to numbers of responses – which includes those from Cluster Coordinators (sometimes more than one per cluster), teachers and Oide TiE Advisors, unless otherwise indicated.

Teachers' stories

At the same time as the post-project survey, Cluster Coordinators were also asked by Oide TiE to create short project descriptions as 'Teacher Stories' for publication on the Oide TiE website. Twelve of these were received and reviewed to provide additional background information on specific projects. They are presented in Appendix 3.

Section 4: Interim evaluation report – Key findings

The Interim Evaluation Report captured a wide range of data from the Cluster Coordinators on the makeup of the clusters and the activities they engaged in. This data was gathered from 35 of the 41 Cluster Coordinators, and Table 1 below captures the distribution of the responses.

The Interim Evaluation Report described the features of the cluster projects: the use of sophisticated technology (such as iPads, Lego WeDo kits, VR headsets), improved learner engagement and problem-solving skills; engagement of young women in STEM; examples of teacher collaboration; developments in curriculum planning; and also provided detailed case studies of the work of three of the clusters. The report also collected evidence that:

- Most of the teachers involved in the project found the experience very positive, that it fostered good links between teachers across different schools, facilitating knowledge and expertise sharing, and that it advanced students' digital skills.
- The SEF Digital & STEM Programme experience helped the move to online learning during lockdown, as learners and staff had developed skills in the use of online learning tools through project activities.
- The clusters involved in the Programme were successful in disseminating their activities to the wider community, using a mix of websites (60%) and social media (mainly using Twitter)[9]. Some clusters also participated in or initiated large-scale public events.

Table 3: Response to Cluster Coordinator Survey.

	All Post-primary	All primary	Mixed post-primary led	Mixed primary-led	TOTAL
All	14	12	8	7	41
Responses	11	11	6	7	35
Percentage	79%	92%	75%	100%	85%

Of the STEM projects 8 out of 9 responded, and of the Digital projects 27 out of 33 responded.

Clusters and schools

A total of 41 Clusters were funded, 33 Digital projects and 8 STEM projects. Of these, 35 responded to the Cluster Coordinator Survey in April 2020, which was reported as part of the Interim Evaluation Report. While some schools struggled to complete their End of Year Progress Reports for 2019, they were still active in the project, while two clusters withdrew from the project by the end of Year one.

[9] Sixteen Clusters used Twitter, though many of these only tweeted a few times, or tweeted occasionally using the general school Twitter account. Some, however, were very active, nine Clusters tweeted more than 70 times in the period from the project start to 1 July 2020, and four of these tweeted more than 200 times.

There were 222 schools in the 41 clusters that were funded. Of these 86 were post-primary (including 1 Further Education and Training College) and 136 were primary[10]. As to the composition of the 41 Clusters, 11 consisted of all post-primary schools, 13 of all primary schools, and 17 were mixed (post-primary led – 10, primary led – 7).

The total number of schools in Ireland in 2019 was: 723 post-primary schools, 3,106 primary Mainstream Schools and 134 Special Schools. These figures indicate that 12% of post-primary schools were involved in the SEF Digital & STEM project, 4% of Primary Schools and 1% of Special Schools.

Across Ireland, 890 schools were included in the DEIS programme[11] in the 2019/20 school year. These compromised of 692 primary schools (334 urban and 358 rural) and 198 post-primary schools[12]. The numbers of DEIS schools in the project are shown in Table 4.

Table 4: Representation of DEIS schools participating in SEF Digital & STEM.

Schools in Ireland			SEF Digital & STEM schools		
Primary (Mainstream)			**Primary (Mainstream)**		
Total	**DEIS**	**% DEIS**	**Total**	**DEIS**	**% DEIS**
3,106	692	22%	135	38 *Urban Band 1 (14)* *Urban Band 2 (13)* *Rural (11)*	28%
Post-primary (not including FET)			**Post-primary (not including FET)**		
Total	**DEIS**	**% DEIS**	**Total**	**DEIS**	**% DEIS**
723	198	27%	85	25	29%

These figures would suggest that the representation of DEIS schools in SEF Digital & STEM was broadly representative of the number of DEIS schools in the national population of schools.

The Cluster Lead School had a DEIS classification in 13 Clusters, that total was comprised of 8 post-primary schools, and 5 primary (1 Urban Band 1, 2 Urban Band 2, and 2 Rural).

[10] https://www.education.ie/en/Publications/Statistics/Data-on-Individual-Schools/Data-on-Individual-Schools.html)

[11] *DEIS - Delivering Equality of Opportunity in Schools: an action plan for educational inclusion* is the main policy initiative of the Department of Education to address educational disadvantage at school level. The Department of Education introduced it in 2005. The plan brought together several programmes which targeted educational disadvantage throughout the public schools' system under the framework of Delivering Equality of Opportunity in Schools (DEIS).

[12] https://www.education.ie/en/schools-colleges/services/deis-delivering-equality-of-opportunity-in-schools-/

Numbers of teachers and learners

The Cluster Coordinator Survey indicated that the number of teachers and students involved in each cluster varied considerably between clusters.

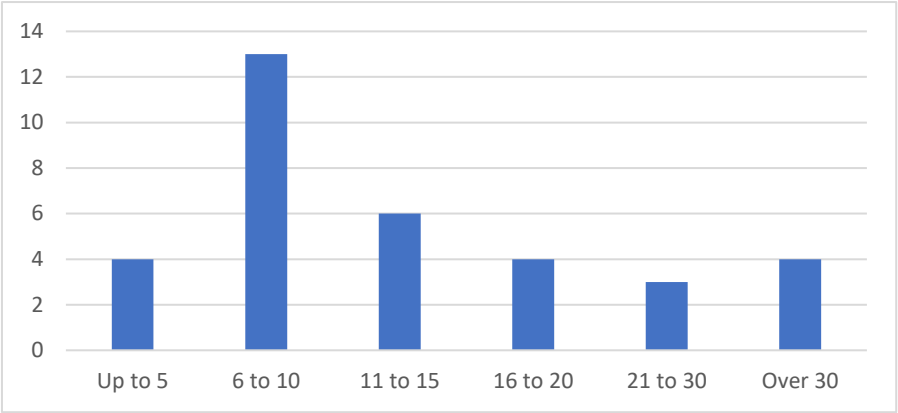

Figure 2: Number of teachers involved in a cluster

Most clusters had 6-10 teachers (see Figure 2) and involved 101-200 students (see Figure 3), which suggests an average class size of 20 students.

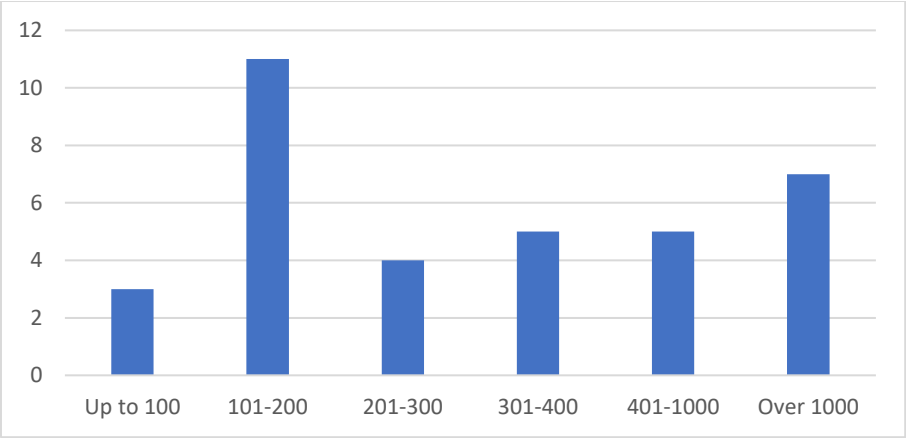

Figure 3: Number of students involved in a cluster.

However, there were outliers with large numbers of teachers and/or students. A scatter plot (see Figure 4) shows that the number of students involved was not closely related to number of teachers, some projects (e.g., those centred around coding) worked intensively with smaller numbers of learners, whereas some (e.g., those producing a database of resources) worked with relatively few teachers and their resources were then used by a large number of learners.

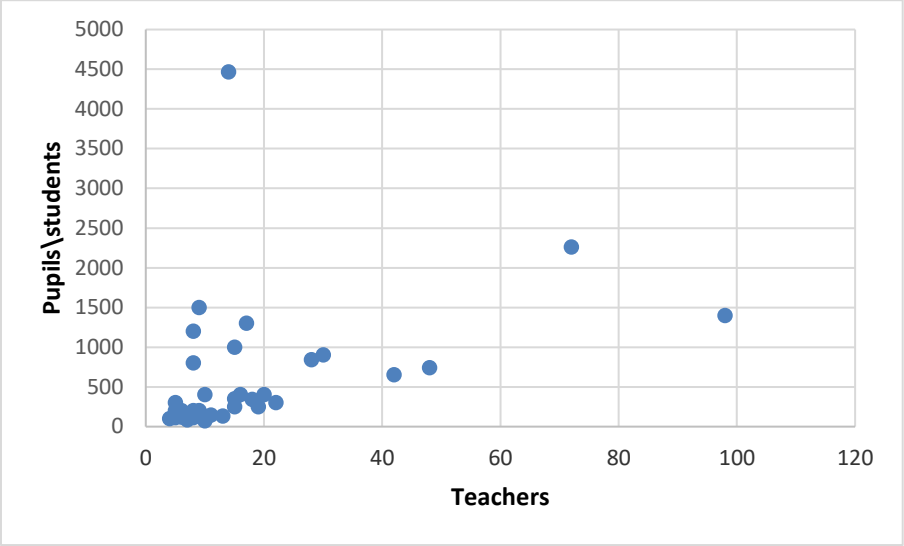

Figure 4: Number of students involved in a Cluster by number of teachers in the cluster.

Most frequent terms used to describe projects

Below is a word cloud for all the keywords used by Cluster Coordinators to describe their projects, where the font size of each word is directly related to its frequency of occurrence.

Figure 5: Word cloud of keywords used by Cluster Coordinators to describe their projects.

The frequency of occurrence of the **five topmost keywords used** is given in Table 5.

Table 5: Frequency of five topmost keywords

Keyword	Frequency
Collaboration	23 times
Innovative	17 times
Creative	11 times
Fun	9 times
Student-centred	7 times

Thus, the majority of Cluster Coordinators from the outset highlighted the collaborative, innovative, creative, fun and student-centred nature of their projects. These elements continued to dominate the active projects over the lifetime of the Programme.

Most innovative elements of the project

The Cluster Coordinators were asked in the Interim Evaluation Report to identify the most innovative elements of their projects. Their responses could be classified in **six** categories:

1. Use of sophisticated technology (e.g., the use of equipment/infrastructure allotted for the project e.g., use of iPads, Lego WeDo kits, VR headsets etc.)
2. Inter-cluster collaboration
3. Teacher collaboration
4. Upskilling of learners (e.g., increase in learners' skills e.g. project management, problem-solving, 21st-century skills)
5. Curriculum planning
6. Engagement of young women in STEM careers.

The following graph indicates the number of responses in each of these categories.

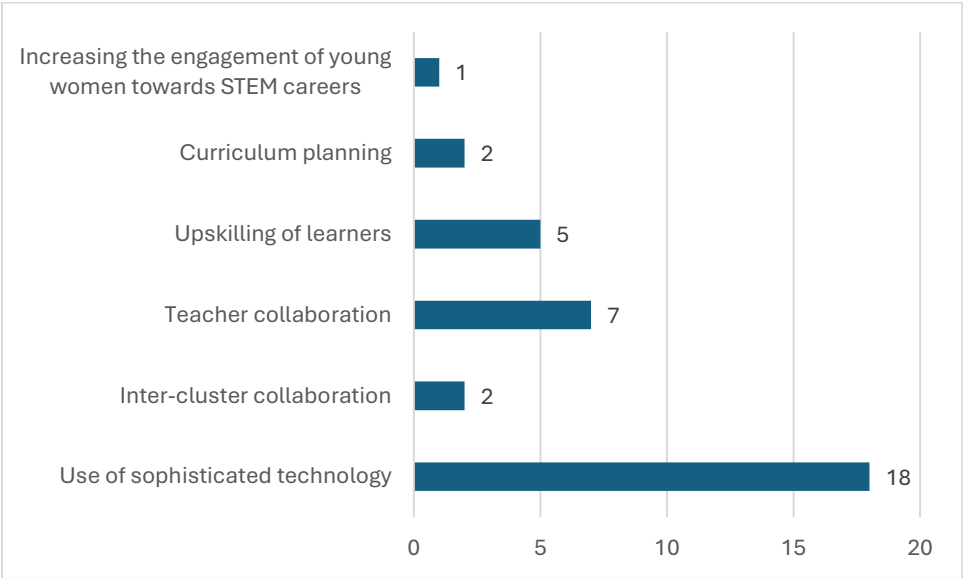

Figure 6: Most innovative element of the project.

Interestingly, the use of specific technologies ranked first, closely followed by the inter-cluster collaboration and teacher collaboration. Other benefits cited included support for curriculum planning and in increasing the engagement of young women in STEM careers possibilities. Collaboration featured strongly in the early stages of the Programme.

Project achievements and issues

Despite the challenging climate in which the project took place, the Cluster Coordinators reported positively on their achievements, while also identifying a range of issues that emerged in the initial phase of the project.

Gains

All 35 clusters reported important gains from taking part in the project. 22 out of 35 clusters (63%) described their experience as very positive. While schools may have overreached in terms of their project activities, the project provided a focus and opportunity for schools to work together and developing teacher knowledge and associated practices.

> *"So far the SEF has given all schools a very positive experience. Despite the fact that we possibly overreached with content all schools report very positive benefits in terms of student advancement in ICT. Strong personal and professional bonds have been established between staff members in all schools. Significant spin off benefits in terms of staff sharing knowledge and expertise in areas well beyond the scope of the project."*

> *"It has been inspirational to see how teachers and pupils from 4 quite different schools have come together into what is now a movement in our local educational community. We were already doing some of these activities in a very occasional and ad-hoc way. SEF gave us an opportunity to do this in a properly planned and structured way, gave us the funding to put the basic resources in place and supported us along the way, in particular with the much-appreciated assistance of our Oide TiE mentor."*

The gains in terms of teacher collaboration and professional networking were mentioned frequently (21 respondents - 60%) and probably constituted the clearest gain for many of the projects.

> *"I have personally enjoyed the professional challenge that coordinating this project has afforded me. Taking on a leadership role has given me great confidence in my abilities and I look forward to embracing similar opportunities in the future. Working with other schools as part of a cluster has been hugely beneficial, sharing knowledge, resources, experience and best practice. We have formed a strong professional network, and I have learned so much from this process. I know that I can call on my STEM Champion colleagues for advice and support at any time."*

Eighteen Coordinators (51%) commented on the innovations in teaching they had made.

> *"Our digital cluster are working together to create a bank of practical and engaging STEM lessons that can be used in Primary Schools. We have analysed the impact of this innovative pilot project on both teachers' and students' learning using qualitative (interviews) and quantitative (surveys) research methods to assess how and if their skills and attitudes have changed as a result of their participation with a view to spreading our project to other digital clusters, schools nationwide and beyond to Europe through involvement in Erasmus+ projects."*

The creation and/or sharing of resources was mentioned by 16 Coordinators (46%). The creation of learning materials is typically undertaken by teachers in their own time and often on their own, but it is interesting to see that more collaborative approaches can work effectively.

> *"Our project is a simple one, but the process has allowed all 6 schools to collaborate in such a way so as to create something of vast importance to all of the schools in our network."*

> *"We have discovered a great source of resources together and really branched out."*

> *"Digital citizenship and the development of digital literacy is very much at the fore of our project so to date participating teachers sourced/created learning resources and shared with the teachers in the other cluster schools by uploading to the cluster email."*

Other gains mentioned by a small number of clusters included development of leadership capacity, increased confidence in use of technology in teaching and learning, and contacts and links made that would continue into future collaborations.

> *"3 out of 4 of the lead teachers participating in this project are not post holders. This group thus improves and strengthens middle management structures in all schools. Training and mentoring empower school leadership."*

> *"Teachers who never really used technology in their pedagogy are now using it more and more. This is an iPad project, and teachers are using the iPads for other areas of learning."*

> *"Looking to the future I hope that I can share the outcomes of this project (with other schools and parents). We plan to continue our learning cluster into the future (after the initial three years) and work on other projects together. This is just the beginning."*

However, not all Cluster Coordinators had a positive experience, one for example stated:

> *"My experience to date has been mixed, to be honest!"*

The next section describes the main issues that were described by the Cluster Coordinators.

Reported issues

Just over half of the clusters (N=18, 51%) reported a range of issues in the running of the Programme, with many reporting more than one issue. A summary of the reported issues is presented in Table 6, followed by a detailed description of each.

Table 6: Issues reported by Cluster Coordinators.

Issues	Number of clusters reporting
COVID-19	8
Time constraints	7
Project management	7
Finance	2
Lack of support	4
School level	5
Technology related	3

As was to be expected, COVID-19 had impacted negatively on many clusters (N=8, 23%).

"The project was somewhat stunted this year (2020) with uncertainty and lack of drive, staffing changes and now the extended school closure."

Time was a big issue for some Cluster Coordinators (N=7, 20%):

"On the flip side, I've found the workload as coordinator extremely demanding, given the huge range of tasks involved ranging from equipment research and procurement, through coordinating across four different schools - with different needs and priorities - to facilitating the students with their various projects. I found the hours allocated for the project to be wholly insufficient for the workload involved."

"Perhaps hours should be pro rata, as clusters attempting a whole school role out should be given more time than if just one teacher per school is participating."

"The substitute days for training are paramount to the future success of the project."

Project management, including organisation of the clusters, management of goals (which were sometimes recognised as having been over ambitious), and uncertainty was an issue for some (N=7, 20%).

"It has been challenging to gain and maintain momentum in a distributed cluster and in a SEF process that seems to be evolving."

"At times I feel that perhaps we have not achieved some of the goals as originally set out in our application, but it has been a learning experience for everybody, and we have had to change our goals as time has passed."

Finance was mentioned as an issue by two clusters (6%).

"In our own case there was much uncertainty surrounding funding right through until January 2020 - 5 months into the school year."

Some clusters felt that they sometimes did not have sufficient support (N=4, 11%). The types of support mentioned included: the frequency and content of professional development on offer from the Oide TiE Advisors, the unforeseen issues related to substitute cover, the possibility of engaging more deeply with other schools within their cluster.

"The Oide TiE Advisor can only provide 1-day CPD per school. This appears to be the same number of days regardless of the scale of the role out. For the high number of teachers (almost 100) participating in the project, not all teachers and classes will avail of this service. Perhaps hours should be pro rata, as clusters attempting a whole school role out should be given more time than if just one teacher per school is participating."

"The Oide TiE need to be providing clusters with the latest training available (on par with our European counterparts) and when schools request training in a particular area related to their project it should be available by Oide TiE team or facilitators sourced free of charge."

"In order to have meaningful collaboration in a school, teachers need to be able to bounce off each other rather than train each other. True collaboration then is not simply to work with someone but to 'think' with someone; to build idea on top of idea with others so you all get to a point you couldn't have reached alone."

School level issues included lack of support from school management (N=2, 6%), and staff turnover (N=3, 9%).

> *"Some members of staff were more enthusiastic than their school management."*

> *"Most participating schools have experienced a high number of new staff and also staff changing class level. A significant amount of time at the start of the year was spent with staff familiarising themselves with the project and developing the necessary skills."*

Technology inevitably had its problems (N=3, 9%) including software changes and delays, and weak technical infrastructure. Many of these issues were outside the control of the DE and the school clusters and involved issues with commercial software and learning platforms, in addition to ongoing challenges around technical support for schools and access to reliable broadband.

> *"Although it was intended to introduce students to the ... app earlier in the project, it is still in development with only intermittent access."*

> *"This project now has undertaken another substantial change since December 2019 when news of the Learner Management Software platform being used ... announced its intention to cease trading with limited functionality until June 2020. This obviously frustrated many of the teachers involved in this project since."*

> *"To deliver a progressive project successfully we found that the necessary infrastructure needed to be in place for it to work throughout school buildings. A strong and stable broadband connection, managed Wi-Fi throughout schools with up-to-date devices."*

While these issues are beyond the control of the project, it is important to note that they did impact negatively on the experiences in schools.

Critical reflection and evaluation

Only four clusters (11%) made a specific reference to evaluation. This issue of evaluation or critical reflection was often ignored or interpreted narrowly by schools, and they seemed to put greater emphasis on project implementation than on systematic review.

> *"We have analysed the impact of this innovative pilot project on both teachers' and students' learning using qualitative (interviews) and quantitative (surveys) research methods to assess how and if their skills and attitudes have changed as a result of their participation with a view to spreading our project to other digital clusters, schools nationwide and beyond to Europe through involvement in Erasmus+ projects."*

> *"These four schools are using the School Self Evaluation process to develop, implement and track oral language, writing and reading in various class groupings."*

Project communication

Six (17%) of the clusters described their use of cluster meetings and training meetings to support the project:

> *"We meet three times each term as a cluster. The first meeting discussed specific dates, outcomes and events for the term/year ahead. The second meeting is usually a training day to upskill teachers or pupils in different digital tools, technology or approaches."*

> *"Teachers continuously engage with each other via the cluster email. Feedback is gathered through this format and also through discussion at staff meetings and reverted to me by the co-ordinators in each school."*

Thirteen (37%) described the organisation of showcase events to share work within the school, with other schools or with the wider community. This is particularly noteworthy as it provided schools with a chance to come together and showcase the work of their learners to the wider community.

> *"At our first annual showcase, each primary school was able to demonstrate BeeBots in action, working WeDo engineering models, working programs written in Scratch on Raspberry Pi computers as well as some additional IT projects e.g. VEX robots, Music programs, Lego Mindstorm Robotics and Python programming on Raspberry Pi computers."*

> *"End of Year Showcase Event - June 2019 - An Róbó Seó. As a showcase event to highlight the work done by the cluster all of the participating cluster teachers and their classes set up in the auditorium of DCU (St. Patrick's Campus) to display their Mission Moon projects. An Róbó Seó was the culmination of 9 months of collaboration between the Cluster Group, Prof. Deirdre Butler, Learnit and the Oide TiE. Over 320 children from the 6 participating schools, divided into 75 groups, were present. It was the largest ever First Lego League Junior event in Ireland or the UK and the largest STEM event to take place almost entirely through the medium of the Irish Language. Reviewers from Gael-Linn, COGG, Foras na Gaeilge and Gaeloideachas were in attendance on the day."*

Seventeen (49%) described their use of websites and blogs to disseminate their activities to the wider community. This figure is broadly consistent with searches at the time, which identified 61% of clusters overall with a public website. Where such websites and blogs existed, they provided staff and learners with multiple learning opportunities, while also enabling the schools to connect to a global audience.

> *"The primary teachers conduct, record and document any practical activities undertaken, and the work is recorded currently on the virtual learning platform 'Edmodo' with the intention being to populate our cluster website with all materials."*

> *"We are also using websites to communicate to the parents/students."*

> *"The cluster website has gone live, and teachers can upload resources/links along with samples of the children's projects."*

The creation of the website itself was also viewed by others as being a useful learning experience for teachers and students:

> *"Creating a website has been a huge learning experience for all involved."*

Supporting online learning

Whilst COVID-19 had a negative impact on many of the projects in June 2021, the work of the projects also supported schools' transition to online learning during the initial phase of school closures from March 2020 to June 2021.

Most respondents (N=26, 74%) said that the SEF Digital & STEM Programme experience helped with the online learning approaches used during lockdown.

> *"Yes, the collaborative features employed in our different platforms have been used extensively. The project has aided remote learning hugely as the students became quite accustomed to using the online learning tools through project activities."*

"It has been most beneficial to have digital portfolios set up and children/parents familiar with them. Seesaw has been the main means for communication used between primary school teachers and their students/parents in our cluster. Microsoft teams is being used by Secondary school staff and students who set up this platform with the support of our cluster Advisor during our project. These digital portfolios are being used across the curriculum not just PE however some of our schools have been busy with on twitter using #ActiveHomeWeek for primary schools and GAA Future leaders' programme with St Nathy's Secondary School. The digital portfolios have allowed teachers to correct & grade schoolwork returned digitally and give encouragement to students with comments. However not all students have the internet and devices to use these."

"Primary Level - Yes, we had been using Seesaw as an e-portfolio for the pupils for use in school but had not extended it to give access outside the school. We have found this an invaluable tool when thrown in the deep end of distance learning and teaching with very little notice on March 12th. The pupils were already familiar and proficient in using this app, so it was less pressure on parents trying to access the information from the school. Teachers can assign work, and pupils can return work through the app allowing the teacher to correct and comment on each pupil's individual work. The secondary school is using TEAMS and PowerPoint screen record which is great for making videos."

However, four of the clusters (11%) commented that they were unable to continue work on the projects because of issues around access to technology:

"No, as the iPads are within the schools which are out of bounds at the moment."

"There is an issue around access to internet devices for many of the pupils in our schools."

"No. Children don't have access to the Lego WeDo kits. In September (if school days are staggered) we hope to work out a rota where children can bring the materials home to create."

Overall, participation in the SEF Digital & STEM Programme enabled schools to transfer many existing practices online during COVID-19.

Curriculum and goals

Curriculum areas

Table 7 shows the curriculum areas which clusters mentioned in their accounts. The total number of mentions is greater than the number of clusters because some clusters indicated more than one curriculum area. The largest grouping was that associated with Science, Mathematics and the related areas of STEM (Science, Technology, Engineering and Mathematics), STEAM (Science, Technology, Engineering, Arts, and Mathematics) and SESE (Social, Environmental and Scientific Education).

The focus on science and mathematics was in part explained by the fact that 8 of the projects were designated as STEM projects. One of the STEM projects did not mention STEM in their response, choosing to focus on problem-solving, but their initial proposal had identified maths and science as important areas for development of problem-solving skills.

Table 7: Curriculum areas mentioned by Clusters

Curriculum area	N	% of Clusters
Science, Mathematics, STEM, STEAM, SESE	19	54%
English, Irish	6	17%
Music	5	14%
Art	4	11%
Computer Science (including AI and User Experience Design)	4	11%
Geography	2	6%
History	2	6%
Physical Education	1	3%
Modern Languages	1	3%
Cross-curricula	4	11%
Curriculum planning	2	6%

Amongst the Digital projects 12 out of 27 (44%) focused on Science, Mathematics, STEM, STEAM, or SESE. The following quotations capture the range of activities that schools undertook from using BBC micro:bit technology[13], to embedding digital technology across the primary curriculum, to enhancing PE classes in post-primary to enhancing oral language and reimagining the Junior Cycle experience for students.

> *"Students are exploring STEM activities using a range of technology: 1. Scratch with Computer Science (CS) first. 2. Micro:bit programming with inventor kits. 3. Sphero programming 4. 3D Design and Printing. The project sees second level students using their skills to help STEM grow at primary level."*

> *The narration is recorded, paying attention to voice pitch and clarity. A Digital Story (DS) is created using programmes such as iMovie which is no longer than 3 minutes. The story provides evidence of learning. The DS assessment has a rubric that includes traffic lights for keywords, the main message of learning, and what it means to the creators. All work is original, can be symbolic, metaphorical, or literal."*

> *"We have enhanced the Physical Education Programme at both Primary and Secondary School level in our cluster through the use of the video app on iPads by students and teachers to give effective peer assessment and feedback in their PE lessons."*

> *"Our project is based on the use of iPads to implement and enhance the oral language curriculum in both English and Irish. In term one in each school, we focus on E-portfolios, namely SeeSaw. Students record themselves using the voice message for both English and Irish. In term two we focus on digital storytelling using apps such as iMovie, Stop Motion Studio, Chatter Pix, Doink Greenscreen and Toontastic. Classes in each school create movies in both English and Irish. In term three the focus is on Coding using Scratch. Students code animations and games and record their own voice within the animation."*

> *"It is the curriculum planning aspect in creating these Units of Learning and the conversations around such. The Framework for JC has required a mind shift change in educational thinking over the past number of years and as teachers grapple with this the*

[13] https://microbit.org/

fruits of their labour are reflected in the high quality of work presented to engage learners in a blended learning approach which is very fitting in the times, we now find ourselves in."

Goals

Outside specific curriculum goals, 22 (63%) clusters mentioned teaching goals related to digital literacies in a broad sense. In this category the analysis included a variety of terms and counted clusters using one or more of the following closely related terms: digital literacy (5 Clusters), digital skills (9), Key Skills (2), 21st Century Skills (6). Related terms were also noted, including: safe and ethical use of technology (3), digital citizenship (2), problem-solving (11), computational thinking (8), critical thinking (3).

"These lifelong skills of investigating, communicating, teamwork, reflecting etc. are explored and developed alongside the development of digital skills."

"Digital Citizenship and the development of digital literacy is very much at the fore of our project so to date participating teachers sourced/created learning resources and shared with the teachers in the other cluster schools by uploading to the cluster email."

Some clusters referred to wider and broader educational goals: gender equality (2), student engagement and motivation (6), student autonomy (3), insight into career paths (1), development of leadership capacity by students (1), involving students in their own learning (1). The following statements capture how the introduction of digital technologies acted as a trigger to develop new skills and insights for students.

"Students of ... are trained to become STEM role models for younger female students in primary schools. The STEM role models have visited all three primary schools twice this year."

"Through the project we are building leadership capacity amongst the students by training students as iHelpers and giving visibility to their role as peer supports for their fellow students."

"The opportunity for students to lead projects and have real and meaningful autonomy over their work both in terms of content covered but also choice around what subjects to attend (Choose your Own Subject -TY)."

"We set out to improve senior cycle students' access to real insight into college and career paths shared by college students and professionals in a variety of fields, so that they can make more informed and confident choices."

Pedagogy

Constructivism

The definition of constructivism in the *Digital Strategy for Schools, 2015-2020*[14] is described as follows:

> *"A constructivist pedagogical orientation supports teachers in effectively using ICT with their students i.e., learners are actively involved in a process of determining meaning and knowledge for themselves."*

The *Digital Strategy for Schools* further states: "constructivist teaching approaches aim to foster critical thinking and to create motivated and independent learners" and:

> *"Where constructivist teaching approaches are used, learning experiences often include the following features:*
> - *Engagement with the learner's prior understanding.*
> - *Active involvement of the learner in the learning process.*
> - *Opportunities for the learner to make decisions that affect the subsequent course of the learning activity.*
> - *A high level of interaction and exchange of ideas between learners."*

The report *Digital Learning 2020. Reporting on practice in Early Learning and Care, Primary and Post-Primary Contexts*[15] by the Department's Inspectorate describes constructivism as follows: "The Digital Strategy for Schools advocates the use of digital technologies in schools in a constructivist way. This involves learners using technologies actively, sometimes in collaboration with others, as part of their learning", and the report then goes on to describe what was observed in schools in terms of 'Active use of digital technologies' and 'Collaborative use of digital technologies'.

The term 'constructivism' was not mentioned by the Cluster Coordinators in the survey, though the closely related concept of constructionism[16] was mentioned by one Coordinator. For ease of comparison, the evaluation used the operationalisation of constructivism given by the Inspectorate and therefore considered 'active use of digital technologies' and 'collaborative use of digital technologies'.

Cluster Coordinators described students making active use of digital technologies in 34 of the 35 projects (97%), the exception was a teacher-based project creating resources for teachers to use. These resources may have involved active use of technologies by students, but this was not described in the response.

In 21 of the 35 clusters (60%) students were described as making collaborative use of technologies.

[14] https://www.education.ie/en/Publications/Policy-Reports/Digital-Strategy-for-Schools-2015-2020.pdf
[15] https://www.gov.ie/en/publication/c0053-digital-learning-2020-reporting-on-practice-in-early-learning-and-care-primary-and-post-primary-contexts/
[16] https://fablearn.stanford.edu/fellows/archive/fellows/blog/constructionism-learning-theory-and-model-maker-education.html

> *"The students primarily work in teams, with a wide degree of autonomy, with teachers helping to facilitate the creative process."*

> *"The children - presented their models, explaining how they designed and tested their rovers to complete a series of exploration-based tasks - compared models and provided feedback to each other - explained why each function allowed the rover to move over fluctuating terrain to complete the assigned/chosen task."*

It seems likely that in a number of other clusters students were working collaboratively but this was not made explicit in the descriptions, perhaps because it was taken for granted as standard practice.

The use of problem-based learning/enquiry-based learning and the use of real-world problems are often associated with constructivism, and the use of such approaches was noted in six clusters (18%):

> *"The Showcase Event in Year 1 saw the most innovation with Projects ranging from classic computer games such as "Pong", "Maze" & "Mole Mash" get re-incarnated in Scratch to real world problem solving using micro:bits to programme Sensors to alert you if your Plant needs watering!"*

> *"We chose a problem (e.g., pollution in the sea) and the students in their groups had to build/create a motorised vehicle/machine that would help solve the problem."*

Learning activities

The two largest categories of learning activities[17] were as follows:

- Student design and creation of artefacts (26 clusters, 74%)
- Coding and control (25 clusters, 71%).

Eighteen (51%) of the clusters mentioned using learning activities in both categories. Looking at the use of these learning activities by type of school, all 12 primary school clusters mentioned coding and control technologies, whereas 11 of the 15 mixed clusters and only 2 of the 8 post primary clusters did so. This indicates that the preponderance of the use of coding and control was with younger children.

> *"Build circuits and complete a range of programming activities on embedded systems (micro:bit)."*

> *"Each school was equipped Raspberry Pi computers for 5th and 6th classes, Lego WeDo robots for younger classes and BeeBots for the youngest pupils. At our first annual showcase, each Primary school was able to demonstrate BeeBots in action, working WeDo engineering models, working programs written in Scratch on Raspberry Pi computers as well as some additional IT projects e.g., VEX robots, Music programs, Lego Mindstorm Robotics and Python programming on Raspberry Pi computers. This year two schools had their code running on AstroPi computers in the International Space Station."*

[17] The responses to the survey did not provide much detail on the learning activities schools engaged in. In some cases, it is possible to get a fuller view of these learning activities via the Cluster websites, though this is often in the form of video and concentrates more on the outputs than the processes engaged in. Descriptions of some learning activities are given in the Case Studies in Appendix 3.

Student design and creation of artefacts included activities such as digital storytelling (3 clusters), digital photography (4), video production (9), student blogs (1), students creating games (4):

> *"In term two we focus on digital storytelling using apps such as iMovie, Stop Motion Studio, Chatter Pix, Doink Greenscreen and Toontastic."*
>
> *"Students code animations and games and record their own voice within the animation."*
>
> *"Children in primary classes 4th-6th and students in secondary classes 1st-3rd, learn a topic in science class. The learning is then independently reflected in the art class. In pairs the reflection is storied using personally created visuals, using any medium to hand render the artwork. These images are then developed as a storyboard, photographed, edited using photoshop, or other chosen programmes. A narration using the 4 Cs scaffold is developed to support the images. The narration is recorded, paying attention to voice pitch and clarity."*
>
> *"Some senior classes have also set up Class Blogs so that pupils can document their learning and also participate in peer assessment by commenting on each other's work."*

The use of e-portfolios was also common (10):

> *"The use of the digital portfolio was allowing the children to become competent and comfortable at expressing how their learning was progressing - language and critical thinking was developing and was evident in their videos."*
>
> *"This year we added in a digital portfolio (Seesaw) to be used by students to log their learning journey and also to help them to assess their own problem-solving skills."*
>
> *"The complete integration of many features of the Google platform (Sites, Slides, Drive, Docs, Classroom) to create eportfolios which function efficiently for learners and teachers and meet their intended objectives."*

Other pedagogical aspects

Twelve of the clusters specifically mentioned assessment in their descriptions, and some provided examples of peer assessment:

> *"The pupil's practical activity allows the teacher to assess the learning that has taken place while engaged in this activity."*
>
> *"The use of shared documents in Google Drive, Docs and Classroom to enable real-time teacher-student and student-student interaction/feedback toward the completion of a CBA in Modern Languages."*
>
> *"We have enhanced the Physical Education Programme at both Primary and Secondary School level in our cluster through the use of the video app on iPads by students and teachers to give effective peer assessment and feedback in their PE lessons."*

The importance of planning for progression was mentioned by six clusters.

> *"Moving from the Getting Started, where they learned to take on different roles when constructing a project, to the Guided Projects and eventually progressing to the Open Projects, supported by the design library and other resources online."*
>
> *"The pupils had lots of fun working through guided builds before moving onto more complex problem-solving builds of their own."*

Other aspects of pedagogy that were mentioned included blended learning (2 clusters), flipped lessons (1), whole school approach (2), TY students working as mentors for younger students (2), student centred learning (3), personalised learning (2), and importance of the student voice (1).

> *"Each of the subjects includes a module where students create subject content which then becomes part of a revision programme (flipped lessons) for the students in younger year groups."*

> *"To explore and disseminate good practice in terms of whole school approaches to ICT in teaching and learning."*

> *"A group of TY students from … visit local primary schools weekly and mentor young students in areas such as coding and G Suite. This year, our second year of the project, the emphasis is on G Suite - primary school students are learning to use Gmail, Docs, Slides and at the same time building a Google Site."*

> *"Each Blue-Sky module is delivered in a student-centred way supported by technology and through the use of teaching methodologies such as flipped classroom, project-based learning and portfolio-based learning."*

> *"A huge part of the monthly meeting is dedicated to student voice where we analyse surveys, focus groups results and example of work of each schools' students to see what we need to improve in our approach to create a Maker Space or STEAM room in the school."*

Schools, teachers and wider partnerships

Shared vision

The development of a shared vision across a school or cluster for the use of digital technology is potentially of great importance, but it was mentioned by only four clusters, two of which mentioned the role of the Digital Learning Framework in their school planning activities:

> *"In formulating our Digital Learning Plans both schools recognised the benefits of aligning the work of the cluster with the standards in the Digital Learning Framework and worked collaboratively on this aspect of their plans."*

> *"These four schools are using the School Self Evaluation process to develop, implement and track oral language, writing and reading in various class groupings."*

Teacher development

As noted earlier, teacher collaboration and professional networking were seen as an important gain from clusters' involvement in the Programme. The Cluster Coordinators were also very positive about other aspects of the project that contributed to teacher professional development – the support and training from the Oide TiE Advisors, cross-phase mentoring, cascade training, and involvement in the development of resources:

> *"The opportunity to engage with other schools and in particular with our Advisor, … [from] Oide TiE, was of enormous benefit."*

> *"Great support as well as hands on training in all areas has been made available to us by our Oide TiE contact."*

"Having a second level Physics teacher available to call to the primary schools to give a lesson or to simply explain the background theory to the teacher has helped to ease any sense of 'lack of knowledge' and has helped to improve teacher engagement."

"These workshops have been central to the model of learning, with the students and teachers trained at them then disseminating their learning to the wider school."

"This digital project addresses a new practical form of continuous professional development for the teachers. Over the course of the project teachers take an active role in their continuous professional development as they develop and enhance their digital technology skills in each area of the project. Initially this entails teachers participating in facilitated workshops but then the teachers create and share resources for their class level with the help and support of their colleagues. These resources are used to scaffold the pupils' learning."

Wider partnerships

In their initial proposals the clusters were asked to propose ways of working with industry and with third level institutions.

Third level

The nature of the supports provided by higher education and by further education and training institutions varied from access to facilities and additional CPD, to inviting institutions to visit individual schools. Interestingly none of the clusters engaged in critical discussions around their projects with local third level and further education and training institutions, and this is an area that is revisited later in the report.

"Additional CPD is made available to the co-operating schools in LIT Thurles to aid building capacity in the schools through separate funding they've accessed. LIT also discount tickets to their annual ICT in Education Conference to the co-operating schools."

"At the end of year one we had a Showcase Day to share the work of the project with pupils, students and teachers participating in the project. Pupils, students and teachers all had an opportunity to showcase work that they had carried out during the year. We also had a tour of Cavan Institute (Third Level) to see the facilities on that campus with particular focus on digital technology and design. An exhibition of furniture and art created with the help of digital technologies were also on display. The pupils had an opportunity to see a 3D printer in action."

"Our 5th and 6th year chemistry classes participated in a workshop called "spectroscopy in a suitcase" ran by PhD students from UCC. Demonstrators walked our students through high tech equipment used worldwide to help solve crimes."

"This showcase invites third level academic centres such as NUIG, CIT, LIT and IT Tralee to give our students the opportunity to gain more information about the different STEM paths available to them."

Industry collaboration

A number of clusters reported that their involvement in SEF Digital & STEM Programme afforded them opportunities to reach out to their local industries and to invite them to visit the school or their cluster showcase events.

"Local industries, such as Gleeson Engineering, Campion Pumps & DevEire have been made aware of their local primary schools' involvement & sponsor some equipment, as well as representatives from Oide TiE, LIT & the local industries to attend the Showcase to observe their projects & findings, as well as present sponsored prizes to the students."

"The opportunity to see a factory such as Combilift, Co. Monaghan at work, from the design and planning stage to the production line and finished product being packed for shipping. To see such innovation at work was an amazing experience for both students and teachers."

"We found that many firms are glad to be associated with schools. Many have a dedicated educational team who will outreach to schools. To date this has included members from the Codeucate, Elan and Microsoft companies."

Clusters also extended their involvement with other projects and organisations focused on digital technologies. Six clusters reported involvement in national and local competitions:

"5th and 6th class pupils proudly represented their school at Dell Technologies National VEX IQ Robotics competition at CIT on Wednesday 12th February. Our team did not come away empty handed on the day. The Judge's Award was presented to our team for special recognition of our team displays and exemplary effort and perseverance at the event. 600 students from schools all over Ireland participated in the National Finals of the 2020 Dell Technologies VEX IQ Primary School Competition hosted in conjunction with Cork Institute of Technology (CIT). The pupils brought Science, Technology, Engineering, Arts and Technology (STEAM) skills to life by undertaking a game-based engineering challenge to design and build robots."

"We are the only 2nd level school in Ireland to have progressed to this stage of the European Space Agency's Astro Pi competition."

Other examples included involvement with code clubs, the Digital Schools initiative, and other digital projects:

"If we can continue this familiarity with the idea of coding to the point where we will have a pool of volunteers from our community to establish and maintain a community CoderDojo, then we will have achieved our aim."

"As part of our Digital Schools Excellence initiative we are continuously delivering, teaching and producing digital projects with our pupils. In Year 1 of our Digital School Excellence project with our cluster, our focus had been to enhance teaching and learning of local history (SESE) using various technological tools."

"Each Blue-Sky module is delivered in a student-centred way supported by technology and through the use of teaching methodologies such as flipped classroom, project-based learning and portfolio-based learning. In addition to a STEM focus each module in the Blue-Sky programme is linked to the Junior Cycle Key Skills."

Some clusters made significant efforts to involve parents and the wider community:

"Our STEM Fair at the end of year 1 was hugely successful. Over 700+ parents, pupils and people from the wider community attended this event."

"It has been most beneficial to have digital portfolios set up and children/parents familiar with them. Seesaw has been the main means for communication used between primary school teachers and their students/parents in our cluster."

"We have excellent links forged with the wider community to provide a wide range of enrichment activities to promote students learning and engagement with the subjects."

"Many parents, especially those of ICT workers were especially taken with the coding element of the project, seeing it as something every child should be given an opportunity to try."

"One of the children's grandfathers has windmills on his farm & his dad came into the class and did a questions and answers session with the class. He brought in photo albums of the construction process."

Thus, schools by their participation in SEF Digital & STEM Programme, found multiple ways to connect with industry and many of these connections have the potential to develop long lasting relationships with industry and the wider community that extend beyond a one-off event.

Technology

Range of technologies used

Nine clusters referred to specific computer technology, eight to iPads and one to iPads and Chromebooks:

"With regard to the project itself, I think each school has found that the introduction of iPads to each school has been the greatest innovation, with SeeSaw being the most valued app."

"This is an iPad project and teachers are using the iPads for other areas of learning."

"In the present school year, we all introduced and used Lego robotics with our pupils through their iPads or Chromebooks."

Given the focus of many projects on coding and control, the most frequently mentioned technology was Scratch (15 Clusters), followed by Lego WeDo (14), Bee-Bots (9), and BBC micro:bits (6). A wide variety of other products were used by small numbers of clusters: Barefoot Computing, Code.org, Java, Makey Makey, Microsoft Dream Space, Minecraft, Python, Raspberry Pi, Rover Robotics, Sensors, Sphero, Vex and three Clusters referred to 'unplugged' activities for coding.

"Our aim is to develop the teaching of computational thinking skills in a constructionist and inquiry-based science learning environment in Irish Primary Gaelscoileanna through the use of robotic materials, namely Lego We Do 2.0."

"In the present school year, we all introduced and used Lego robotics with our pupils through their iPads or Chromebooks. The Lego kits motivate students to collaborate, build, problem-solve, and explore, all while deepening their STEM learning and developing 21st-century skills. The pupils had lots of fun working through guided builds before moving onto more complex problem-solving builds of their own. Our theme throughout the Lego Robotics project was climate change and how to prevent it."

"As a group, we discussed issues that had arisen from year one. The team decided that the best way forward would be to focus on three core sets of equipment - Bee-bots, Lego WeDo and micro:bits. We will supplement these resources by exploring unplugged activities with some additional resources that we bought such as Lego Mindstorm and Sphero."

"One particularly exciting module is our Minecraft School Build where a group of students have built the school on Minecraft. This is now being used by our new intake students as a 'map' of the school as they negotiate the building in their first few weeks of the school."

"The goal this year was to try to incorporate the use of the wireless sensors in a more meaningful way. The intention is that teachers will try to use the technology in at least one lesson per week so that the students become familiar and accustomed to the possible applications of wireless sensor technology in their own lives as well as becoming more familiar with interpreting data when presented graphically."

Outside coding and control, the most mentioned technology was the Google suite of apps used by 14 clusters, followed by SeeSaw (11), Microsoft Teams (5), Twitter (5), Virtual Reality (4 - 2 of which were using Google Expeditions with VR), Animation (4), and small numbers also using: Adobe Spark, 3D Design, Drones, e-twinning, Flickr, iMovie, and Video conferencing.

"Four schools are using Virtual Reality technology as a teaching and motivational tool to develop many curricular areas and in particular the Primary Language Curriculum. These four schools are using the School Self Evaluation process to develop, implement and track oral language, writing and reading in various class groupings. The focus for year 1 of the project was Oral Language and 'Oral Reports' in particular. This year's focus is Writing, and we zoned in on 'Report Writing'. We use the VR technology and Google Expeditions to explore many topics in many curricular areas and as a stimulus to develop the aspect of oral language/writing/reading that we are working on. We plan to use the 360 Panoramic cameras to develop our own expeditions based on local areas of interest or topics to further develop oral language and writing in the coming months."

"Students are now using Digital portfolios in all schools including Seesaw at primary level and Microsoft Teams at secondary level which are now being used not only in PE but across the curriculum."

"Most class groups have met each other via video calling (Skype/Google Hangouts). During these video calls they have shared their thoughts on various aspects of digital citizenship and the assigned project they are undertaking."

Other technology related topics

Two clusters used some of their funding to improve their IT networks. Coordinators of projects that were heavy users of new hardware such as VR found that they had new responsibilities, and other clusters devised rotation systems for scarce equipment.

"All five schools have made improvements to their IT networks."

"The school project coordinators found themselves "responsible for the upkeep and maintenance of the Virtual Reality equipment. To keep all equipment updated and in good working order is vital to the success of the project."

"We will rotate the equipment differently so that when individual schools have equipment, they will have enough for children to work in pairs rather than having two sets per class and children working in much larger groups."

Thus, schools used a wide range of technologies in innovative ways to enhance learning, prepare students for a digital world, and improve access to digital technologies.

Online presence

In the Interim Evaluation Report, which spanned the period March 2019 to June 2020, most of the information collected about the project websites and social media accounts used by clusters and the schools in the clusters came from the survey responses. This description was later expanded using previously collected data and further online searches for the projects. An examination of all the cluster project websites in June 2020 found that while some were excellent, 8 projects had provided no information, 2 stated their website was in development, and 6 provided links to closed sites (password required). At that time, 25 of 41 clusters (60%) had a public-facing website.

Social media use was principally via X (formerly Twitter) with only a few clusters using other platforms such as Instagram, Facebook, YouTube, Flickr. Sixteen clusters used X, though many of these only tweeted a few times, or tweeted occasionally using the general school X account for project related posts. Some, however, were very active, with nine clusters having tweeted more than 70 times in the period from the project start to 1 July 2020, and four of these tweeted more than 200 times.

Oide TiE Advisor feedback

In addition to collecting data from the Cluster Coordinators, the Interim Evaluation Report collected data from nine Oide TiE Advisors. Each advisor reported supporting either two or three clusters. In total the respondents were supporting 23 clusters.

Support provided by Oide TiE Advisors

The team of Oide TiE Advisors supported the clusters in a variety of ways:

- Face-to-face meetings (whole cluster meetings, school meetings, and individual meetings with the Cluster Coordinator).
- In-school support (workshops, co-teaching, planning, capacity building).
- Training sessions and workshops.
- Advice and support regarding materials, resources and purchases.
- Support via email.
- Phone – mostly for liaising with the Cluster Coordinator, and for follow-up of other activities.

Some Advisors gave detailed accounts of how they carried out their support roles:

> *"Having the opportunity for whole cluster meetings at key points in the year have proven to be very effective. Checking in with progress, reviewing and setting new targets and actions have in my view been very beneficial. Further supports such as workshops and in school support stem from these meetings. In school supports have also been very much appreciated by principals and teachers. The ongoing and sustained nature of this support has been very effective and appreciated by the lead teachers at each school."*

> *"Planned meetings were arranged at various points throughout the year. These were used to plan, reflect, assign tasks, define roles, adjust timeline etc. These were facilitated as a whole cluster meeting in the main with lead coordinator, lead teachers and key stakeholders."*

> *"CPD takes place in many forms. Some support is workshop based where all the schools involved came together to receive support and training. This can be complimented with school support visits to each of the cluster schools involved. In school support has been delivered in various forms, including workshop, co-teaching, planning capacity building, and engaging with principals and in school management teams."*

For many Advisors, the principal point of contact was the Cluster Coordinator and most communication with the cluster took place via the Cluster Coordinator, either face-to-face or by phone.

> *"Continuing phone and e-mail correspondence and liaison with the co-ordinator regarding on-going progress and obstacles, they may have encountered."*

The Advisors felt that face-to-face meetings, whether in the form of whole cluster meetings or individual meetings with the Cluster Coordinator were the most effective type of support. They valued these face-to-face meetings for:

- Greater engagement and motivation
- Keeping clusters on track
- Building relationships among cluster members.

> *"The ongoing sustained support and opportunity to meet up as a cluster keeps cluster schools motivated and engaged."*

However, the organisation of such face-to-face meetings was identified as a challenge by some advisors, who highlighted the importance of having structured meetings to ensure they werepurposeful.

> *"Regular meetings to build up relationships amongst the cluster members. It is important that these are highly structured, and each meeting achieves a purpose."*

Programme progress

Overall, the Oide TiE Advisors were positive with the kind of support they provided to the clusters and felt that their support had been appreciated. They were satisfied with the clusters' performance during the first year in achieving their project aims. The advisors described their collaboration with clusters as 'enjoyable', 'wonderful', 'unique', and 'engaging'.

> *"This has been an extremely enjoyable experience. Every cluster has had a different path but many share similarities at different points. Being involved in a number of clusters means you can advise on things that you have seen working in other clusters."*

Programme challenges

There were changes in expectations and targets for some clusters after Year 1, as circumstances changed, and difficulties emerged. Most projects had initially anticipated that all schools in a cluster would progress at the same pace, but this did not always occur.

> *"Some of the schools may be operating at a slower pace. This can be for various reasons including location, teacher skills and confidence, resources, personnel changes etc. Although it may have been envisioned that each of the schools would work at the same pace with regards to initial expectations and outcomes alterations were necessary."*

This is an interesting finding, as all schools are different and were often at different stages in relation to their use of digital technologies.

Other challenges included:

- School closures due to COVID-19 made it difficult to track progress
- Technical support was sometimes needed
- Lack of availability of substitute teachers sometimes made it difficult for teachers to attend meetings
- Change of Cluster Coordinator or other personnel involved in the project
- Time constraints.

> *"Technical support can be a barrier at times. While pedagogical support and advice i.e. teaching learning and assessment is our main goal sometimes this can be hindered by technical issues which can slow down or inhibit some clusters targets and actions."*

> *"Sub covered days have been used up by the lead schools usually quite early on in the year."*

> *"Availability of substitute teachers has affected both clusters at different times with teachers finding it challenging to attend cluster meetings."*

> *"... there have been changes in leads and school personnel in both clusters so there was a short lead time at the start of the year."*

Some Advisors felt there was a lack of clarity in the administration of the Programme, and in the expectations associated with the role of Cluster Coordinator. Their observations highlight several areas that are revisited in the final section of the report in relation to providing greater clarity around Programme reporting and in articulating expectations with clusters in advance and during the project.

> *"There is a lack of clarity, consistency and expediency regarding changes or dissemination of information to clusters and advisors about projects."*

> *"Clear guidelines on reporting, expectations, supports etc are essential to have at the start of the process."*

> *"I also think the cluster should have been provided with a roadmap at the beginning of the process. Clusters would have benefited if clear expectations were outlined from the onset. Clusters would have benefited if they knew when there were showcase days and the type of website/online presence required."*

Suggestions and recommendations

While the clusters were working well in the main, the Advisors made the following recommendations to encourage greater collaboration between clusters:

> *"Creating a community of practice of the cluster leads would be very beneficial. I put two of my leads in touch as they were geographically close and had similar aims. They shared many of the same challenges and were able to help each other out."*

It was felt that a shared SEF Digital & STEM Programme platform, and an online collaboration space would have been useful from the start of the project:

> *"An initial consistent website/platform for all clusters to view progress and development of each cluster over the course of the programme - to enable reflection on each cluster*

> *progress, learn from each other and have a record of what had worked and what needed adaption."*

> *"Establishing an online collaboration space for all clusters to share ideas and their work would have been beneficial. At the moment all their work is hosted in different places."*

A further recommendation related to the start of the project concerned the need for project management training:

> *"Project management training at the beginning of the process would have been very useful. This would have supported the project lead with the management of the project. Some cluster leads had never managed a project like this before."*

The issue of project administration was also highlighted by the Cluster Coordinators and like the Oide Advisors, they suggested that there was a need to provide clearer central guidelines and supports to all involved.

Finally, one Advisor noted the limitations of relying on email for communication:

> *"Sometimes email isn't really the best form of communication as some cluster members do not check their email regularly. A more informal method of communication e.g., WhatsApp or Slack might be better."*

These constructive observations provide valuable insights for future programmes and are addressed further in the final section of the report.

Section 5: Post-Project Survey – Experience of the project

The previous section described the Interim Evaluation Report which was developed at a time when the schools were actively engaged in their projects, and the data was collected during the first phase of school lockdowns in May/June 2020. Subsequently schools encountered further disruption to their project plans, and this proved challenging for some clusters. However, the Interim Evaluation Report captured the various impacts that the project had on life in schools and how teachers used digital technologies in a range of innovative and new ways. The next three sections of this report move forward to May/June 2022 when the Programme was concluding, and many of the issues that appeared in the Interim Evaluation Report remained present.

Overview of the survey

The post-project survey (see Appendix 4) was sent to Cluster Coordinators, teachers and Oide TiE Advisors, with a different selection of questions for each of the three audiences. The survey consisted of both closed (multiple choice) questions and open questions. The open questions explored three themes:

1. Experience of the project.
2. Views on lessons learned (related to good classroom practice, infrastructural models, support models and embedding digital education).
3. Views in relation to possible future digital initiatives.

In total 61 responses were received, 36 from Cluster Coordinators, 16 from other teachers and 9 from Oide TiE Advisors. The responses related to 34 clusters from the original 41 clusters in the SEF Digital & STEM Programme.

The results of the quantitative data from the closed questions are summarised as bar charts and are presented in this section.

The responses to the open questions were analysed in relation to each of the three themes described above, supplemented by a thematic analysis to identify sub-themes, and representative responses are provided in this final report. The results related to experience of the project are presented in Section 5, those related to lessons learned are presented in Section 6, and those related to possible future digital initiatives in Section 7. Quoted frequencies in the reports in these chapters usually refer to numbers of responses – which includes those from Cluster Coordinators (sometimes more than one per cluster), teachers and Oide TiE Advisors. Where the report separates out these responses by type of respondent, this is indicated in the text.

Experience of the project

All clusters reported important gains from taking part in the project. Respondents from 22 out of 35 clusters (63%) described their experience as very positive.

Responses to the closed questions

The responses to the questions related to the Cluster Coordinators and teachers experience of the project are shown in Figure 7. Responses were strongly positive, with less than 10% negative responses on all issues.

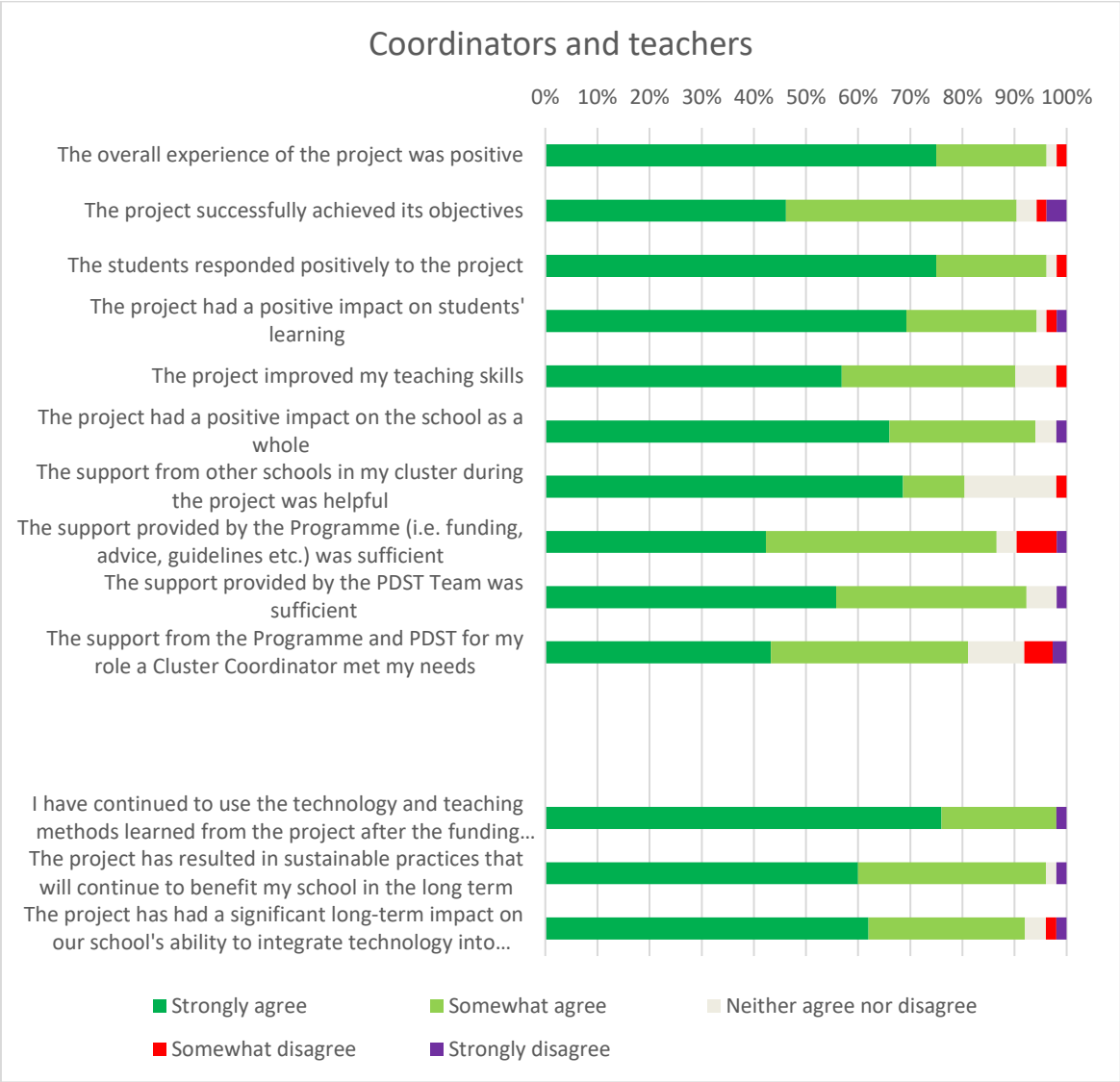

Figure 7: Responses from Cluster Coordinators and teachers

The highest level of agreement was found for the following statements:

- The overall experience of the project was positive.
- The students responded positively to the project.
- I have continued to use the technology and teaching methods learned from the project after the funding period ended.

The lowest level of agreement was found for the following statements:

- The support provided by the Programme (i.e., funding, advice, guidelines etc.) was sufficient.

- The support from the Programme and Oide TiE for my role, as a Cluster Coordinator, met my needs.

Most respondents (90%) indicated that their project successfully achieved its outcomes, with 46% strongly agreeing and 44% somewhat agreeing. This is a very positive response considering the challenges the Cluster Coordinators and teachers encountered during COVID-19. Respondents reported that students responded positively to the project, with 75% strongly agreeing and 21% somewhat agreeing with this statement, while they also reported that the project impacted positively on student learning (94% either strongly agreed or somewhat agreed). Thus, Cluster Coordinators and teachers observed a positive impact on student learning because of their involvement in the project.

Respondents also noted a positive impact on their teaching skills with 57% strongly agreeing and 33% somewhat agreeing (i.e. a total for 'agree' of 90%). Similarly, they noted that the project had a positive impact on the school, with 66% strongly agreeing and 28% somewhat agreeing (i.e. a total of 94%). A large number, 69%, strongly agreed that the support from other schools was helpful, while 12% only somewhat agreed with this statement. However, 18%, which is almost a fifth of responses, stated that they neither agreed nor disagreed with this statement.

In terms of supports, 42% strongly agreed that the supports they received from the Programme (i.e. funding, advice and guidelines) were useful, while 44% somewhat agreed, with 8% somewhat disagreeing with this statement. Many (93%) of the respondents (Cluster Coordinators and teachers) noted the support they received from Oide TiE was sufficient, with 56% strongly agreeing and 37% somewhat agreeing. For the Cluster Coordinators specifically, 81% reported that the support they received from the Oide TiE was sufficient with 43% strongly agreeing and 38% somewhat agreeing.

Encouragingly, 98% of respondents reported that they had continued to use the technology and teaching methods developed during the project itself, with 76% strongly agreeing and 22% somewhat agreeing. Furthermore, 90% of respondents indicated that the project resulted in sustainable practices and 92% believed that the project had a lasting impact on their ability to integrate digital technologies into their teaching, learning and assessment practices.

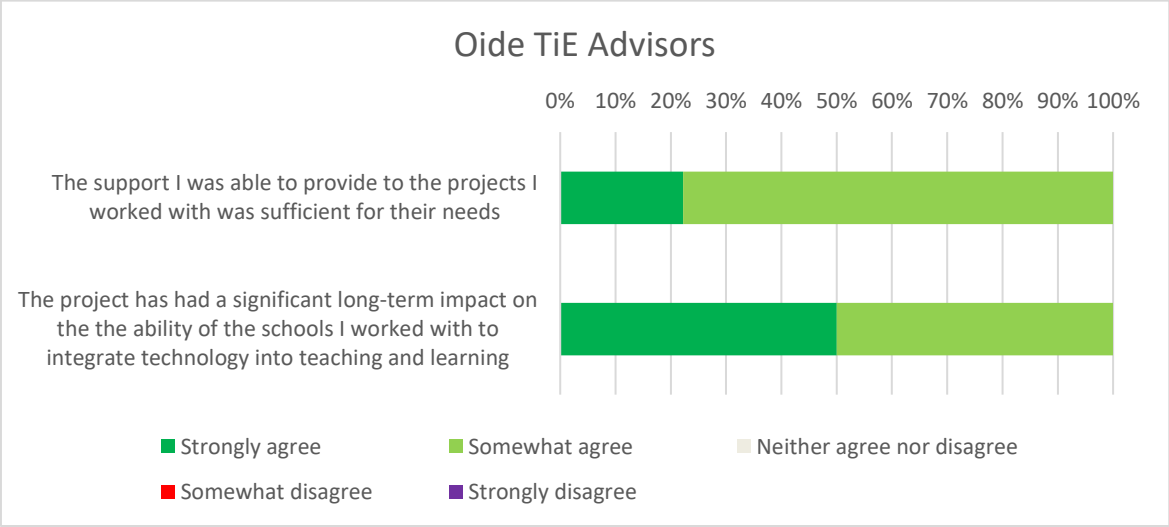

Figure 8: Responses from Oide TiE Advisors.

Figure 8, above, captures the overall responses of the Oide TiE Advisors who also had a positive view of the achievements of the Programme. Interestingly, 78% of the Advisors somewhat agreed that the support they provided was sufficient for the needs of the cluster, while 50% strongly agreed that the project has had a significant long-term impact on the ability of the schools they worked with to integrate technology into teaching and learning practices, with the other 50% somewhat agreeing with this statement. Overall, the advisors were very positive in their reflections on the impact of the Programme on the schools they worked with.

Responses to the open questions

In addition to the closed questions, respondents were asked a series of open questions about their overall experience of the project. In this section the report presents a high-level synopsis of the key findings from the qualitative data analysis on this topic in Table 8 below. This is then followed by a detailed account of the supporting evidence for these findings including quotations from the respondents.

Table 8: High-level synopsis of open-ended survey responses.

Overall Experience	• Most participants had positive experiences of the project, noting high student engagement, collaboration between schools, and the development of innovative teaching approaches. • Coordinators and teachers identified challenges that came up in some clusters, which included high staff turnover, problems around project communications, and some technical and infrastructure Issues.
Impact of COVID-19	• All clusters were badly disrupted by the pandemic and school closures, though some were more able to adapt than others.
Collaboration	• Collaboration between schools was a major element of the success of the clusters. It is notable that a third of the school clusters continued to collaborate after the end of the project, this was most often in STEM or computing related areas.
Long-Term Impact	• The Advisors noted the positive impact of the projects on digital skills, on the integration of technology into teaching and learning processes, and on the collaborative working that had developed with other schools. They also noted that where staffing was consistent and project goals were clear, digital practices were sustained after the project.
Funding	• Some clusters reporting issues over funding, and in particular the impact lack of funding had on the sustainability of their projects, especially in terms of ongoing procurement and repair of equipment.
Regulation and Data Privacy	• The survey did not ask specifically about regulation, ethics or data privacy, but it was perhaps surprising that only one respondent raised issues here, this was in relation to “support for managing data and compliance with data protection regulations”.

Overall experience

When asked about their experience of the project and the SEF Digital & STEM Programme as a whole, the most common responses were positive, highlighting increased student engagement, innovative teaching methods, and collaboration among schools.

> *"The programme worked very well, and we achieved excellent outcomes. We hoped that when our female junior cycle students engaged in the programme, by becoming STEM role models for younger, primary school students, that this would improve their attitudes towards STEM. This was greatly achieved, and in our school, we have seen a 500% increase in female students taking chemistry at senior cycle."*

> *"This was a fantastic experience! Our cluster members worked really well together, and the sharing of ideas and information was outstanding. We had an incredible Cluster Coordinator, and I believe the success of the Cluster was a testament to the huge effort that she put into the whole project."*

> *"The collaborative project was indeed a great initiative. It allowed us to pool resources together, both human and material, to create learning experiences for our students that we could not have created individually. The experience also allowed us to learn from each other. We had several staff development workshops that were based on sharing good practice observed in our partner schools."*

> *"The collaborative cluster project worked very well for us. We were able to bring together expertise from several schools to put together a curriculum that would not have been possible in our individual schools. The project allowed for sharing of resources and ideas and helped to build a community of practice amongst the teachers involved."*

> *"The collaboration was really beneficial for our school. We managed to create new interactive learning materials, to conduct workshops with our teachers, and to exchange our experiences with other schools. It was a great way to learn and to improve our practices."*

> *"The project was an amazing experience for our school. It led to a great exchange of ideas and best practices between the schools. The collaborative efforts helped us develop new teaching methods and approaches to learning that are still being used in our school today. The project also helped to foster a sense of community among the teachers, which I believe is very important in education."*

The above responses highlight the positive impact the SEF Digital & STEM Programme had in these clusters, by, for example, improving STEM attitudes and facilitating greater collaboration among cluster schools so that staff could share resources, ideas and ultimately learn from one another. The comments in relation to the collaborative experiences of cluster participants are most encouraging and capture the key role of the Cluster Coordinator in developing a collaborative cluster culture, which in turn led to the creation and implementation of a range of innovative approaches.

Coordinators and teachers, however, also identified some challenges, and among these the most mentioned was the impact of COVID-19, and the second most common reason was the high staff turnover in certain clusters.

Several responses pointed to issues associated with the administration of the Programme, specifically issues around communication and coordination. These included lack of feedback, unclear project deliverables, concerns over budget handling and too much emphasis on

administrative tasks like creating websites or providing feedback, which were felt to detract from the main goals of the project.

> *"Support from Oide TiE coordinators was very good but towards the end I couldn't get answers about leftover money etc."*

> *"The only negative of this project was the need to produce a website which was a huge workload and turn up a huge amount of time in the product of the cluster instead of focus on the more important process of the project."*

> *"However, the amount of administration involved in drawing down and managing funds was quite a lot for a small sum (3000 or so)."*

These comments highlight the need to clearly outline what is expected of participants, both the Cluster Coordinators and the schools, at the outset so that everyone is aware of what is expected of them in terms of project expenditure, project deliverables and project management. The Department did outline what was expected from Cluster Coordinators at various junctures of the project, specifically at the opening event in University College Dublin (UCD) and in writing on several occasions. Thus, clusters were aware of their responsibilities, however, they either underestimated what was required or they lacked the skills and/or time to fulfil these obligations for the Department. In some cases, this also highlighted the importance of clusters adopting a team approach, where multiple teachers took responsibility for aspects of the project.

While COVID-19 undoubtedly impacted on cluster plans, the issue of project duration should be considered in future, as this has implications for Coordinators, teachers and Oide TiE Advisors. Over time, teaching staff move to other positions, they retire or take on new responsibilities and thus this should be factored into future programmes, as staff turnover is a reality in all aspects of education. Furthermore, these issues suggest that there needs to be a level of flexibility in terms of amending project timelines and plans if such issues arise.

Some respondents encountered challenges around technical and infrastructure issues, and whilst this issue was only raised by a minority of respondents, when it was raised it was argued quite forcefully.

> *"More often than not, the iPads wouldn't work, the internet was down, the sound wouldn't work, the screens were wobbly. Eventually, we pushed them to one side, utterly frustrated and brow beaten by the whole thing."*

> *"The ideas underpinning the collaborative project are laudable. However, the same difficulties remain, and it is enormously frustrating. Namely, there is NO support for technical repairs and ongoing upkeep of technologies. A huge amount of the money invested into technology supported learning and cluster initiatives is poorly considered. We don't need experts with ideas. WE NEED TECH SUPPORT. Most of the initiatives that our cluster decided upon never happened in our school. They couldn't happen because the technologies cannot be relied upon to work."*

> *"Teachers are not qualified technicians, engineers, repairs people and yet that is what constitutes one's role a Cluster Coordinator or technology enthusiast. If the cluster idea is really going to work, we need qualified people to visit schools on an ongoing basis. The cluster initiative was largely frustrating and dispiriting because most of what was planned could not happen."*

These comments highlight the need for reliable technical support in general in schools and specifically in future programmes where schools are exploring the uses of emerging technologies. The comments note that teachers, and Cluster Coordinators, are not technicians and that they need ongoing technical support to ensure these technologies are in working order. Thus, there is a strong rationale for considering the inclusion of additional technical support funding and supports in any future programmes to address such issues.

When asked to assess their overall experience of the Programme, the Oide TiE Advisors highlighted the following positive experiences.

> *"I was in the fortunate position to have three projects which were similar in structure (lead post primary school with primary schools working collaboratively with them). This helped in terms of guiding the lead teacher in each project as they had similar needs at similar times. Two clusters were geographically close to one another, so this helped when organising external CPD for those teachers."*

> *"In some clusters buy in from all schools wasn't there from the offset. This hampered progress after cluster meetings when schools returned with goals not complete. From experience when buy in was there from all schools and clear communication and ways of working were communicated these clusters ran smoothly and had a lasting legacy."*

These comments address the issue of how schools were clustered and where there was buy in from the outset, the project appeared to work better. Thus, the issue of how clusters are formed is hugely important and it is essential that clusters in future programmes develop such a shared approach, in terms of project design, communication and implementation issues, from the outset.

They also described some of the challenges, particularly where they felt they did not always have the expertise required to meet the needs of a particular cluster:

> *"Some of the clusters could have more success if they had project management training but this was not something I was in a position to provide."*

> *"I was comfortable in working with the clusters largely. Some clusters had quite a specific focus in programming. I was not in a position to provide high level training specifically, so outside expertise was sought and this worked well."*

The first quote above highlights the issue of project management support, which surfaced regularly during the project. While the project did provide project management supports at the outset (for example see Appendix 2 to view a sample of the guidance provided), it seems some advisors struggled with providing the necessary supports to their clusters. There is a need to provide greater clarity and supports for those operating in the role of the Cluster Coordinator and looking to see if some of their tasks could be outsourced to other colleagues. The second quote relates to content knowledge in relation to programming, not an area many teachers would be comfortable with. This quote suggests that, as with technical support, there may be occasion for schools to buy in outside expertise, and that the Oide TiE Advisor might support the school in finding appropriate supports but not necessarily deliver the training themselves. The report further expands on the 'lessons learned' in Section 6.

Impact of COVID-19

The COVID-19 pandemic had a significant impact on the project, resulting in changes in priorities, challenges in coordination, and a negative impact on its overall success.

> *"Hugely positive experience overall. Covid, unfortunately made a mess of the whole process. The children in my school benefited enormously from the whole experience with the Lego WeDo and Vex Robotics. A great initiative overall."*

> *"Also, after a great start to our cluster, COVID impacted hugely on it, and it never regained its momentum."*

> *"Covid destroyed it quite simply. It just wasn't a priority for other schools and with an extension to the programme it only led to teachers moving on or taking up new roles including within our own school that demanded different responsibilities. Not the programme's fault, completely the pandemic's fault."*

While the Interim Evaluation Report captured a range of issues in relation to COVID-19, the above selection of quotes highlights how the pandemic impacted on project momentum and how priorities changed overnight. Projects were only getting up and running in March 2020 and the impact of school closures, the move to online learning and other challenges negatively impacted project implementation in most schools. Despite these challenges, schools continued to do their best, but this was often challenging.

Collaboration

An important element of the SEF Digital & STEM Programme was the collaboration of schools within clusters. Cluster Coordinators and teachers were asked whether this collaboration had continued after the end of the project and, if so, in what ways. Over a third of respondents said that collaboration with other schools had continued (37% Yes, 63% No).

While the level of collaboration dropped from the levels during the project as captured during the Interim Report, it is interesting to see the kinds of activities that schools continued to collaborate on.

The most common form of ongoing collaboration (9 respondents, representing 37% of those saying that collaboration had continued) was where schools collaborated on STEM, digital learning, and technical subjects such as coding, robotics, digital literacy, and computer science. These involved both online and in-person collaborations.

> *"Continued collaboration continues to occur around the implementation of new subjects like JC Coding and LC Computer Science. There is ongoing upskilling of teachers across all the schools with representatives from each school responsible to coming up with training material."*

Resource sharing was mentioned by some clusters.

> *"We meet every term in the Tralee Education Centre. We exchange the shared resources that were purchased using the funding. We continue to collaborate and share information with regard to the use of the resources in each school, along with any other projects or opportunities that we feel would be of interest to each other."*

Some clusters continued to work on shared projects.

> *"Our cluster has decided to continue to collaborate and expand our project. We meet in-person in the Education Centre once a term. We continue to work on our shared google drive folders. We communicate regularly through email and WhatsApp. Our current project is 'Superpowers'. We are looking at how energy is sourced and distributed to homes, schools and businesses. Children are identifying issues and design solutions to create better energy journey for their community with regards to energy sources, energy distribution and energy usage."*

Others had maintained more casual and infrequent links.

> *"Sharing links to courses and webinars. Plans to meet socially."*

> *"I haven't had to get in touch with anyone but if an issue arose, I would not hesitate to reach out."*

The comments indicate that the purpose for collaboration is important and that other organisations, such as the education centres, could play an important role in bringing schools together. This was particularly true in the case of a number of the primary clusters, and this approach could be further developed in any future programmes.

Long term impact

The evaluation asked the Oide TiE Advisors about sustainable practice and the long-term impact of the projects and of the Programme as a whole. They emphasised the positive impact on digital skills, on the integration of technology into teaching and learning processes, and on the collaborative working that had developed with other schools.

> *"When clusters had agreed ways of working and had clearly outlined and agreed goals a clear impact was made on the ability of schools to support themselves independently and look for further improvement. It became part of the school culture."*

> *"Teachers have told me about instances where students who were involved at primary level have since moved on to the lead school for post primary and have benefitted from their involvement in the project in terms of their digital skills."*

> *"The communities of practice within large and small primary schools was particularly effective. There were consistent staffing arrangements and teachers were assigned the time and space to work collaboratively with teachers from their collaborative schools."*

These comments further support the importance of having shared goals and ways of working to ensure that clusters engage in meaningful collaborative activities. They also showcase the benefits of connecting primary and post-primary schools in terms of developing digital skills over time. The final comment references the need to provide time and space for collaborating schools to engage in communities of practice.

However, the advisors made the point that in those schools that committed time and effort, and where staffing was consistent, greater progress was made in embedding digital technologies.

> *"For those schools that were in a position to commit the necessary time and effort, as well as having consistent staffing arrangements throughout the duration of the project, significant advances were made in the embedding of digital technologies in teaching, learning and assessment."*

The above comment also highlights the need to invest in digital education over time and the importance of consistent approaches to embed digital technologies into teaching, learning and assessment practices. This is not a once off activity, but something that schools need to prioritise and build over time.

The Cluster Coordinators and teachers were asked to describe the barriers they faced in sustaining the practices they developed during their project after the SEF Digital & STEM Programme had finished. The chief barriers mentioned were; time constraints (10 respondents), the impact of COVID-19 (10 respondents), concern about the obsolescence of current equipment and so the need for continued funding (6 respondents), staff changes (3 respondents) and the need for ongoing training, particularly for new teachers or upskilling existing staff (4 respondents), and poor internet connectivity (4 respondents).

> *"For sustainable practice & Long-term impact, the main reasons are Funding, Staffing & Time / Timetabling...Teachers need to be properly trained & resourced... I propose a post-grad similar to the one that up-skilled existing teachers to become Maths Teachers."*
>
> *"Time constraints. Our main area of focus was coding. As this is not yet on the curriculum, making time for coding in a 4 class multi- class and an overloaded curriculum is our biggest challenge."*
>
> *"Time needs to be given specifically to development of projects in order for them to be sustainable. There is a need for sustained support from outside agencies in order to drive projects and coordinate projects and not just an email at the start and end of the year or when there is a change in personnel in Oide TiE."*
>
> *"The pandemic shelved our ability to interact with the technology for two full years. The high cost of the technology also provides a barrier to allowing the students to fully integrate with its practices."*
>
> *"Over time new robotic kits suitable for primary school pupils will become available or the kits we currently use will become obsolete. Funding replacement equipment may become a problem in the long term."*
>
> *"In our project, we (the secondary school) ran STEM workshops for primary school students... The cost of transport has increased hugely, and so, we have not been able to continue running these workshops since the funding and the programme finished."*
>
> *"Shockingly bad Internet. And we are in a town!!!"*

The issue of time appeared several times in the comments from teachers and Cluster Coordinators, as does the issue of curriculum relevance and ongoing support. Respondents noted that there was a need for ongoing support in keeping programmes such as SEF Digital & STEM alive and evolving. Other issues such as the cost of technology and lack of broadband were also a reality for some schools.

Funding

The survey also asked teachers and Cluster Coordinators "has funding (for replacement/ updating of equipment, training and support) been an issue for sustainability and long-term impact of the project?" In response 57% said Yes and 33% said No. Respondents highlighted

the following issues in relation to project financing: funding for replacement or updating of equipment and infrastructure, ongoing maintenance and repairs.

> *"Expensive equipment that requires updating like any other piece of technology. It still works but at some point, it will become obsolete or unable to function due to continued updates, much like an iPad."*

> *"The initial funding bought a limited number of resources that could be used among the vast number of teachers (close to 100) and students (almost 1400) taking part in the Cluster Project."*

> *"Funding for ICT is always a challenge when considering the maintenance costs of equipment."*

> *"Lack of spend for music now that SEF fund no longer available. As music departments grow or are established in schools, we do not have the capacity to scale the equipment needs."*

> *"Although we are a town school, our internet is shockingly bad. Due to the fact that we purchased extra devices, more internet was required. We are prevented from embracing Digital Technology to its fullest due to poor internet."*

These comments capture a spectrum of the issues schools faced in relation to funding and they highlight the costs associated with procuring and updating a range of digital technologies. While the cost of some digital technologies, such as computers, has come down over time, there is still a significant outlay for schools in terms of procuring sufficient equipment for staff and learners and ensuring it is regularly maintained. A number of these issues pertain to the sustainability of certain technologies and their suitability for schools, an issue that could be explored further in the future. Finally, the issue of access to quality internet provision again was also raised, as more and more technology requires access to reliable fast internet services.

While most of the comments related to the additional funding to procure digital technologies or related services, it is interesting that no one commented on the need for additional discretionary funding to release staff from teaching duties or to attend additional professional learning activities.

Regulation and data privacy

Whilst the survey did not directly ask for comments related to ethics and regulation, it is nevertheless surprising that there were almost no comments made in relation to ethics, data privacy, GDPR or other data regulations, neither in relation to wider societal concerns with respect to the use of technologies, nor in terms of the duties of teachers. One Cluster Coordinator noted the need for "support for managing data and compliance with data protection regulations." This is an issue that could be further explored in the future, as more and more digital platforms, tools and applications are capturing and using teacher and student data.

Experience of the project – concluding comments

Participants' experience of the SEF Digital & STEM Programme was overwhelmingly positive, though around 10% of Cluster Coordinators and teachers had quite negative experiences. The Programme led to a range of positive outcomes, it contributed to better student engagement, innovative teaching methods, and collaboration among schools. The Programme was impacted by several factors which were outside of its control, particularly the COVID-19 pandemic and

changes in personnel (an issue made worse by the epidemic and by the overall length of the extended Programme). The disruptions caused by COVID-19 appeared to have a major impact on project activities, as was the case in many other areas of education. The pandemic impacted negatively on face-to-face collaborative events, school project plans and seriously disrupted momentum.

Other specific issues of concern that were raised most frequently in the post-project survey related to communication, coordination, and administration of the Programme; technical support, infrastructural issues; and funding.

Section 6: Post-Project Survey - Lessons learned

This section summarises the responses of the Cluster Coordinators, teachers and Oide TiE Advisors when asked to comment on lessons learned in relation to the four themes that the SEF Digital & STEM Programme sought to address:

- Classroom practice in relation to the embedding of digital technologies and constructivist approaches to teaching, learning and assessment at primary and post primary levels.
- The infrastructural models that can best support innovation in relation to teaching, learning and assessment, communication, and collaborative planning.
- The most appropriate support models that can be put in place to facilitate innovation in schools.
- How the embedding of digital technologies in teaching, learning and assessment within and across all schools can be achieved.

In this section the report presents a high-level synopsis of the key findings from the qualitative data analysis on this topic in Table 9 below. This is then followed by a detailed account of the supporting evidence for these findings including quotations from the respondents.

Table 9: High-level synopsis of the key findings.

Good Classroom Practice	• Effective strategies included student-led projects, collaborative learning and personalised learning. In such cases teachers often moved from structured to more open approaches over time.
Infrastructural Models	Need for: • Strong broadband. • Physical space for housing and securing technology. • Hardware and devices, for both technologies for general use, but also specialist equipment for specific areas. • Software and platforms (i.e. Google, Microsoft and learning management systems). • Standardisation and clear policy.
Support Models	• Importance of continuing professional development and training and peer-to-peer collaboration. • Participants wanted practical, up-to-date training tailored to their needs. In-school and peer-led models were preferred. Some questioned the value of private providers (i.e. those who developed certain technologies or software products providing training). • Technical support. • Administrative and institutional support. • The sustained Oide TiE support model was widely praised.
Embedding Digital Technologies	• Embedding required long-term funding, training, time, collaboration, curriculum alignment, • Relation to assessment needs consideration. • Evaluation of effectiveness of technology integration is required.

Good classroom practice

Coordinators, teachers and Oide TiE Advisors were asked to describe what they saw as effective strategies and techniques for creating an engaging and effective learning environment for students when employing digital technologies.

Many (50% of responses) stressed the importance of student-led projects, with students working collaboratively in pairs or groups.

> *"Student-led and engaging practices worked very well for us. We trained secondary school students to become STEM role models who then ran workshops for the younger primary school students. This student-led model ensured engagement at both primary and post-primary level."*

> *"Project based learning. The new leaving certificate computer science syllabus uses applied learning tasks. We wanted to give the students a set of foundational skills and have them develop projects in teams. Students were given access to online platforms that would facilitate them developing their own solutions to the challenges given."*

> *"Allowing children to be creative in their learning so that they have an input into what and how they learn. Having children working in pairs or small groups to help each other and learn from each other is hugely beneficial. Having children sharing their expertise with other children ensures much more effective learning."*

Respondents highlighted a range of innovative approaches they engaged in with their learners, from students leading learning to actively collaborating with their peers and teachers in their learning activities. Learners, both in primary and post-primary settings, used digital technologies to engage more actively with their own learning, something teachers noted led to more effective learning.

Some respondents described specific ways in which digital technologies could be integrated into classroom practice.

> *"Clearly defining where technology can be of benefit in the classroom and not including it for the sake of using technology. Teachers have to be smart in how it is used and incorporated into lessons to ensure the maximum benefit is achieved. technology is a tool to be used but is not the be all and end all of the educational experience."*

> *"We started out with very structured and well-defined lessons for the children to follow. When the children became familiar with the resources, we broadened out the scope of the challenges that we set them, allowing them to start with the basics and then encouraging them to develop and extend their knowledge through experimentation and investigation. Finally, we were able to integrate the use of the resources with the topics that were being followed in class at the time and gave the children the freedom to follow their own initiative when using the resources."*

> *"Digital technologies can provide opportunities for personalised learning experiences, where students can work at their own pace and level. Teachers can use digital tools to differentiate instruction, allowing students to engage with content at their own level and pace."*

> *"There needs to be a clear framework where children develop and progress their skills on a year-on-year basis."*

These comments capture the critical reflections of teachers in relation to how they and learners used technology. Teachers critically reflected on how they were using technology and noted that it should never be used just for the sake of using it. Technology should be used for specific purposes, and it is important to capture and share these with colleagues. The comments also capture the journey that staff undertook with digital technologies, and their role and that of their learners evolved over time. Specifically, teachers gave students more freedom to engage in self-regulated learning over time and moved from highly structured to less structured activities, while others used the technology to support more differentiated learning. Finally, some teachers noted that there was a need for more supports around developing student digital skills over time and this was addressed within the *Literacy, Numeracy and Digital Literacy Strategy* developed by the Department of Education in 2024[18]. These lessons could be captured and shared more widely with all teachers in the future.

Several respondents stressed the importance of reflection and self-assessment in the learning process.

> *"Digital technologies can be used to enhance the learning experience by incorporating them into the curriculum. Robotics can be used in the classroom to promote active learning - this can help students develop problem-solving skills and critical thinking skills. Coding skills introduces coding concepts and principles through games and puzzles. A constructivist approach to teaching and learning encourages student-centred learning, this can be facilitated by providing opportunities for students to collaborate, problem-solve, and share their ideas."*

> *"Formative assessment strategies allow teachers to assess student progress throughout the learning process. This can involve providing feedback on student work, using self-assessment tools, and providing opportunities for peer assessment. Pupil reflection is an important part of the learning process pupils are encouraged to write reflections, share their thoughts with peers, and engage in class discussions."*

> *"Students were engaged in the process and technology played a big part in assessment of learning."*

The first comment above is particularly interesting and highlights the opportunities for digital technologies to enhance existing learning experiences. It also highlights a range of approaches and activities that can promote student-centred learning by encouraging learners to be more reflective. The comments also emphasise the potential role of digital technology in assessment, specifically formative assessment, where students can use data to reflect on their progress with their peers.

Infrastructural models

When asked about the most appropriate infrastructural models to support innovation in relation to teaching, learning and assessment, a few respondents provided quite comprehensive overviews:

> *"The best infrastructure and technological framework for supporting innovation in education will depend on the specific needs, goals, available resources and budget of the individual school. Cloud based infrastructure provides an efficient and flexible way to store, manage,*

[18] https://www.gov.ie/en/department-of-education/publications/literacy-numeracy-and-digital-literacy-strategy/

and access educational content and services. This allows for scalability as resources can be scaled up or down as needed. Learning analytics can provide insights into student progress, engagement and preferences and can support personalised and adaptive learning. Good wireless network infrastructure provides reliable and secure connectivity for pupils and teachers to access educational content and services. One-to-one or one-to-two device ratio, iPad or laptop, supports personalised and self-directed learning as well as collaborative learning and assessment. Mobile device management allows schools to manage and monitor mobile devices. It provides features such as app management, content filtering and remote device management."

"The following is essential in a digitally emerging school: Robust speedy internet connection with adequate bandwidth Safe internet monitoring for pupil and wider school community Adequate internal network points with up to date switching managed Wi-Fi of APs is a must in larger school areas where there is a steadily increasing number of mobile devices in use from year to year There needs to be a stronger more collaborative link between the Schools Broadband Service and schools internal network provision. Dual provision of internet providers so if one provider fails a second is available to provide continuity. Data storage and cloud providers are an essential part of digitally enabled schools. A dedicated portal for all government, local authority, 3rd level colleges, 2nd level schools, 1st level primary schools, libraries and preschools would be a very aspirational wish."

These comments highlight the need for robust internet connectivity to the door of the school and corresponding technologies within the school to enable teachers and learners to access digital tools safely and efficiently. They capture the complexity of these evolving digital technologies and how these tools can enhance student learning. Schools indicated a desire for more advice on these technologies and more connected platforms across multiple education levels and stakeholders.

Broadband

Broadband, Wi-Fi or network connectivity were mentioned as issues by 48% of those responding to this question, highlighting the need for high-speed, reliable broadband connections to enable digital learning and support the use of technology in schools.

"One of the biggest restrictions there is at the minute is the speed of internet coming in to the schools. The fact that schools are being supplied with only 300mb when the area has up to 1GB capacity is laughable."

The issue of robust broadband provision has been highlighted by schools for many years and is also a concern at EU level, where it is regarded as a key enabling factor in the provision of digital education[19]. There is an increased need for reliable fast broadband in schools with the growing reliance on cloud computing services and an ever-increasing number of digital devices being deployed in schools.

Physical space

The survey captured the need for greater investment in locating and using digital technologies.

[19] See the Digital Education Action Plan, https://education.ec.europa.eu/focus-topics/digital-education/action-plan/action-1?/focus-topics/digital-education/action-plan/action-1=#Proposal

> *"Space is vital for any project. I think if schools are to deliver anything they have to consider security of tech, housing of tech and so on."*

Hardware and devices

Many teachers indicated the need for technologies for general use such as iPads, Chromebooks, or laptops, whereas others pointed to specific technologies for specialist use such as micro:bits and LEGO kits for STEM projects.

> *"There needs to be sufficient hardware and software (ideally 1 per pupil)."*

Software and platforms

Many teachers mentioned the value of specific software tools and platforms such as Google Workspace (including Google Drive, Docs, Slides, Classroom), the Microsoft suite, Learning Management Systems, and various educational apps.

> *"Cloud-based infrastructure provides an efficient and flexible way to store, manage, and access educational content and services."*

> *"The use of GSuite has really benefited our school with the input of everything from Gmail to Drive and constant use of Docs and Slides."*

These comments highlight once more the use of cloud-based solutions that rely on robust internet provision.

Standardisation and policy

Several responses advocated for a standardised model of digital infrastructure.

> *"The department should provide a one for all model of infrastructure as at the moment it differs greatly between schools. Some schools with no IT experienced teachers can struggle. Some schools rely on very expensive IT companies to help them. Having one model for all schools would help to train teachers nationally and would help collaboration between schools. The main issue with this now is that most schools will be reluctant to change to a new system having put a lot of effort and money into developing their current model."*

This comment suggests that some schools in the Programme have struggled with procuring and implementing digital infrastructural models. However, the respondent noted that this may not be feasible in the future as schools have already invested heavily in their existing systems. Once again it highlights the issues schools face in procuring, installing and providing staff with adequate supports to use these expensive technologies effectively. This issue might be further explored in any future programmes.

Support models

One of the Cluster Coordinators provided a comprehensive overview of support models (presented in the table below for ease of reading). The Coordinator had given this issue considerable thought and provided a detailed response paraphrased below.

Table 10: Support model suggestions.

Professional Development	• Provide ongoing professional development for educators on innovation, emerging technologies, and new pedagogies. This can be done through workshops, seminars, online courses, and other approaches.
Technical Support	• Provide technical support to ensure that the necessary infrastructure and hardware are in place and working efficiently. This includes providing access to software and applications that support innovative teaching and learning.
Administrative Support	• Provide administrative support to teachers and school leaders to facilitate innovation. This includes streamlining administrative processes and providing access to funding and resources.
Curriculum Design	• Support the development of innovative curricula and lesson plans that promote creativity, critical thinking, and problem-solving.
Peer Collaboration	• Encourage peer collaboration and networking opportunities to share best practices, innovative ideas, and successful implementation strategies.
Parental Involvement	• Involve parents and families in the innovation process by creating opportunities for them to learn and contribute.
Partnerships with External Agencies	• Establish partnerships with external agencies, such as universities, businesses, and community organizations, to support innovation in schools. This can include providing access to experts and resources, as well as funding opportunities.

Such a comprehensive support model, that integrates all these components, could be more effective in facilitating digital innovation in any future schools' projects.

> *"A technical support team could provide assistance with hardware and software issues and maintenance. This team could also provide guidance on the appropriate use of technology in the classroom and ensure that devices and software are up to date. Professional development can support teachers in developing the necessary skills and knowledge to effectively use technology in the classroom. This could include training on new technologies and pedagogical approaches as well as ongoing support and mentorship. Administrative support could include guidance on procurement, budgeting and policy development related to technology in the classroom. This can also include support for managing data and compliance with data protection regulations. Technology suppliers provide customer service and support for the devices and software used in the classroom. This can include support for setting up and configuring devices, troubleshooting issues and providing maintenance and updates. Pedagogical support could provide guidance on how to integrate technology into the curriculum in a meaningful and effective way. This could include support for developing lesson plans and assessments that incorporate technology as well as support for evaluating the impact of technology on student learning."*

> *"The success of the 'support model' does have a very high dependency on the 'human resource'. A lot has to be carefully put in place so that the area of successful digital technology development can take place in schools. These are different from school to school. Different countries use different models, working at government level, local authority, school districts or municipalities and even schools operating in an insular way providing their own support. There is no doubt but there are certain support structures that*

must be considered by schools so they can successfully engage and develop their own digital development."

The above comments are particularly comprehensive and identify a range of supports or enabling factors that ideally should be in place to support innovative digital practices. Respondents felt that these resources should span a range of areas, such as professional development, curriculum design, technical support and administrative support. The suggestions around administrative support are notable, as they include data compliance and protection. They suggest providing support on establishing partnerships with industry and third level institutions. There is also a suggestion that any innovation support model should have sufficient human resources, to enable full engagementwith digital education in a meaningful way.

Most of the Oide TiE Advisors referenced the importance of a sustained support model.

"In my opinion the sustained support model is an ideal support model to facilitate innovation in schools. This model allows for multiple support sessions building on targets and achievements over the course of an agreed timeframe."

"The sustained support model provides ongoing professional development opportunities for teachers, which is critical for supporting the embedding of digital technology in schools. Teachers need to develop their knowledge and skills to effectively integrate technology into teaching and learning practices. Ongoing professional development ensures that teachers are continuously improving their skills and staying up to date with the latest digital technologies."

Here again the advisors highlighted the need for ongoing regular support interactions with schools over time. The embedding of digital educational practices takes time and commitment from schools, and this view was shared regularly by respondents. They noted that the nature of the support depends on the school context.

Professional development and support

Many respondents to this question (50%) highlighted the need for continuous professional development and the provision of practical, manageable, and regularly updated training materials.

"Training and professional development for teachers needs to be free and training provided needs to be in areas of new digital technology and skills – rather than lagging behind the rest of Europe we need to be on par or ahead of other European countries using new up-to-date materials."

"During this programme, I became familiar with the Bridge 21 model through Trinity College Dublin. I would recommend teachers to complete similar training for CPD."

"Profession development courses need to change on a regular basis rather than the same courses being delivered year after year and very little access to new material."

"Training with tech experts. Tech companies to visit schools and give practical solutions to each building i.e. broadband/Wi-Fi still a big issue in most schools as well as solutions for tech i.e. iPad and app management systems etc."

> *"I have found that the simplest support networks work best. We have used a WhatsApp group to work with other teachers in the local schools. I have found that a training day works well but if you have a group that has a wide level of abilities when it comes to using technology then a training day will not get the benefits that you want. Support after an initial meeting is a very effective way of seeing improvements."*

They expressed a desire for a wide range of courses that would constantly evolve to include new or emerging technologies. Where possible they would prefer these courses/events to be free of charge. The reference to Bridge 21[20] is interesting, as this is a project in Trinity College and highlights that there is a wide range of providers in this space that schools can avail of as well as the supports available from Oide TiE, the education centre network and others. Ultimately schools indicated a need for quality professional learning experiences that would help them utilise and maintain their existing digital technologies.

There were differing views on the value of 'external' training.

> *"For example, we purchased our iPads from Wriggle and so received training on app management through JamSchool and whole school training on device and app use."*

> *"Private companies have a mixed success rate in our experience."*

> *"We rarely use the Oide TiE. At times we consult the company who supplied out iPad/laptops."*

Again, these comments capture the complexity and variety of professional development options schools can avail themselves of depending on the topic and issues they require support with.

Some wanted more explicit professional learning opportunities at post-primary level, rather than generalised 'support'.

> *"A subject-by-subject teaching plan with online webinars on how to implement the teaching plan. Just like the book publishers do with their teaching plans for teachers. Or equally, add to the book publishers' plans, by showing what and how you can use digital technology with their books/scheme. Keep it small, practical and manageable with the heavy teacher workload taken out of it, speaking as a teacher with 4 classes."*

> *"In-school support through modelling in the classroom."*

Some respondents expressed a preference for very practical hands-on suggestions on how to use digital technology in specific curriculum contexts, while others wanted to see digital practices modelled in real classroom settings.

[20] https://tft-project.scss.tcd.ie/index.php/the-bridge21-model/

Peer-to-peer collaboration and support

Many (20% of respondents) emphasised the importance of networking, sharing knowledge, observing and showcasing the work of other educators or schools. Suggestions included local and regional collaborations, communities of practice, teach-meets, and the use of digital platforms for communication.

> *"Facilitated collaborative models like communities of practice. Opportunities to share/ observe the work of other schools/ digital leaders."*

> *"There is a need for a more local support person in an area for schools to work with schools on integrating technology into learning. Having a person in a school that can work with other schools in a geographical area would be of huge benefit. building a relationship of trust takes time and needs to be worked on over a number of years rather than changing from year to year."*

The idea of a 'local support person' for digital education existed in the past, as the 1998 digital education plan, Schools IT2000[21], made provision for ICT Advisors to perform such a role and they were located in education centres. Ultimately the idea of capturing and sharing digital practices with colleagues was strongly valued by those responding to the survey, particularly during the project phase when they had time and resources to sustain such activity.

Technical support

The issue of technical support was mentioned frequently (23% of teachers and cluster coordinator respondents), specifically in relation to maintaining and deploying technology hardware.

> *"Redirect the funding from people with the ideas towards qualified technicians. Employ a company that will visit a cluster of schools on a rotational basis. They can then fix the computers, laptops, iPad, Wi-Fi, boards etc. The teachers can then focus on using the technologies and the technicians can make sure that they work."*

> *"Expecting teachers to keep a technology infrastructure up and running is every shade of crazy. It is infuriating, pointless and dispiriting. Teachers teach. Managers in the school try to manage. There is no space for keeping hardware working."*

> *"A school or any institution that relies on hundreds of devices, a complex network as well as hundreds of other ancillary hardware needs a full-time technician to be effective in its management. At present, teachers maintain these complex systems in their few non-teaching hours."*

Respondents also highlighted well known issues related to technical support and broadband infrastructure.

> *"A formal, structured and possibly centralised technical support system is essential."*

> *"Ongoing repairs. Get this part correct and everything else will follow."*

The issue of technical support emerged a number of times in this report. There was a strong message that teachers teach, and administrators manage, and that there needs to be additional support to schools to ensure that devices and networks were working effectively. The

[21] https://www.gov.ie/en/department-of-education/publications/schools-it-2000/

comments suggest a range of possible solutions, and this is an area that could be further explored in future programmes.

Administrative and institutional support

Several responses reported that they felt they lacked access to sufficiently detailed guidance on how to approach the management of their project finances. Others reported a need for more guidance on their projects in general.

> *"Administrative support could include guidance on procurement, budgeting, and policy development related to technology in the classroom."*

> *"Oversight in all areas. The whole process was very loose, with respect, infrequent check-ins, left to our own devices (excuse the pun)."*

These comments suggest some people wanted more contact with the Department, other clusters and Oide TiE to ensure they were on the right track.

On a different issue related to institutional support, several responses reported difficulty in obtaining substitute cover, even though funding was available for it.

Embedding of digital technologies

The Cluster Coordinators, teachers and Oide TiE Advisors were asked for their opinions on how best to achieve the embedding of digital technologies in teaching, learning and assessment within and across all schools.

Funding

Many respondents (29%) highlighted the need for adequate funding for hardware, software, internet connectivity and maintenance.

> *"Through appropriate funding to schools so that all pupils have access to digital technology. At present schools may have to share a set of 16 tablets in classes of 30. Funding is necessary to provide more hardware and software to embed digital technology into the classroom."*

> *"The Digital Strategy for Schools to 2027 provides a framework for the integration of digital technologies in teaching, learning and assessment across all schools in Ireland. Adequate funding and resources will be necessary to support the implementation of the national strategy."*

> *"Completion, sharing and familiarisation of the agreed digital framework for schools will greatly help to embedding digital technologies coupled with the correct funding to aid development and creative skills."*

> *"A department-led, broadscale funded model that includes technicians, and a central procuring system would be the only system that can ensure this."*

The provision, upkeep and replacement of digital technologies are costly and requires ongoing funding. Schools are acutely aware of the costs involved in procuring and maintaining these technologies and called for adequate funding.

Professional development and time

Many respondents (49%) stressed the need for appropriate teacher professional development and time. This was referenced in relation to professional development and in providing more time for Cluster Coordinators.

> *"Teacher training is key to embedding digital tech in reading, learning and assessment. Continuous Funding is also required as technology is constantly requiring updating."*

> *"Professional development for teachers: Teachers need to be equipped with the necessary knowledge and skills to effectively integrate digital technologies into teaching and learning practices."*

> *"More time (with sub cover) for IT co coordinators in the schools and clusters to meet, plan and implement change."*

Though some thought that compulsion was necessary:

> *"By making it compulsory to include such practice within your teaching and learning."*

There was a recognition that staff, at all levels of education, require ongoing professional development supports and that adequate time and funding should be made available to enable teachers to meet and share their practices.

Collaboration and consistency

Some (15%) respondents noted the need for collaboration across schools and with other institutions.

> *"The most important thing in embedding technology across a school is a whole school approach and teachers working together and sharing their knowledge."*

> *"Consistency. Two opposing schools running oppositional technologies doesn't benefit the educational landscape. It should be collaborative within geographical areas also."*

> *"Common sharing platform (e.g. Teams/OneDrive) common and shared outcomes which are measured."*

> *"The model of colleges/ universities providing support to secondary schools and secondary schools supporting primary schools is definitely a model that needs to be built upon. It is a model that gives huge collaborative support to teachers."*

The comments suggested that it is better if schools adopt a whole school approach and that clusters, where possible, use common platforms, devices and applications. Such approaches helped promote a collaborative culture within and between schools. The final comment suggests that the involvement of higher education institutions (HEIs) could provide collaborative support to teachers. However, it is not clear how this might occur, as there was limited evidence of such approaches in the project.

Curriculum redesign

Some respondents stressed the need for curriculum change.

> *"Curriculum integration: Digital technologies should be integrated into the curriculum across all subjects and grade levels."*

> *"Align curriculum objectives with digital technology (DT) best practice - teachers have a curriculum to deliver and must see the value in DT in helping achieve this."*

> *"Focus on student centred learning, fostering problem-solving, creativity and differentiation for all learners. Technology used as an effective tool for learning across a range of subject areas."*

These comments highlight that teachers are primarily tasked with implementing the curriculum in their school and thus digital education should be embedded within the curriculum and not seen as something additional. While this is increasingly the case, some respondents felt that it needed further strengthening, through more explicit guidance on where and how to use digital technology across the curriculum.

Assessment and evaluation

Respondents highlighted both the need for involving technology in assessment of student learning, and the need for evaluation of the effectiveness of technology integration.

> *"Schools should explore the potential of using technology in assessment to a great extent."*

> *"Ongoing evaluation and assessment of the impact of the integration of digital technologies in teaching, learning and assessment will be necessary to determine the effectiveness of the national strategy and to identify areas for improvement."*

The first comment reflected earlier themes regarding curriculum integration and the use of formative assessment. The issue of ongoing evaluation of the national digital strategy is an important issue that many other countries are also facing.

Section 7: Post-Project Survey - Possible future digital education initiatives

In this set of questions, the survey asked the Cluster Coordinators, teachers and Oide TiE Advisors to reflect on the ways in which future Department of Education and Youth initiatives in the area of digital technology might be organised.

In this section the report presents a high-level synopsis of the key findings from the qualitative data analysis on this topic in Table 11. This is then followed by a detailed account of the supporting evidence for these findings including quotations from the respondents.

Table 11: High-level synopsis of feedback on future initiatives.

Application, funding, support and monitoring processes	• The present model worked well overall, but there were concerns about monitoring and about guidance on creating deliverables. Some respondents needed more guidance on funding, and some suggested more flexible and phased funding would have been better. • Some found the application and monitoring process complex, and suggested alternative approaches, which included simplified applications, predefined project options, and curriculum-linked guidance.
Project communication and support	• More support and training for Cluster Coordinators was needed. Communication from Programme leadership was sometimes an issue.
Monitoring and deliverables	• Some respondents felt that there was little feedback from the monitoring process. • Some respondents felt that the required deliverables were somewhat ill-defined, and the use of local showcases was suggested as an alternative.
Insuring sustained impact	• Recommendations included long-term planning, use of open-source tools, continued technical support, and regional collaboration through education centres.
Programme funding	• For sustained impact there should be sustained, additional, or maintenance funding after the initial project funding period. This should include support for replacing obsolete equipment and resources for repair and upkeep of the technology equipment, as well funding for hosting face to face events or workshops to share what they have learned with their peers.
Ongoing training and support	• Respondents suggested continued support in the form of Oide TiE support for digital skills development, and short-term support for a few years after the project.
Future Initiatives	• There was strong interest in future programmes, especially in AI, maths, and digital literacy. It was also suggested that it would be good to propose a mixture of future initiatives which offered either high autonomy for those who wanted it or more structured options for those who felt they needed the additional guidance.

Application, funding, support, and monitoring processes

The Oide TiE Advisors felt that the process worked quite well, and that clusters had adequate funding and were well supported by them.

> *"I feel the (application) form provided schools ample opportunity to outline what they hoped to achieve in the three years. I feel clusters were adequately funded and the model of releasing funding annually worked well. Clusters had to outline their budget in the application form which worked well to monitor how much they needed annually. Having Advisors assigned to support the clusters was a very good model and allowed us to develop a relationship with clusters. The secondment model led to changes to the Advisor assigned to the cluster which negatively impacted upon the support provided. Project management training would have worked in favour of these clusters if it was provided at the beginning of the project. It would have worked well if the lead coordinators had an opportunity to meet throughout the project as a way to support each other through the project."*

The issue of project management was highlighted by an Advisor and was something that should be revisited and built in from the outset in any future programmes. Furthermore, it was recommended to bring project/Cluster Coordinators together more frequently, either online or face-to-face, to further support each other and to learn from one another.

> *"We were not involved in monitoring."*
>
> *"Monitoring and reporting was difficult in my role as an advisor as it wasn't part of my remit."*
>
> *"As I wasn't involved in monitoring perhaps an element of project management or support in this regard for teachers might have been helpful."*
>
> *"As monitoring was not part of my role, it is difficult to comment on this area."*

Several Advisors stated that they had no role in the monitoring and reporting activities of the Programme. This is an important finding, as the role of the Advisors was primarily to support the clusters, and they performed well in this role. However, the role of monitoring or providing critical feedback to the projects did not appear to be assigned to any organisation or individual in the Programme.

Some Cluster Coordinators and teachers were satisfied with the existing processes.

> *"I have to admit very little change. I believe the scheme ran very well and the open-ended approach of the scheme meant schools had the opportunity to explore how the funding could be allocated."*
>
> *"The current model worked really well. Even though we were geographically challenged, IT afforded us the opportunity to keep connected."*

However, others pointed to a number of problems and made suggestions for improvement.

Process of application and setting up projects

Some respondents found the application process overly time-consuming and suggested a less formal application process, while others suggested the need for clearer guidelines and expectations.

> *"Give direct examples and a working plan on how these digital initiatives can be directly implemented into the curriculum, and everyday teaching and learning."*

"A step-by-step plan: 1. What to teach. 2. What to use. 3. How to teach it. For example: To use Book Creator in English. The options/ scope of what schools can do is way too broad, in primary school anyway, and there is so much time wasted trying to figure it out ourselves. I really felt as if I were scrambling in the dark for a long, long time before we got up and running."

"Previous application process extremely time consuming!"

"I would like to see a less formal approach to applications for funding."

"Pick initiatives from a menu. Pick a max number of maybe 3 from 12. Run for an initial 3 years with funding to purchase anything that's required. Maintenance funding yearly after that. After the first three years pick 3 more initiatives from the list of 12 and repeat the same process. funding based around how many schools; colleges are involved in the cluster. The most expensive part of these clusters is setting them up. They could run very cost effectively once setup."

"The tech should be small scale. Even our project was too high end."

There are some particularly interesting suggestions here in relation to how future innovative digital projects might be organised. Some schools expressed a preference for less choice and more direction in the future so that cluster plans would be more manageable in the future. This suggests that the Department might initiate specific calls in the future in areas such as Generative AI or in digital literacy and provide ongoing guidance and support to the successful clusters.

Management of funding

Some felt that there was a need for more advice on managing funding issues, and some argued that there should have been more attention to contingency planning. Others suggested a phased funding approach and potentially funding to extend some projects into the future.

"More advice regarding how to split funding. We were a school of 430 students and as there was no advice re how funds were to be split, it was split evenly with the other schools in our cluster. Both much smaller schools."

"Drawing down of funds should be a lot easier and the amount of reporting and accounting should be handled by school management."

"Obviously, with COVID, the project went on for an extra year. This of course was unprecedented; however, we budgeted for a three-year programme."

"I think that the funding should be released on a phased basis so the school can better assess what technologies are needed."

"Funding needs to be available over a much longer time period at least 5 years possibly 10 to fully embed in the initiative in schools."

"Making sure that those involved are being reimbursed for expenses they incur."

The issue of project administration has featured earlier, and it is clear that funding and budgeting issues presented challenges for some clusters. Cluster Coordinators would have benefited from greater clarity on how to manage this process and this aspect might have been outsourced, in some cases to an education centre. Clusters often struggled with how to budget

and manage their funding, as the Coordinator was often a classroom teacher and this aspect may be revisited in any future projects.

Project communication and support

Whilst Cluster Coordinators and teachers expressed appreciation for the support provided by Oide TiE, concerns were raised in relation to general Programme administration communication activities and the level of support provided to Coordinators to manage their clusters. The following quotations capture these two issues from the perspective of the Cluster Coordinators and teachers.

> *"Luckily, we had a super cluster coordinator who appeared to manage these processes efficiently. Personally, I would have found the role of coordinator very daunting, challenging and time-consuming - it would put me off taking on the role. I feel the process would need to be simplified considerably."*

> *"Training for co-ordinators would have been a great support & improved the monitoring piece also."*

> *"Very poor communication throughout the project. End of year reports were sent in and never acknowledged. Emails were ignored. But for the support of the Oide TiE Advisors, we would have been lost."*

> *"Overall communication from the project leader was poor. We regularly sent in reports which were not acknowledged. Emails were not answered. This was very disheartening."*

> *"There was very poor communication from the top down. (I do not include the Oide TiE advisors in this, they were fantastic) Reports were sent in and not acknowledged. This was very disheartening especially given the significant amount of time given to the project and the reports."*

These comments highlight a number of issues in relation to how project reporting and requests for ongoing support might be handled in any future project. It may be worth revisiting the kinds of reports that clusters were asked to submit and determine whether the information requested was essential, while also considering how best to acknowledge such reports and determine if additional information was needed. Secondly, it may be worth exploring how best to respond to cluster email correspondence in any future potential projects. Ultimately, a more simplified reporting approach could work better in the future, and communication with clusters could be outsourced.

Monitoring and deliverables

Monitoring was recognised as important, and some respondents felt it had generally worked well, but some felt that the definition of deliverables was unclear, and that greater clarity in this area would have helped the monitoring process.

> *"Implement regular monitoring and evaluation of the project with clear objectives and deliverables."*

> *"Everything was great except the lack of a clear list of deliverables from the beginning. This would greatly help teachers not to be demotivated every time a new ask is being made. This lack of clarity likely contributed to a lack of whole staff buy-in."*

> *"Monitoring/review completed immediately on completion of the initiative."*

> *"No need to have to make a website for the project, this was a waste of time and money when a free twitter page was much more useful."*

> *"I think a more localised approach to showcasing projects would be beneficial. Provide an opportunity for schools to meet and view each other's work."*

These comments suggest that future programmes should clearly outline what deliverables are required from the start, what supports are available and what is expected from projects. In future, other organisations, such as the education centre network, could play a role in this process in hosting regional showcasing events for clusters in a particular region.

Ensuring sustained impact

In this set of questions, the survey asked the Cluster Coordinators, teachers and Oide TiE Advisors to reflect on how the Department might ensure that any such future initiatives have a sustained impact beyond the project's funding period.

It should be acknowledged that several projects were already able to evidence the sustained impact of their projects.

Existing good practice

> *"We have decided to continue with our project. It is important that resources that are initially invested in are sustainable and can be reused. I think that it is a good idea to stagger the funding over a period of time. This gave us the opportunity to see what the resources were like within a classroom setting and allowed us the chance to monitor and evaluate the next phase of our project."*

> *"Allow for projects to be extended if there is room to do so. We loved our project so much we extended it into an ERASMUS project called STEAM teams in order to go deeper and wider."*

These comments suggest that future programmes should include options for extension and that funding should be released in stages over time, rather than dispersing significant funding at the outset. It is notable to see that some projects found ways to extend their activities through programmes such as Erasmus+ and this option could be included in future programmes. Furthermore, some of the structures that exist within Erasmus+ projects, such as clarity on project deliverables could be considered for inclusion in future programmes.

Suggestions for improved sustainability included sustainability planning, communities of practice, continued funding, ongoing support and training, and the need for attention to teachers' existing workloads.

Sustainability planning

Some responses focused on the need for clear sustainability planning from the beginning of the project.

> *"Consider a whole department approached around open-source technology to ensure that the initiative is sustainable and can be maintained beyond the project's funding period."*

This suggestion, though a post-primary example, suggests that where possible clusters should use open-source technologies, so that they can continue to use these technologies after the

project has ended. This is a reference to sustainable funding and the cost of digital technologies, an issue raised earlier in the report.

Communities of practice and local involvement

There were numerous calls to build in opportunities for clusters to meet and share their practices and issues with a range of stakeholders including project teachers, NGOs and local companies, and coordinating teacher meetups. Events such as local competitions, school showcases, and teachmeet-style events were suggested to build local community engagement.

> *"Local Education Centres are ideally suited to encourage sustained impact. School showcases, local competitions and similar activities would be very beneficial."*

> *"Ensuring that a central element of the project is the development of a sustainable community of practice for each cluster that can be kept running after the funding period has ended."*

> *"Give schools power to build local capacity by training and empowering local communities to take ownership of the initiative. (b) Give schools the opportunity to foster partnerships and collaborations: industry, NGOs. This can be done by having a framework already in place at the beginning of the project."*

> *"Create digital communities of practice in local ed centres or in local schools whereby schools can meet and work on projects or shared initiatives and help each other to implement digital strategies in their schools. Assisted by the Oide TiE Advisors for support/training."*

> *"Establishing strong communities of practice amongst cluster teachers could ensure sustainability by providing a strong foundation for learning amongst school leaders and teachers. Facilitating coordinating teachers to meet up and collaborate face-to-face during the project could also prove beneficial in this regard to support one another."*

These comments suggest that local education centres could play a role in organising and facilitating future communities of practice associated with digital education projects. These communities of practice could also include the Advisor working with the cluster and other organisations. Furthermore, the creation of such local networks could enable projects to disseminate what worked well and potentially lead to the creation of teacher professional learning courses and possibly new projects in the future.

Programme funding

Many respondents (52%) emphasised the need for sustained, additional, or maintenance funding after the initial project funding period. This included support for replacing obsolete equipment, funding for hosting events or workshops, as well as resources for repair and upkeep of technology infrastructure.

> *"Access to funding to replenish obsolete equipment in time."*

> *"If face-to face workshops are accepted in the future as part of the project - to sustain these, it would be great to receive small additional funding after the programme finishes to allow the continuation of these workshops. Even once a year to cover travel."*

"Arrange technology repairs people to visit schools regularly so that they can keep the hardware working from day to day. There is not a single day in this school when some device isn't working. There is no teacher qualified to fix the computers, interactive panels, Wi-Fi, iPads, laptops etc. They then lie idle. This is shocking and awful. Ongoing repairs is essential. It must form part of the reality of schools."

"One of the best ways to ensure sustained involvement in a project is that in a few years it would be checked back in on and if it is still running successfully that a further allocation of funds would be available to build upon that work. Incentives continued involvement."

Many of the comments address the issue of ongoing equipment maintenance and repair, and this is typically funded via the Digital Strategy ICT infrastructure grant[22], which is provided on an annual or biannual basis to schools. In addition, there were calls for funding to enable clusters to share their learnings, via face-to-face learning events, allowing them to share experiences with peers. Such events could take place in an education centre. The final quote raises the question of how to ensure future projects are incentivised to continue innovating and building on their initial findings.

Ongoing professional development and support

Respondents suggested continued support in the form of Oide TiE support for digital skills development, short-term support for a few years after the project, and ongoing training from established frameworks.

"Provide short term support for a few years after."

"Continuing Oide TiE support for digital skills development for teachers and creating a suggested equipment list for schools."

It seems clusters wished to continue learning and therefore sought to retain the connections they had built with the Oide TiE team. This also raises the issue of identifying projects that had the potential to be further developed over time and the possibility of providing them with extra supports.

Consideration of teacher workload and time

A few respondents stressed the importance of greater consideration of the existing workload of teachers when planning future initiatives.

"Lighten the workload in "figuring it all out". Give the teachers a plan/ options of plans per subject give a certain amount of funding for upgrades replacements each year."

"A time allocation for the teachers involved."

"I believe it comes down to time and funding. As coordinator, the release days with sub cover were essential to making the project a worthwhile experience. Without that it is hard to keep the original initiative going beyond the 3 years."

"The release days for the Cluster Coordinator allowed excellent transfer of skills and ideas between schools. A small number of release days on an ongoing basis would improve the sustained impact of the project."

[22] https://www.oidetechnologyineducation.ie/technology-infrastructure/ict-grants/

> *"Some sub relief for school coordinator to meet annually."*

There are some particularly interesting suggestions here in relation to how Cluster Coordinators and teachers can be better supported in the future. While time and funding are mentioned, there is also a suggestion to disperse project management duties among a number of individuals or organisations. These suggestions should be considered in any future programme rollout, as issues around teacher time and workload[23] have been increasingly highlighted in public discourse.

Possible further initiatives

All respondents were asked to provide suggestions related to similar future initiatives that they would see as valuable. A wide range of suggestions were made.

Some were happy to take part in similar initiatives in the future.

> *"Keep them coming, for me it was one of the best initiatives I have ever been involved with. The personal and professional growth was hugely beneficial. The schools involved are now passionate about enhancing STEM education."*

> *"We really enjoyed the project that we undertook, and we would love to revisit this area and expand where possible."*

Schools clearly recognise the benefits of collaborating with other schools, the Oide TiE team, and other partners in implementing innovative digital and STEM practices. These schools want to be part of such collective experiences where they can explore the potential and challenges associated with embedding new or existing digital technologies into their classroom/school practices.

However, some wanted more autonomy.

> *"Allow schools to have greater autonomy, flexibility and freedom when trying out any new initiatives."*

And some wanted less autonomy.

> *"I think it should be planned by you and delivered to schools en masse. As an example I've been exploring LEGO coding elements and robotics. It's so simple but can be upscaled for higher years so easily. If the Dept/Oide TiE offered a set of technology criteria and set amount of it there wouldn't be such disparity between the breadth of projects. You define the project, then ask schools to sign up. This will allow for more rigid deadlines rather than each project having its own unique set of criteria and goals. Ours were too lofty, happy to admit that now."*

Future programmes could offer both flexible options for schools with well-developed proposals and more structured, guided projects for schools seeking additional support at the design stage. There appears to be value in both approaches, depending on what the Department wishes to achieve in any future digital innovation projects.

Some specific suggestions were provided.

[23] https://www.independent.ie/irish-news/education/study-reveals-teachers-stretched-to-the-limit-with-workload-demands-and-lack-of-resources/a1957320861.html#:~:text=Some%2083pc%20of%20teachers%20%2D%20when,hours%20or%20more%20each%20week

"Coding and Online pupil profiling are two excellent digital programmes that could be funded by the dept."

"I would like to see something similar around the new Digital Strategy for Schools. A shared vision of where we are going nationally is important to both set clear goals for the future and to ensure a minimum standard is achieved by both students and teachers."

"Something creative could be done with the Teaching, Learning & Assessment of MATHS to improve the Classroom Based Assessments & promote Higher Level up-take. As enjoyable as the initiative was it was niche. Something creative in a core subject as important as Maths should aim to encourage students to keep on Maths to HL in LC and to purse course & careers in STEM."

"STEM hubs of excellence at regional level where schools could opt in to participate and receive funding to help showcase innovation in their school."

While much has been achieved in the current project it is clear that many more areas could be developed using a similar approach. Over time, emerging technologies, such as AI and immersive technologies are likely to become more widespread and new curriculum areas may require further exploration. Approaches such as SEF Digital & STEM could provide spaces for the Department and its support services to work with clusters of schools to explore how these technologies might be implemented in schools within a supported environment.

Section 8: Programme recommendations

In this section a number of recommendations are outlined that could enhance any future digital education action research projects that take place in schools. At the outset it should be noted that this Programme was an extremely positive experience for most of the Cluster Coordinators, teachers and Oide TiE advisors who responded to the interim and final report surveys. They reported that the Programme impacted positively on student learning experiences and that, in most cases, they continue to embed digital technologies into their classroom practices. While this positivity must be balanced against the challenges that were outlined in previous sections (i.e. funding, advice, timely communications and project implementation guidance), the Programme had a long-term impact in many schools. This has taken place despite the significant disruption the Programme experienced due to rolling school closures during the COVID-19 pandemic.

The education system should continue to assess how emerging technologies affect administration, teaching, learning, and assessment, so that policy and practice adapt to these developments. Many technologies are not specifically developed for education but have the potential to enhance existing teaching practices and student learning experiences. These technologies require further exploration of how they might be used and their positive and negative impacts evaluated. Dedicated 'safe spaces' and structures should be established to allow schools, support services, and other stakeholderscan engage in experimentation to see how these technologies can be used in schools.

The European Commission[24] has been actively encouraging ministries of education to engage in such experimentation in recent years, as part of their Digital Education Action Plan[25], through the Erasmus+ Key Action 2 policy experimentation instrument[26]. Ministries of education, particularly those involved in the Digital Education Learning Teaching and Assessment Working Group (DELTA WG)[27] are keen to learn how schools are using these new technologies and what lessons have been learned. There are always new or emerging technologies that have the potential to impact on the work of schools, and today these include Artificial Intelligence, digital education content, immersive learning environments and much more.

Programmes such as SEF Digital & STEM can complement larger EU funded programmes and help inform future policy creation and implementation practices. This is an on-going activity that should be taking place in all countries. The following specific recommendations are based on the evidence presented in the previous sections.

[24] https://commission.europa.eu/about/departments-and-executive-agencies/education-youth-sport-and-culture_en
[25] https://education.ec.europa.eu/focus-topics/digital-education/action-plan
[26] https://erasmus-plus.ec.europa.eu/programme-guide/part-b/key-action-2
[27] https://wikis.ec.europa.eu/spaces/EAC/pages/23461992/DELTA

Establish clear programme goals

In the document *Description of Scheme – Schools Excellence Fund – Digital*, the purpose of the Programme was set out as follows.

> *The purpose of the scheme is to demonstrate the innovative use of digital technologies in teaching, learning and assessment, through clusters of schools collaborating on specific projects.*

The call was seeking projects that would demonstrate the innovative use of digital technologies in teaching, learning and assessment in the first instance and secondly looking for clusters of schools to come together to achieve this. The focus on innovation had an influence on the application process, in that projects framed their proposals as innovative, when on occasion replication of existing practices might have been a better approach.

Many proposals came from teachers with strong experience using technology, supported by both their schools and partner organisations. These applicants were well positioned to innovate. However, some proposals appeared more ambitious than appropriate, given the applicants' own level of experience with technology and project management, the readiness of their schools, the support available within their institutions, and the strength of their partnerships. This is reflected in the expressed desire from a number of the survey respondents for a much more structured process, with the Department proposing detailed implementation and teaching plans. Some specifically stated that their projects were too ambitious and thus should have been scaled back over the course of the programme.

Whilst 'innovative use of digital technologies in teaching, learning and assessment' was a reasonable goal for this initiative in 2018, other goals may be more appropriate in the future. An obvious future goal might be:

> *The implementation of a wider uptake of already established use of digital technologies in teaching learning and assessment and taking advantage of advice from teachers who have already implemented the proposed use.*

However, teachers are faced with a rapidly changing technological environment and such straightforward solutions are not always available. It may therefore be appropriate to focus on supporting teachers in becoming more flexible, evolving their technical and pedagogic skills as well as developing new approaches to the curriculum, through initiatives aimed at:

> *The development of teachers' professional skills through communities of practice in the area of using digital technologies in teaching, learning and assessment, with a focus on personal reflection, and taking advantage of the advice of critical friends from partner organisations.*

Such an approach might call on ideas of teachers as designers and researchers, and the success of such an initiative would be measured in terms of teachers improved skills, rather than the continued use of specific technologies in the classroom.

The second element that was mentioned in the call was for schools to come together and form clusters. This aspect seemed to work well where clusters collaborated and worked together towards a common goal. However, there were issues with some clusters that were created after the application process had closed and this is, perhaps, something to avoid in the future.

Initial organisation of the programme

The responses suggested that where projects encountered problems, particularly in the later stages (i.e. Year 3 onwards), these were sometimes due to decisions made at the initial stage of setting up the Programme. For instance, the application form, the types of projects that were sought, the supports that would be provided and the deliverables that were expected from schools.

Whilst many respondents were happy with the original application process, there were some who expressed a wish for a more structured approach with explicit guidance on what they were to do. There were also issues with how clusters applied for and managed their finances particularly in the face of unanticipated costs for replacement equipment and repairs.

It was suggested that a two-stage application process would have helped clusters avoid these problems. Clusters might have been asked first to develop an initial idea outlining the area they wished to explore, and then those that were selected at that stage could have been provided with support and training to develop more detailed plans that were realistic in terms of time commitment, finance, and the support available in their schools.

It is also likely that a three-year project duration (which ultimately was four years because of COVID-19) was too long, and that yearly funding cycles, followed by a review before further funding was released might have helped. For example, in France where they run innovation competitions with industry, they evaluate the projects at the end of Year 1 to see which ones will be funded in Year 2[28]. A similar approach might have worked well in this Programme, particularly for projects that would have benefited from annual funding rather than receiving the bulk of their funding at the outset. Such a staged funding and review process could also have identified any projects that were encountering implementation issues early on.

Programme flexibility

The COVID-19 pandemic had a notable impact on the project, causing disruption and leading to a shift in priorities. Whilst this could not have been foreseen, the lack of preparedness for disruption highlights the need for flexibility and adaptability in project planning and implementation. Future programmes should include contingency plans for possible school disruptions, such as future pandemics or natural disasters, as well as for potential issues with chosen technologies, including licensing, data privacy, and technical faults. Responsibilities should also be clearly defined from the outset. Some respondents noted the need for greater flexibility in amending project goals and workplans. When dealing with digital technology there is always a possibility that things will not work out as planned and future initiatives should should include contingency plans to adapt goals and timelines when unforeseen technical or organisational challenges arise.

Project administrative requirements

The Department provided guidance to schools in writing and at the initial launch event in UCD, on what was expected of them and of Oide TiE. The documentation, entitled *Managing your*

[28] Édu-Up system, support for the production of digital resources for schools: https://eduscol.education.fr/1603/le-dispositif-edu

Project, outlined the role of the coordinator, the role of the participating schools and the reporting requirements for each project. The documentation outlined four types of reports that were expected to be completed at various intervals, as follows:

1. **Termly Reports:** The production of termly reports was a key deliverable from each project. These reports were expected to detail progress, or challenges encountered, in relation to the milestones and deliverables contained in the project plan required to deliver the project's overall objectives.
2. **Biannual Expenditure Reports**: These reports detailed the purpose and level of expenditure, and the procurement procedures followed. Note: All quotes and invoices were to be retained on behalf of the cluster for the duration of the project**.**
3. **Annual Reports:** These were intended to be more complete than the Termly Reports and were to indicate progress achieved against the overall project objectives and outcomes over the course of the year.
4. **Making project documentation available** to the Working Group upon request.

Clusters were also provided with templates for reporting and a note relating to financial management. Despite this guidance and direction provided by the Department and Oide TiE to schools at intervals during the programme, several respondents indicated their frustrations with the monitoring and reporting procedures that were in place during the Programme. They sometimes felt they were overly controlled, and they were tired of being asked for reports. This raises issues around the suitability of these reports and the benefits they brought to the overall project management. It also suggests that classroom teachers may not be best placed to carry out this work due to their existing workloads and their lack of experience in project management.

While future programmes will require monitoring of expenditure and activity, these should not be the only kinds of reporting that clusters would need to engage in. Clusters were also asked to capture and demonstrate project progress, and in the process capture student role and voice. However, this element was largely missing from the submitted reports that were reviewed as part of this evaluation. While clusters were directed to some examples of such reports on Scoilnet and elsewhere[29], this is an area that might usefully be further developed in any future project.

The evaluation attempted to address this lack of detailed reporting by developing a series of case-studies of three clusters (see Appendix 3) and by capturing Teacher Stories (Appendix 4) and these highlight many innovative practices schools engaged in and the lessons they learned.

Cluster Coordinators, despite receiving written and oral guidance, often lacked clarity regarding project management responsibilities and reporting obligations. Many struggled in this regard and any future projects should consider the role of such individuals. These issues might be resolved by providing clearer initial guidance and training for Cluster Coordinators, where their responsibilities are even more clearly defined and supported by guidelines and exemplars for the deliverables required. However, it should be noted that the Department felt they provided such clarity to all Cluster Coordinators. Furthermore, there may be occasions where a Cluster

[29] Some reference sites that were shared:
https://sites.google.com/a/sbclara.ie/midlands-digital-cluster/
http://westcorkrobotics.edublogs.org/

Coordinator could be supported on the administrative tasks by another colleague or an outside organisation (e.g. an education centre), as often these individuals were the creator of the proposal idea and were highly competent in using digital technology, but may have lacked the necessary time and project management skills to successfully meet the procurement and reporting requirements.

Sometimes there were issues with how the Department acknowledged and responded to the submission of these reports and other queries in relation to a cluster and its project. The Programme generated additional work for the Department, and they often lacked the personnel and/or time to respond to such requests in a timely manner. Any future projects should carefully consider the additional workload for the Department and if necessary, identify other agencies or organisations that could carry out this work.

Funding of digital technologies

To ensure effective integration of digital technology in teaching, learning and assessment significant ongoing funding is required. It should not be the role of initiatives such as this Programme to provide all the funding, but rather to provide temporary additional funding to enable new opportunities to be explored and evaluated. However, it appears that some schools perceived the Programme as an opportunity to draw down funding to fund core digital activities in their school or cluster. While the Department does provide separate digital education funding to schools, any future projects should focus on additional digital technologies for specific programme purposes and avoid funding regular school digital activities. This relates back to an earlier point regarding the difference between encouraging innovative uses of digital technology as opposed to more mainstream applications of digital technologies.

The results suggest that even with very willing teachers, the lack of resources, equipment (including replacement and updating), software, internet connectivity, and technical support had negative consequences for some projects. Future programmes might consider breaking funding down into two over-arching categories:

- Capital Expenditure
- Current Expenditure

Furthermore, these might be broken down into categories, such as:

- Digital equipment procurement.
- Broadband Connectivity (to school and internally) to enable project activities.
- Teacher Release Time (Human Resources).
- Technical Support to maintain project equipment[30].
- Outside training/support on specialist technologies.

The provision of funding to procure and install digital technologies was most welcome, clusters also needed funding to free up teachers to write reports, to meet and to plan/share their digital activities. Time is always cited as a major barrier in implementing projects in schools and this issue should be addressed more holistically in the future.

[30] While this issue applied to project digital infrastructure, it is a much wider system issue that impacted on the project activities of some clusters.

Projects that required significant capital funding, for example purchases of €20k or over, should be carefully considered and evaluated against the proposed project deliverables. Where possible funding should be released in stages, rather than at the outset of the project, as has been suggested elsewhere. Such an approach reduces the risk of clusters drawing down funding and not completing their project activities.

Ongoing support structures

Clusters consistently identified a need for ongoing support in the following areas:

- Pedagogical support – using technology tools with learners.
- Technical support – maintenance of digital equipment.
- Monitoring and Evaluation support – administrative reporting and critical friend approach.

Where such supports were provided clusters reported satisfaction, but when support was limited, they faced significant challenges.

Oide school support (pedagogical)

The support from Oide TiE appeared to have worked very well. For some specialist training requirements extra training was procured from outside bodies, and this is reasonable, as Oide TiE cannot be expected to cover all specialist training needs. Where such gaps arise, it might be prudent to consider how the Oide TiE Advisor team can participate and benefit from such specialised training themselves.

Technical support and project monitoring were outside the remit of Oide TiE and are addressed in this section.

Technical support

As highlighted in this report, technical support presented a significant challenge for some clusters. There was no clear mechanism for addressing the issue, and responsibility for it appeared to be undefined. Sometimes teachers took on the responsibility themselves, sometimes external private support was procured. This approach was unsustainable, highlighting the need for clear expectations and defined responsibilities. If this responsibility is to rest with schools, it should be addressed at the outset of the project, with additional funding allocated where necessary to support it.

Monitoring and evaluation (administrative support and the role of critical friends)

As noted above, Cluster Coordinators were expected to complete a range of reports on their project's activities. These reports, in the main, monitored project activities and expenditure. There is clearly a need for such reporting, as they may need to be shared with different units or sections within the Department, depending on their purpose and on how they are to be utilised. Future projects should consider what information should be collected, when and by whom and ensure that it is made as easy as possible for schools. Where information is incomplete or non-existent, there should be some processes in place to gather this information in order to take

further action. The creation of such monitoring systems creates a workflow for the Department and additional supports should be provided to implement them effectively.

In addition to the monitoring element, projects were required to propose evaluation plans in their initial proposals. Feedback on these evaluation proposals could have helped clusters to refine them and enhance their overall project plans. A number of these plans appeared somewhat unrealistic, in demanding too much in the way of data collection, and some projects proposed creating weekly evaluation reports. While self-reporting, as an approach to project evaluation, is a good idea, it appears that many clusters struggled with how best to implement such approaches, and greater guidance should be provided in the future. For instance, in previous digital innovation programmes (i.e. Schools Integration Project - SIP) clusters received training on how to conduct action research on their innovative activities[31].

The interruption caused by the COVID-19 pandemic led to few project level evaluation plans being implemented or reported. If such evaluation plans are required in future, all projects should complete them, and the Department should provide guidance on what should be contained in such documentation.

A couple of initial proposals had included the idea of a 'critical friend' as an approach to obtaining critical feedback on their projects. Such an approach could have added value to the Programme, as it would have enabled clusters to engage in critical discussion with an objective third party around their project activities. However, the provision of such a service would have required additional funding, and this was not included in the programme budget. Clusters were advised that funding could not be used to procure the services of a critical friend on this occasion. Oide TiE did not see this as their role and felt it might undermine their relationship with the schools if they were to provide such critical advice. Thus, there was an expectation that projects would self-report on their own activities and decide if they were successful or not. In future it would be worth considering learning models, such as double loop learning[32], where organisations modify their goals based on feedback and this could be provided by input from critical friends to the project.

The funding of 'critical friends' to play a role in future initiatives may be a good way of ensuring some kind of external criticality, without burdening schools with labour intensive data collection for evaluation purposes. There is a need to separate such reporting from the administrative reports that were specified in the early stages of the project.

Programme dissemination

Some clusters felt that what was being asked of them in terms of dissemination was unclear. They wanted clear expectations to be set out from the beginning, but other clusters asked for the ability to negotiate these, as they felt that some forms of dissemination were better suited to their project and the strengths of their team than others.

Some projects had active websites or social media accounts – some detailed descriptions of this are given in the Interim Evaluation Report, however, as reported, the level of engagement with social media accounts was often low.

[31] https://doras.dcu.ie/17938/
[32] For more see: https://hbr.org/1977/09/double-loop-learning-in-organizations

The attempt to collect the *Teacher Stories* at the end of the project was only partially successful. Part of this is no doubt due to the length of time since the projects finished (projects finished in June 2022 and data was collected in September 2023), but it is perhaps also due to the fact that teachers perceived this as an additional burden, not something they are keen to get involved with. Clusters seem to have been happier in sharing their projects through events such as the SEF Digital & STEM Programme meetings in Athlone in 2019 and Croke Park in 2022. Other possible approaches that might be tried in future initiatives could include someone from outside the school interviewing teachers involved in a project and writing up the story for them (maybe a role for mobile journalism as explored in one the cluster projects). The use of Oide TiE webinars and podcasts is another possible approach.

Final word

Overall, the Programme was deemed an overwhelmingly positive experience by the majority of those who actively participated in it. The Programme enabled schools to work together, to use digital technologies in a supported way across the curriculum. The majority of teachers who participated in the Programme continue to use these technologies after the end of the Programme and they believe it has had a lasting impact in embedding digital practices in their schools.

While digital technology has been used in Irish schools for over fifty years there has been tremendous progress in terms of access to technologies in recent years and schools are on a journey of discovery to identify digital practices that work for them. This work should continue, and the lessons learned from SEF Digital & STEM should influence any future experimentation projects, big or small, to support schools in evolving and meeting the needs of society, today and into the future.

Appendix 1: List of SEF Digital & STEM clusters

This is a listing of the clusters that participated in the Schools Excellence Fund Digital and STEM Programme. The Oide Technology in Education website[33] contains a full list of clusters including details of participating schools, a short outline for each project and links to the project websites.

Digital Learning Clusters - General

- Integration and Enhancement of the PLC through Digital Learning in a Mindful Environment
- Integration of Virtual Reality (VR) in 21st Century Teaching, Learning and Assessment
- Creative Problem Solving & Digital Education in 21st Century Teaching, Learning & Assessment
- Multimedia Cluster Project
- Use of embedded digital video strategies to enhance teaching and learning in post-primary schools
- Developing Innovative Pedagogies and ICT Tools for the Assessment of 21st Century Skills in Second Level Schools in Ireland
- Consolidated approach Integration and embedding of technology to enhance teaching, learning and STEM practices in CMETB schools
- Using Digital Mobile Technologies for Teaching, Learning and Assessment of Fundamental Movement Skills in Physical Education
- Demonstrating Excellence in the use of digital technologies in the creation, presentation, assessment and storage of CBAs across different curricular areas within the new Junior Cycle
- Oral literacy development in science through an art medium of digital storytelling
- FLAIR – improving senior cycle student's awareness of how key skills relate to diverse study and career paths using artificial intelligence
- Dreams to Reality: Harnessing the Potential of the Digital World
- Virtual Reality Explorers / Mindful VR Development
- Embed Digital Technologies in Teaching, Learning and Assessment for the Community of Practice of Music teachers "Pobal Ceoil" of Waterford and Wexford ETB
- Building Digital Bridges on the Wild Atlantic Way
- Digital GMP
- Junior Cycle Curriculum Planning, Content Creation and Resource Sharing' – Teacher Collaboration

[33] www.oidetechnologyineducation.ie/projects-initiatives/digital-clusters/

- UAV Technology as an innovative learning tool for local and national Geography teaching

Digital Learning Clusters - Coding & Computational Thinking

- Learn-Code-Share-Assess-Learn
- Developing Computational Thinking Skills in a Constructionist and Inquiry-Based Science Learning Environment in Irish Primary Gaelscoileanna through the use of Robotic Materials.
- How educational robotics and coding can enhance the Irish Primary Curriculum
- Developing Digital Literacy Skills: Computational Thinking and Creativity in a Digital World
- WeDo 2.0/ Mindstorms EV3 construction and robotics – transition through primary and into post-primary
- Coding & Computational Thinking in a Primary & Post-Primary Co-educational Context
- Primary Computer Science – Towards A Curriculum
- A collaborative approach to Computational and Design Thinking to improve Primary School Problem Solving Skills
- Developing a continuum of computational thinking and programming activities/ initiatives in a cluster of schools in Mayo/Sligo
- Coding/Computer Science Across the Curriculum
- Bridging Your Transition to Education

Digital Learning Clusters - STEM

- North-East Stem Champions
- STEM for Girls at all levels
- STEM ROOTS
- Machines and Mechanisms
- STEM/Coding Progression from Primary to Secondary School
- STEAM Hub and Centre of Technology Excellence in St. Brendan's CS for the region
- A cross-sectoral collaboration to promote student interest and engagement in STEM subjects
- Technology to Collaborate, Create and Innovate!

Appendix 2: Department of Education SEF Digital & STEM documentation

This section contains the guidance provided to clusters on managing their projects and the confirmation agreement they signed committing to adhere to the terms of the programme governing project management, and expenditure, and to report to the Department per the agreed schedule and as required.

Managing your project

The purpose of this document

This document is intended to provide guidance and support to project co-ordinators in implementing their Digital Clustering Project within their cluster. The projects operate as part of **The School Excellence Fund-Digital** and are designed to facilitate and support innovation in the use of digital technologies in teaching, learning and assessment in primary and post-primary schools. Effective project co-ordination at local level is essential in ensuring the successful implementation of the various projects.

The projects operate under the aegis of a Working Group established by the Department of Education and Skills (DES) and their implementation in schools is supported by the Professional Development Service for Teachers (PDST) and through the provision of funding, training and substitution cover to facilitate project meetings and attendance at training and other project-related events. Consideration should be given to establishing a steering group to support project management at the local level and, therefore, to ensure:

- The development of a project implementation plan with clear schedules and deliverables.
- Effective communication with all of the project partners and support services
- Delegation of project roles and responsibilities to the participating schools
- Optimum use of project resources, including compliance with purchasing procedures and guidelines.
- Maintenance of records of project activities, achievements and challenges for the purpose of informing project reports.
- Open and effective collaboration with the external evaluation team.

The role of the Coordinator

The co-ordinator plays a key role in enabling the successful implementation of the project and takes responsibility for its day-to-day operation, on behalf of the steering committee, if one is established, the co-ordinator takes responsibility for:

- Managing the development and maintenance of the project schedule, by;
  - Adopting overall responsibility for ensuring adherence to the agreed schedule
  - Assigning tasks to collaborating teachers
  - Ensuring evaluation elements are integral to the project reporting
- Maintaining clear lines of communication with the PDST, Working Group, external evaluator, and the participating schools in the cluster.
- Identifying and communicating to the PDST the project's support and CPD requirements
- Developing and submitting agreed reports on project implementation in line with the requirements of the Working Group
- Facilitating communication and collective decision-making within the cluster.
- Organising meetings and events within the cluster.
- Planning for project requirements, including training and infrastructure.

- Overseeing outlay of project funds, maintaining records of same, and providing expenditure reports to the Department as required.
- Ensuring compliance with the terms of the programme.
- Identifying challenges to project implementation and escalating issues to the PDST if required.

The role of participating schools

It is envisaged that the clusters will operate in a spirit of partnership and collaboration with each participating school taking responsibility for agreed elements of the project's implementation. Details of each school's particular role should be agreed with the project steering group. In addition, each participating school will:

- Contribute to the development of the project, including the project schedule, budget, communications plan.
- Take responsibility for specific elements of the project implementation and provide reports on their activities to the project coordinator.
- Identify challenges to project implementation and escalate those requiring particular attention to the project coordinator.
- Adhere to the expenditure and governance requirements of their project.
- Maintain ongoing communications with the other schools in the cluster.
- Participate in the CPD and training provided to support project implementation and evaluation.
- Engage with external evaluator as required.

Reporting requirements

Each project will be required to maintain all project documentation and submit a number of reports during the project's lifetime, these include:

1. **Termly Reports:** The production of termly reports is a key deliverable from each project. These reports should detail progress, or challenges encountered, in relation to the milestones and deliverables contained in the project plan required to deliver the project's overall objectives. A reporting template is attached in the appendices.
2. **Biannual Expenditure Reports**: These reports will detail the purpose and level of expenditure, and the procurement procedures followed. Note: All quotes and invoices should be retained on behalf of the cluster for the duration of the project. An information note relating to financial management is attached in the appendices.
3. **Annual Reports:** These are intended to be more complete than the Termly Reports and will indicate progress achieved against the overall project objectives and outcomes over the course of the year.
4. The production of project documentation to the Working Group upon request.

Note: Reports issuing from the clusters will be submitted to the PDST for review. A summary report will subsequently be compiled by the Working Group and submitted to the ICT Policy Unit of the Department for sign-off.

The Role of the PDST

The PDST will play a central role in facilitating project implementation and in supporting schools in meeting targets and overcoming challenges. Each cluster will have an identified PDST advisor who will be identified as the point of contact between the service and the cluster. It is envisaged that the clusters, through the project co-ordinator, will maintain ongoing communication with their PDST advisor throughout the project's lifetime. The PDST, in turn, will maintain communication with the Working Group

and the ICT Policy Unit of the DES and the external evaluator. In supporting project implementation, reporting and evaluation, the PDST will

- Engage with clusters to identify their training or professional development needs and facilitate or directly provide the requisite training.
- Provide ongoing advice in relation to project management advice and support on an ongoing and reporting.
- Review project reports.
- Liaise between clusters and working group regarding any challenges or issues that may arise and suggest alterations to project implementation plans to resolve any such issues.
- Escalate any issues, where appropriate, to the Working Group and ICT Policy Unit for resolution.
- Supported by the ICT Policy Unit of the DES, maintain overall records for the programme.
- Engage with the external evaluator as required.
- Produce information and reporting on the programme to the Working Group and the ICT Policy Unit of the DES as required.

Confirmation Agreement

Name of Project

Roll Number & Name of Lead School

Roll Number & Names of participating schools

Confirmation of agreement to participate in School Excellence Fund – Digital Programme

Further to the sign off and approval of your project plan and budget by the Department of Education and Skills, the above referenced cluster, undertakes to participate in this programme for the duration of three years.

The undersigned on behalf of the cluster schools undertakes to adhere to the terms of the programme governing project management, and expenditure, and to report to the Department per the agreed schedule and as required. The cluster also undertakes to engage with the Department supports, in particular the PDST, and to communicate and escalate any issues arising throughout the duration of the programme. Participants in the clusters are also expected to attend Department organised cluster events.

The cluster schools understand and will comply with the financial oversight requirements of the programme, including procurement and maintenance of records.

The cluster also undertake to engage with the external evaluation of the Programme.

The ICT Policy Unit on behalf of the Department of Education and Skills and the School Excellence Fund Digital Steering Group, undertakes in turn to provide the supports agreed and outlined, to the cluster.

This will commence with the year 1 funding to issue on receipt of this signed agreement, of XXXX, of the total approved budget of XXX.

Please return your signed copy of this agreement, for counter-signature. A scanned copy will be sent to you for your records.

Signed on behalf of the cluster by the cluster co-ordinator

Signed on behalf of the Department of Education and Skills by

Appendix 3: SEF Digital & STEM case studies

These case studies were constructed in the summer of 2020 and presented as part of the Interim Report. They are based on the documents from the Clusters that the Department has made available to the evaluators, the Cluster Coordinator Survey in April 2020, the Department published statistical data, and the Clusters' use of public online platforms. Direct quotations from original documents are given in italics, and abbreviations are used to indicate sources (see *Abbreviations* section at the end of the report).

Key factors for enabling innovative practices

One aspect of the analyses in these Case Studies relates to supporting factors for innovation. These factors are discussed under the following headings:

- Students
- Teachers
- School leaders
- Shared vision
- Wider partnerships

These factors are derived from the key findings of the EU report *Supporting School Innovation Across Europe. Final Report to DG Education and Culture of the European Commission*[34] (PPMI 2018). The detailed description of these factors given in that report is reproduced below:

> "Our analysis of school innovation success stories reveals many entry points to the process of innovation in schools. Certain conditions must, however, be met to ensure that change is managed effectively and leads to positive outcomes:
>
> Students are at the centre of learning, and their engagement is crucial to ensure the relevance of school innovation. They can be active actors in the learning process, e.g., through self- and peer- assessment, as well as in school governance and partnerships with external stakeholders.
>
> Teachers' preparedness, commitment and capacity to manage change are key factors in ensuring that innovations are implemented effectively. Professional development and collaboration initiatives, such as professional learning communities, teaching teams, action research projects, teamwork and formative feedback mechanisms, can help the implementation process, collectively focus on achieving success in teaching and learning, and developing innovative mindsets among teachers.
>
> School leaders are a dynamic and influential force in the school community. They can guide schools in creating a culture that initiates and supports innovation. Good leadership can ensure a supportive environment for teachers; promote collaborative practices such as peer-learning, mentoring, formative feedback, reflective and inquiry-based practices; and provide relevant opportunities for networking and professional development. Distributed leadership constitutes one of the conditions for sustainable

[34] https://op.europa.eu/en/publication-detail/-/publication/2e440297-4ce9-11e8-be1d-01aa75ed71a1/language-en

innovation by encouraging teachers to assume increased responsibilities, to engage in individual or collective initiatives, and to build their capacity to take on leadership roles.

A shared vision and clear action plans help to structure the implementation process, ensuring the sustainability of school innovations. This begins by identifying the school's needs, then by agreeing on the way to move forward, and extends to providing feedback and reflection throughout the process. This also helps to instil a sense of commitment and stability among school staff, students, parents and local stakeholders. Innovative schools promote an active role for students, teachers and school leaders, parents and other stakeholders, helping to build a system of horizontal accountability.

Creating wider partnerships is an intrinsic feature of innovative schools. Engaging with parents can be a necessary step in overcoming their potential resistance to changes. Schools also need to draw on the various resources and opportunities provided by other schools, as well as municipalities, teacher education providers, local businesses, civil society organisations, and other local stakeholders. Recognising and supporting the horizontal connectedness of schools with their communities is a crucial step towards overcoming school isolation, facilitating different forms of learning, and bringing learning closer to real-life contexts."

Case Study 1

A Cross Sectoral collaboration to promote student interest and engagement in STEM subjects

Lead School: C.B.S. Thurles (65450W)

School context

The project involves eight schools (one post primary and seven primary) most in Tipperary, with one in Waterford. The lead school is C.B.S. Thurles (65450W). Two of the primary schools have very small numbers of pupils (37 and 57). Sixteen teachers are involved in the project and 400 pupils (from a total of 875 pupils in the primary schools). All the schools have a Catholic ethos. In one of the primary schools all subjects are taught through Irish, and in all the other schools no subjects are taught through Irish. One of the primary schools is classified as DEIS.

External involvement

- **Education Centre Network:** *Kilkenny Education Centre* - assistance with training (**IP**); *Thurles Education Centre* – meetings venue and collaborative space to contact other Education Centres. (**3YP**)
- **Third Level:** *Limerick Institute of Technology (Thurles campus)* - provisions of a collaborative learning space, IT support for creating a 'remote laboratory', support for cluster presentations at ICT in Education Conference. (**IP**, **3YP**)
- **Industry:**
  - *Lennox Educational*: advice on initial sensor purchases as well as training requirements and ongoing maintenance of hardware, queries or technical issues with the sensors.
  - *Elite Drone Services*: Launch event support. (**Y1_Mid_PR**)
  - *Wriggle*: Apple training for staff and students as part of the overall package. (**IP, Y1_Mid_PR**)
- **Evaluation partners:** None identified.

Proposed project

The project and its aims

The Cluster website says that the project aims to increase interest and engagement in STEM subjects among students, by providing opportunities for:

- Primary pupils to develop their problem-solving skills by participating in inquiry-based learning activities.
- Transition Year (TY) science and mathematics students to be involved in creating tutorial videos and facilitating workshops for primary-school teachers and pupils.
- Greater levels of cross-sectoral collaboration between teachers.
- Embedding digital technologies in the curriculum in an innovative manner.

The website says the project aligns with the Digital Learning Framework, focussing on Domain 2: Learner Experiences Standard 2.4 and Domain 3: Teachers' Individual Practice Standard 3.1.

The **IP** indicates an important role for competitions: Our Project will provide opportunities for each school to partake in STEM focused competitions during the 3 years. This will embed interest and encourage students to engage purposefully and meaningful with STEM learning activities both inside and outside the classroom.

Grant allocation

Y1_Mid_ER indicates a grant of €24,239.00. The **IP** indicated that three iPads would be allocated to each primary school and 20 to the post-primary school.

Outcomes and deliverables

Year 1	Year 2	Year 3
• Collection of baseline data on participating schools' attitudes to STEM subjects • Schools adequately equipped with wireless sensor technologies and software • Training on the use and applications of wireless sensors delivered • Schools engaging with project (online collaboration) • Recording and delivery of initial remote laboratory lessons.	• Continuation of remote laboratory through live video feed • Data collection into the teaching of STEM subjects • Increased willingness and confidence in using wireless sensor technologies for teaching and learning • Resource bank available through an online space will have an increased volume of resources • Training on Sphero, micro bits • Schools' engaging with Coding element (online collaboration)	• Primary students' increased enthusiasm for STEM subjects and competence in using wireless sensors for a variety of self- directed investigations • Moodle and video links to show outcomes in schools • Training on use of GeoGebra • Schools' engaging with mathematics (online collaboration) • Transition Year students develop leadership skills. • Increased uptake of STEM subjects in senior cycle for pupils in the project

(**IP**)

Cluster collaboration

IP describes a variety of forms of collaboration including email, Twitter, Blogs, monthly meetings, and workshops.

Dissemination

One of the proposed outcomes is a "resource bank available through an online space". The Cluster will participate in the Final Showcase Event and intended to participate in an ICT in Education Conference at LIT Thurles in May 2020. (**IP**)

Evaluation

The Initial Proposal (**IP**) presents quite ambitious plans in relation to evaluation, stating:

- *Our project actions will be monitored on an ongoing basis to evaluate their impact on learning, teaching and assessment practices.*
- *The final evaluation process will involve the participating schools providing feedback on areas such as student involvement, student classroom participation, teacher confidence levels in STEM subjects etc.*
- *We would also hope to make some evaluation of the success of this project in terms of primary students' attainment levels in STEM related subjects through the early years of secondary school.*

Project progress

Most innovative elements of the project

The Cluster Coordinator identified the following as innovative aspects of the project:

> *"The cross sectoral collaboration between the secondary lead school and the primary cluster schools. This collaboration has huge potential in terms of streamlining the transition for students between primary and secondary school in all subjects, not just science."*

> *"Many of the primary teachers involved in our project would admit to a certain level of unease in using the wireless technology with their students as well as a lack of in-depth knowledge of the background scientific explanations. Having a second level Physics teacher available to call to the primary schools to give a lesson or to simply explain the background theory to the teacher has helped to ease any sense of 'lack of knowledge' and has helped to improve teacher engagement."*

> *"The use of wireless sensor technology at primary school level is in itself an innovative approach and has definitely helped to increase student engagement in the lessons." (CCS2020)*

Example of a learning activity

> *"Students used the knowledge of Lego WeDo to design, build and code a racing car, taking into account factors to improve the speed of their designs. They then used the Pasco wireless motion sensor to monitor the speed of their cars. The students then had the chance to alter their designs before finally using the motion sensor again to find the speed of their designs and to race one another. This activity involved a lot of elements, including research, design, programming, measurement, analysis, reflection and interpretation and was a great example of how the teacher used the topic of motion and speed to link two separate initiatives into one lesson." (CCS2020)*

Reported progress

The End of Year Report (**Y1_Final_PR**) is positive about the achievements of the project, and highlights some specific achievements:

> *"We are happy that year one has been a success on many fronts, most importantly in the fact that the primary school children have shown great enthusiasm and engagement with the project to date."*

> *"Very successful launch event held in October 2018 with over 400 local primary school children attending. The event was covered in local media as well as on social media."*

"Schools are actively engaged with the project and are currently working on an agreed timetable of rotation of sensors between each school."

"Weekly visits by Cluster Coordinator to primary schools has been successful as a means of providing assistance/expertise to primary teachers."

"Twitter account set up by Oide TiE Advisor to promote the project and the great work being done by teachers within primary classrooms."

"The Cluster presented at the SEF event in Athlone IT on May 20th, 2019."

The Cluster Coordinator sums up progress as follows:

"Each of the seven primary schools involved in the cluster is following an agreed timetable to use one of our wireless sensors (motion, pH, voltage, current, weather station, pressure, temperature, conductivity) with their students on a monthly basis. The primary teachers conduct, record and document any practical activities undertaken and the work is recorded currently on the virtual learning platform 'Edmodo'. Sample videos are to be recorded by the lead school as a means of empowering primary teachers and to provide assistance in terms of sensor use. The goal this year was to try to incorporate the use of the wireless sensors in a more meaningful way ... Currently, the primary pupils involved are from 4th, 5th and/or 6th class. Our focus cohort are 5th class as these students will be exposed to the project for the full three-year duration." (CCS2020)

"However, time is an issue: As with any school-based endeavour, time is the big issue for everybody. Time to plan the activities, time to collect the sensors, time to record and document results and findings, time to visit schools, time, time, time." (CCS2020)

It is clear from both Progress Reports that many action points had not been achieved and that original timeframes had been too optimistic:

"The main issue that we have. It has proved difficult to find times that all members are available for, it has also proven quite difficult to have a consistent access to our TY students due to the many demands placed on them during transition year." (Y1_Final_PR)

External involvement: This took place as planned.

Cluster collaboration

This proceeded much as planned: Cluster coordinator has visited each school to offer assistance/expertise to the primary school teachers, on a weekly basis. Communications are ongoing through 'posts' on Edmodo classroom and WhatsApp messenger. Cluster coordinator will maintain contact with cluster schools with weekly visits. (**Y1_Mid_PR**)

Dissemination: The Cluster held a launch event and participated in the SEF Digital & STEM event in Athlone in May 2019. (**Y1_Final_PR).** Details of online dissemination are given in Section 4.4 below.

Evaluation: The evaluation activities are not yet well developed: Measure of progress to date has been observation of student engagement during school visits by Cluster Coordinator. (**Y1_Mid_PR**)

Budget: Almost all the funds had been spent in the first term - see Table 12. (**Y1_Mid_ER**)

Table 12: Budget expenditure for Cluster 20.

Item	Cost
Sensors	€4,090.44
iPads	€17,936.98
Launch meeting	€1,475.48
TOTAL EXPENDITURE	**€23,502.90**
GRANT	**€24,239.00**

Use of public online platforms

The project has maintained a website (https://tippstem.weebly.com/) and a Twitter account (@TippSTEM) since the start of the project. The project website contains a Home page which provides an outline of the project and its aims; a Cluster Launch Day page which provides a description of launch day with photographs, and related tweets; pages for each of the eight schools – one of these (Gaelscoil Durlas) describes activities with the pH sensor, motion sensor and voltage sensor, accompanied by photographs and video (the later were produced by the Lead School). The posts present an outline of the activities and showcase what was done, but provide little detailed pedagogic or technical description, analysis of results, or critical reflection.

Figure 9: Screenshot from project website for Cluster 20.

As of 2 May 2020, the twitter account had 78 followers, and had published 81 tweets, the first on 2 Oct 2018, and the latest on 19 March 2020. The plot of the number of tweets per day, shows two clumps of tweets, the first around the start of the project and the launch day, and the second around a joint training day. The total engagement score (i.e., number of favourites and retweets) over the period was 220 – which is quite low. The highest engagement for a single tweet was 21 for a tweet in October 2019 announcing the launch of the Cluster website.

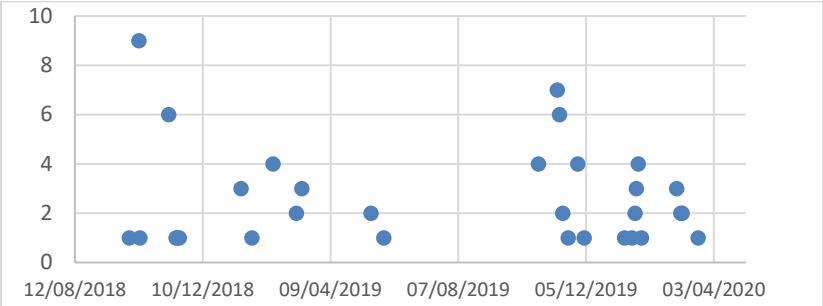

Figure 10: Number of tweets per day by date for Cluster 20.

Observations

Highlights

The DES and Oide TiE support in terms of finance for equipment and training, in the support for the Cluster Launch Day, and the ongoing support provided by the Oide TiE Advisor were clearly very valuable.

The Cluster Launch Day was a major and very successful event, involving the schools (including a significant role for the Transition Year students from the post-primary school), Oide TiE, and several companies, and it made a significant and important start in addressing the aims of the project.

As reported in Section 4.1, the Cluster Coordinator identified the following innovative aspects of the project: cross sectoral collaboration; support by a second level Physics teacher; use of wireless sensor technology at primary school level. He sums up his view of the project in this way:

> *"I am happy however that the pupils involved in the project across the seven primary schools in our cluster project, are enjoying the experience and opportunity to utilise the wireless sensors in class and to design practical activities that mean something to them. Once the primary pupils are engaged in scientific enquiry, prediction and analysis, I believe that the project is a success." (CCS2020)*

Issues

The Initial Proposal was too ambitious, and the proposed evaluation unrealistic, including an examination of impact on primary to post-primary transition, whilst not planning for an evaluation partner. With hindsight, it might have been helpful if more detailed feedback had been given to the Cluster about these aspects of the proposal before commencement. Despite its genuine achievements the project did not do all it had hoped, made little progress on evaluation, and found that the demand on teachers' time was greater than expected.

The evidence we have suggests that the project may have worked better in some primary schools than others. The educational gains for the post-primary partner were unclear. The Transition Year Science and Mathematics students played a useful role in the Cluster Launch Day, but further involvement seems to have met practical problems which should have been foreseen. The allocation of 20 iPads to the post-primary partner in the proposal does not therefore seem to be justified.

The data we have does not provide a detailed account of the collaboration that would enable replication of the practice, and it is hoped that this account will be developed later in the project.

The comments with reference to Digital Learning Framework on the project website might suggest a 'checklist' approach, rather than a use of the DLF to examine, critique and develop current practice.

Whilst the Cluster Launch Day and the Athlone event were significant dissemination activities; the website has little content and there is little engagement with the Twitter account.

Lessons learned

The economic value of working in a cluster in order to share access to quite expensive technology that is only used occasionally (the sensors) seems to be clear.

An issue raised by this Cluster project is the potential value of cross sectoral support for professional development. Is this useful, or would professional development based in the primary schools (perhaps supported by external agencies) be more effective? What are pros and cons, and how might it be designed to achieve maximum impact? Will this collaboration have the added value of impacting on the primary/post-primary transition for these pupils? At present we are not capturing enough information to be able to answer this question, and existing internal evaluation is weak.

Key factors for enabling innovative practices

- **Students** - Some evidence of student as active actors in the learning process, but not in school governance or external partnerships.
- **Teachers** – Professional development and collaboration is the chief focus of the project, though issues have arisen around capacity. No evidence about formative feedback by the teachers.
- **School leaders** – No evidence presented of the role of school leaders.
- **Shared vision** – The relationship of the project to schools' shared vision and action plans is not presented. It is not clear how much of the vision of the project is generated by the Lead School rather than arising from an analysis of needs by the primary school partners. This may have implications for sustainability.
- **Wider partnerships** – There was a strong involvement by some businesses in the initial stages, but it is unclear how this will continue (beyond training which forms part of the hardware acquisition package). Collaboration with other organisations seems not to have worked as planned, and there are no plans for wider community involvement (including parents).

Case Study 2

Use of embedded digital video strategies to enhance teaching and learning in post-primary schools

Lead School: Banagher College (76105Q)

School context

The project involves four post-primary schools, two in Offaly, one in Westmeath and one in Roscommon. The lead school is Banagher College, a LOETB school. The other schools are secondary Catholic schools. None of the schools teach any subjects through Irish. None are classified as DEIS. The project involves eight teachers and 200 pupils.

External involvement

- **Education Centre Network**: *Athlone Education Centre* - Initial cluster meeting and training sessions. (**IP**)
- **Third Level**: *Limerick Institute of Technology (L.I.T.)* - guidance for technology and training, input from 'The Youth Media Team'. (**3YP**)
- **Industry**: *RTE/RTE Digital* (MoJo Training); *Titanium Media* (MoJo Training); *Frontline Films* (MoJo Training/Storyboarding); *TV3* (Journalism); *AIT* (Digital Marketing). (**IP**)
- **Evaluation partners**: None identified.

Proposed project

The project and its aims

The project aims to use industry leading training in MoJo (mobile journalism) video content creation to enhance teaching, learning and digital literacy among educators and students in the cluster schools, while also being central to AFL (assessment for learning) and blended learning approaches in the respective schools. (**IP**)

This would seem to break down into two different, if related, strands. Firstly, access to high level training in mobile journalism leading to the creation of numerous video artefacts, and then in later years of the project this leads on to more specialist training for the creation of more complex artefacts using a wider range of skills, techniques and equipment. Then, secondly, a wider use of these skills in video production in the wider curriculum:

> *"It provides numerous opportunities for student collaboration. Short videos can be used to enhance the teaching experience, to develop learning concepts, to allow for student immersion, to develop problem-solving skills, to enhance computational skills, to develop social skills, to build and develop teamwork, encourage social interaction through different age groups, to "flip" learning, to provide new channels for both formative and summative assessment and much more." (IP)*

The **IP** states that project ties in with the STEM Action Plan 2017 – 2019 under the Implementation phase in providing support for schools and teachers to work together on innovative and collaborative initiatives in STEM, scoping out the partnership with STEM business and industry, and supporting effective and engaging STEM teaching.

The **IP** states that the project aligns itself with Statements of Highly Effective Practice in the Digital Learning Framework (DLF) under all four of the Teaching and Learning domains and three out of four of the Leadership and Management domains (omitting 2 – Managing the Organisation).

Grant allocation

The total grant requested was €30,000. This includes €1,800 on teacher meetings and €8,700 on training, €7,830 on 20 iPads and another €4,695 on 8 iPhones, the rest of the money is for other equipment (mostly specifically relevant to mobile journalism). (**IP**)

Outcomes and deliverables

Year 1	Year 2	Year 3
• One industry-led training workshop for coordinating teachers and for each school (Topic- MoJo) • 'MoJo videos' by each school - 9 short (3/5 min), 1 medium (15/20 min) • One social media platform for the project • Year 1 Total = 45 short videos & 5 medium videos; 50 students and 10 teachers trained by industry experts in video production.	• One industry-led training workshop for coordinating teachers and students (Topic - Journalism) • Purchase of supplementary equipment to allow for enhanced technical training, teaching and learning experiences • 'MoJo videos' production by each school - 9 short videos (3/5 min), 1 medium (15/20 min) • Social media platform - for the project and for each school • Inter-cluster video challenge between clustered schools (short video) • Year 2 Total = Additional 45 short videos & 5 medium videos.	• One industry-led training workshop for coordinating teachers and students (Content - deployment) • Purchase of supplementary equipment to allow for enhanced technical training, teaching and learning experiences • 'MoJo videos' production by each school- 9 short videos (3/5 min), 1 medium (15/20 min) • Social media platform - for the project and for each school. • Inter-cluster video challenge between clustered schools (medium video) • Year 3 Total = Additional 45 short videos & 5 medium videos.

Total 3-Year Project Output = 135 short videos and 15 medium videos available online via various platforms. Project completion showcases by all schools involved. (**IP**)

Cluster collaboration

The **IP** says that collaboration in the Cluster will take place through key meetings (in Athlone Education Centre and the participating schools, in rotation) and the use of Google hangouts and Slack to communicate and share resources.

A mentoring system is envisaged to allow any cluster schools encountering problems, or struggling to meet targets, to be assisted by the other schools in the cluster. (**IP**)

Dissemination

Videos will be disseminated via a dedicated Twitter MoJo account for this project, Twitter MoJo Account set up by each participating school; available online via School websites; and. Available online via a dedicated You Tube channel. (**IP**)

Evaluation

The **IP** describes a range of evaluation measures: initial and annual surveys to assess technological skills (for participating students and teachers), teacher observation, peer observation, self-assessment, reflective journals, and monitoring of number of videos produced and the use of social media platforms by each school. Further: Independent feedback will also be taken from school principals, MoJo trainers and DES representatives as part of the Schools Excellence Fund. (**IP**)

Project progress

Most innovative elements of the project

The Cluster Coordinator identified the following as innovative aspects of the project:

> *"Students effectively operate as they would in the wider media world. They get training, use industry standard equipment, plan stories schedules and shoots, edit, appraise and work to deadlines. They also regularly publish their work and monitor public feedback, all while working autonomously. They also build the 'mojo' (mobile journalism) brand across the cluster schools." (CCS2020)*

Example of a learning activity

> *"Students have attended a number of workshops by leading video content professionals, where they have had the opportunity to develop a wide range of digital skills. These workshops have been central to the model of learning, with the students and teachers trained at them then disseminating their learning to the wider school." (CCS2020)*

Reported progress

The project reported that it had met its objectives for the first year:

> *"The key milestones that have been achieved in the project is that all schools have received their basic training in MoJo and all schools have been supplied with two different tranches of equipment at this point. All schools also met their production targets in terms of the quantity of digital artefacts that were created." (Y1_Final_PR)*

However, the Cluster Coordinator felt that the workload had been very heavy:

> *"The nature of the project also lends itself to time demands. This is primarily because the students work needs constant supervision and guidance. The students need ongoing input at all stages of production, the vast majority of which has to take place outside of regular class time." (Y1_Mid_PR)*

> *"It's been an incredibly demanding but rewarding experience for the many students and teachers involved. The students have gained a huge range of skills and the culture of content creation - of not seeing a device as a passive thing but a creative tool - is now firmly embedded." (CCS2020)*

External involvement: The initial Cluster meeting was held at Athlone Education Centre, however when training sessions were held there it was felt that the active learning nature of the workshops which involved use of the buildings and grounds was possibly interfering with other centre users, and there were also concerns around GDPR implications, and so it was decided not to run any future training in the centre (**Y1_Mid_PR**). There was no Third Level involvement in the project (**Y1_Final_PR).** The industry-led training was greatly appreciated: *The training so far is of a professional standard, provided by leading Mobile Video industry leaders. This has been a successful strategy, and it is hoped to continue down this road by also utilising a variety of industry experts in Year Two* (**Y1_Final_PR**).

Cluster collaboration: Training events and Cluster meetings held (**Y1_Mid_PR**). A number of teachers attended the 3-day MojoFest (Mobile Journalism Conference in Galway) for training. Next training event and cluster meeting were planned for September 2019 (**Y1_Final_PR**).

Dissemination: **Y1_Mid_PR** reported that dedicated project social media platforms for promotion and dissemination of content had been established.

Evaluation: **Y1_Final_PR** says *Both learner knowledge and attitudinal surveys are scheduled to be conducted at both the start and the end of the school year and* also sets out plans for a thorough assessment of year 1 and planning for year 2 to be discussed at a meeting in September 2019.

Budget: The project is running within the budget allocated, and some first-year funds (for training and administration) have been transferred to the second year (for equipment). **Y1_Final_PR**

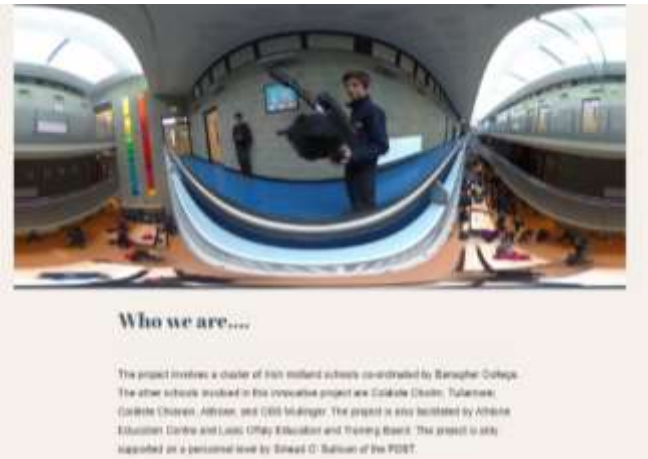

Figure 11: Screenshot from project website.

Use of public online platforms

The project main website is at https://edmojoireland.wordpress.com/ and the project also uses the main social media channels of the different schools in the cluster (**CCS2020).** The main website was set up on 9 April 2020, and as of 2 June had four posts which are a series of short introductions to the project. We identified some related websites: the TY Edmojo Project page (https://www.bccns.ie/ty-edmojo-project/) which contains a link to the Banagher College Mojo YouTube channel (https://www.youtube.com/channel/UChHGvznqRGPPuEynHQYHLWQ/) with 36 videos added since the start of the project in October 2018 (some with several hundred

views) and another 16 older videos, and the Edmojo related YouTube channel for Coláiste Mhuire (https://www.youtube.com/channel/UCnQqycsxF82aSEeMJLCMSEg) with five videos.

The Twitter account @EdMojo1 "Cluster of four schools in the Midlands embarking on a Mobile Journalism journey!" had 52 followers as of 2nd May 2020 and published 24 tweets between 29 May 2018 and 19 Jan 2019. The English Department at Banagher College Twitter account (@BCCNSEnglish) had 59 MoJo related tweets between 28 Nov 2015 and 19 March 2020. Other school accounts (@CbsMullingar, @ULTIMojoCBS; @colaiste_choilm; @CChiarain) had small numbers of MoJo related tweets, and the schools' Instagram and Facebook accounts had a small number of videos related to this project.

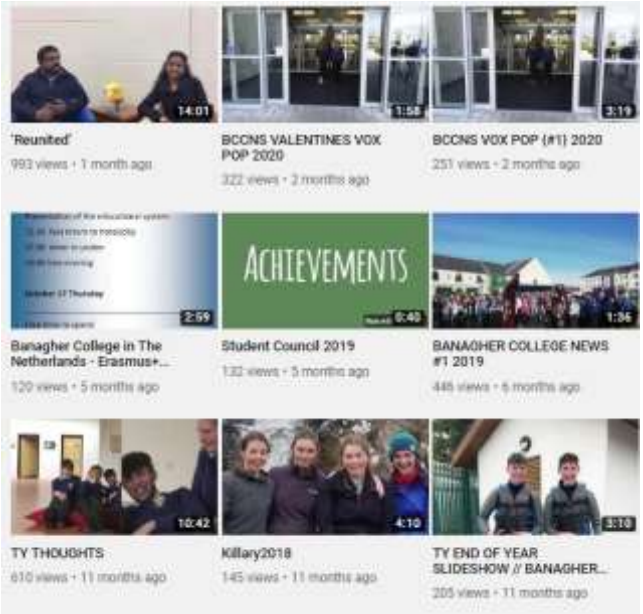

Figure 12: Screenshot from YouTube channel for TY Edmojo Project.

Observations

Highlights

The opportunity for students to engage with mobile journalism as they might in the wider media world, using professional equipment, benefitting from professional training and having their work published is clearly exciting and potentially a very valuable experience. The link with industry partners for training seems to have gone very well and been a highlight of the project. Banagher College has produced a large number of videos as a result and made them available on their YouTube channel. Banagher College has been involved with mobile journalism for several years, and this project enabled them to obtain additional specialist equipment, increase their own expertise through high level professional training and extend the work to a wider cluster of schools.

Issues

The project aim of obtaining access to high level training in mobile journalism leading to the creation of numerous video artefacts, seems to have been largely achieved. However, the project claims to have also achieved its aim of embedding these skills in video production in the wider curriculum, though this is not evidenced in the published materials. Most of the published videos are of general interest (news, school events, sports events) and journalistic in tone,

seemingly somewhat cut off from the mainstream curriculum. This disjunction is illustrated by the comment that:

> *"The students need ongoing input at all stages of production, the vast majority of which has to take place outside of regular class time." (CCS2020)*

Whilst the Banagher College YouTube channel hosts a significant number of videos, we did not find evidence that the other schools met the targets the project had set. The engagement with social media in general (particularly Twitter) was much less than anticipated in the proposal. The present outputs fulfil the goal of demonstrating the students' skills in creating materials, but it is important that the project finds some way of capturing the process as well as the output, in order to enable other schools to learn from the project. The existing project video of a training session simply provides a view of some classroom activities accompanied by a music, with no overview of what was happening or explanations.

Lessons learned

An important question for schools potentially interested in this approach, is how useful is mobile journalism in education outside the field of journalism as part of language and literacy studies? So far, this project does not seem to have demonstrated its relevance to other parts of the curriculum, and other digital storytelling approaches based on simpler technologies and production methods may well be more successful in that respect.

The coordinator's strong comments on excessive workload are an important consideration for the SEF Digital & STEM Programme. This raises the question of whether to invest in additional time to pursue ambitious goals or to set goals that are more realistic and manageable within existing time constraints.

Key factors for enabling innovative practices

- **Students** – Some evidence of student as active actors in the learning process (though difficult to be sure from existing evidence), though not in school governance or external partnerships.
- **Teachers** – The teachers involved with the project are very committed to it, but there is a concern on whether they have the time capacity. It may be that it was because the project was not fully integrated with the curriculum that so much work was done outside class time.
- **School leaders** – The school leaders may be pleased with the outputs of the mobile journalism in terms of publicity, but it is it is unclear how much support they provided.
- **Shared vision** – It is unclear to what extent this project arises from a shared vision of the school's needs, rather than reflecting the (important) concerns of one curriculum area.
- **Wider partnerships** – The link with industrial partners through the paid training is clear, but it is not clear whether any sort of connection continued after this. Whilst parents and other stakeholders will be interested in the news outputs (and they do get a lot of views), it is unclear to what extent this manages to engage them with the innovation in the schools.

Case Study 3

Machines and Mechanisms

Lead School: Firies N.S (15945T)

School context

The project involves four primary schools- all in Kerry. The lead school is Firies N.S. (15945T). All the schools are Catholic, none teach any subjects through Irish, and none are classified as DEIS. The project involves 48 teachers and 741 pupils. One school is particularly small with just 68 students, three classroom teachers and one Special Education Teacher.

External involvement

- **Education Centre Network**: *Tralee Education Centre* – will host Cluster meetings, provide STEM related courses including coding, robotics, games and animation. **(IP)**
- **Third Level**: *Tralee Institute of Technology* – will host pupil visits, provide input on STEM careers, involvement in STEM week, and pupils will showcase their work there. Students will visit the schools and work with the children. (**IP**)
- *Mary Immaculate College Limerick* – will provide support in embedding STEM, and student teachers will visit schools and work with children. (**IP**)
- **Industry**: *Liebherr*- will make school visits for class demos and provide collaboration with the Co-ordinator of Learning and Education in Liebherr in piloting an educational link Programme between Liebherr and the Primary School sector. (**IP**)
- **Parents:** We have a fantastic cohort of parents with unlimited skills and talents. We have involved these parents at planning stage of this project, and they are eager to continue to work with the schools. (**IP**)
- **Evaluation partners**: None identified.

Proposed project

The project and its aims

The Initial Proposal sets out the project aims as:

Students:

- Actively engage in STEM learning – both in school and out of school (home, Liebherr, fire station (Killarney), Kerry airport, train station Killarney and/or Farranfore).
- Students have an increased awareness and appreciation for the importance, value and operation of STEM.
- Use STEM to open up new forms of learning and collaboration.
- Children will be engineers, designers and artists.
- Use ICT to support different styles of learning and children with special educational needs.
- Develop 21st century skills such as problem solving, inquiry-based learning and teamwork.

Teachers:

- Collaboration between teachers and collaborative planning across class levels in areas like literacy, numeracy, STEM as project has cross curricular focus.

- Facilitate teachers in cluster schools to share exemplars of good practice in using digital tools.
- Improve instruction and student engagement in STEM by expanding STEM courses.

School:

- Three-year plan for school approach that will help develop a clear rationale for embedding digital technology in teaching and learning and inform subsequent planning and decisions as to types of infrastructure.
- Help support the principles and standards of the school curriculum in other areas for example maths and literacy
- Use technology to enhance the implementation of the curriculum for example arts education and music.
- Source products and solutions that support inquiry based and active learning through hands-on learning opportunities in STEM.
- Advise students on the safe and responsible use of technology for learning and life.

Parents and wider community:

- Parents have an increased awareness and appreciation for the importance, value and operation of STEM.
- Offer parents a chance to come to school for a parents evening and work on STEM projects.
- Parents will have the opportunity to come to school and work with their children.
- Involve people in the wider community e.g., Donal Lyne from Firies (Designer Minds).
- Teach parents the importance of online safety.

Grant allocation

Total grant of €29,000.

Outcomes and deliverables

Year 1	Year 2	Year 3
• Training workshops on Computer and coding workshop for teachers.	• Training workshops on Scratch-MIT/Harvard Syllabus.	• Training workshops on Expanding and the future.
• Introduce children in three classes to the STEM activities (participation rate: 33%) – Junior Infants, Senior Infants, First class and children attending special educational unit/classes.	• Increase the participation of children in STEM club (participation rate: 66%) – Second, Third and Fourth classes.	• STEM club available to all children (peaking at above 95%).
• Themes: – Term 1: trains – Term 2: cranes – Term 3: airplanes.	• Themes: – Term 1: cars/buses – Term 2: tractors – Term 3: helicopters.	• Themes: – Term 1: – Term 2: – Term 3: wind turbines.
• Teaching and learning: – Junior and Senior Infants (Simple machines) – First and Second class (Engineering and Robotics).	• Teaching and learning: – Third and Fourth (Junior Robotics): covers Information and Control Technology, Science and Maths, Language and Literacy.	• Teaching and learning: – Fifth and Sixth class (Senior engineers): cover a range of topics, engineering, mechanics and design.

Cluster collaboration

Introductory meeting for the principals and nominated project co-ordinators in the cluster schools. Whole school session. Nominated project co-ordinators presentation to staff. Time allocated for questions, discussions and input. Identify areas of concerns/need for learning (**IP**). Use of WhatsApp groups, Facebook, Skype meetings, and school visits for project communication (**3YP**).

Dissemination

Dissemination of project outcomes by: Creating good practice videos, making resources and lessons available online, publishing articles and photographs on school websites, Blog posts, sharing ideas and resources with schools, use of digital photography, developing personalised electronic stories about project, using presentation software and presenting project to an audience, Recording, videos and pod cast. (**IP**)

Evaluation

> *"We will assess the outcome by looking at parental/student/teacher surveys and questionnaires at the end of the year. We will assess student's performance in STEM activities. We will monitor the uptake of children taking chemistry, physics, technology and engineering in Secondary School (special emphasis on females) ... The impact will be evaluated by the comparison of results obtained from the beginning and end of the period. General feedback from children/parents/teachers regarding all aspects will be collated. There will be self and peer reflection of STEM lesson." (IP)*

Project progress

Most innovative elements of the project

Leaders in participating schools have been supported and valued as professionals (by Oide TiE, Department of Education and our local education centre in Tralee). We have access to high quality Continuing Professional Development (CPD) thus improving learning outcomes for our school communities.

3 out of 4 of the lead teachers participating in this project are not post holders. This group thus improves and strengthens middle management structures in all schools. Training and mentoring empower school leadership. Lead teachers help to support the design, development and delivery of quality CPD for teachers in their own schools. This helps to foster a culture of engagement with CPD amongst school staff.

> *"Empower Teamwork - It also fosters teamwork (both internal and external partnerships). Schools work together to develop best practice. The project has improved morale and encouraged innovation. We are working as a team to embed digital technology in teaching, learning and assessment. We are working as a group to collaborate on ideas based on our local experience, perspectives and industries. Collaboration on this project (both within and between schools) is focused on distributed leadership." (CCS2020)*

Example of a learning activity

> *"Our children in our four cluster schools got the opportunity to come together and visit Tralee Institute of Technology. They attended and participated in a number of workshops for example coding, animation and drone education. They spoke with students who attend the*

college and learned about the career opportunities that are available to them. The children were able to showcase projects that they have been working on to date. This visit integrated real-life situations with technology. They gained knowledge and understanding of engineering. They also experienced the possible place of technology in their future education and world of work." (CCS2020)

Reported progress

"The project is 'improving learning experiences and outcomes for students, strengthening school planning, strengthening the impact of effective leadership on teaching and assessment practice, enabling schools to engage with school communities through exchanging ideas, sharing experiences and innovative practices and building leadership capacity of teams in schools."

"We are embedding digital technology into the school curriculum for all class levels from Junior Infants to Sixth. Our project-based activities involve inquiry-based learning opportunities that are pupil-driven. It supports multiple and holistic approaches to teaching, learning and assessment. Children are becoming developers, creators and change makers of technology. STEM disciplines of knowledge enable pupils to measure, analyse, design, understand and advance the world around them."

"The resources and digital technology that we are using for example blue bots, robotics, coding, drones, K'Nex, Mag formers, iPads and LEGO, allow for quality teaching and learning outcomes. These tools are suitable for all ages and apply hands on, cooperative learning experiences to solve real world problems. The project is cross curricular and encourages the development of 21st century skills such as problem solving. The resources are shared between the schools." (CCS2020)

External involvement: Courses and meetings held at IT Tralee. Third level students came to school and worked with children on animation. The industry partner *Liebherr* arranged to visit to schools to provide class demo on a mini robot. **(Y1_Final_PR)**

Cluster collaboration: 'Teach Meet' sessions Croke Park meetings and STEM/digital technology notice boards allow us to share practical and creative ideas with the whole school community … We have created folders, a Team google drive, group website, Facebook page and Twitter account to share teaching and learning ideas, successes and challenges. (**CCS2020**)

Dissemination: Much of the dissemination activity has been directed at the school communities and is described above. The use of online media for wider dissemination is described in Section 4.4.

Evaluation: The **Y1_Final_PR** states that the outcomes of the project have been achieved as proposed: Parent / Teacher and student questionnaires distributed and compiled to get a base line, Information displayed in pie charts, Digital Technology notice boards on display in schools, Staff informed and updated on the project at Croke Park meeting.

Budget: €10,160 of the €29,000 grant spent in year 1, however, surprisingly (we do not have access to the original project budget), the project says that it will need to raise additional funding:

"This year we worked with LEGO Duplo. We have realised that even though pieces are big they can get lost in transportation. Originally, we thought that we may be able to share LEGO We Do kits for year 2 of project. We now realise … that this will not be a possibility. We will

need to have sets in each school. ... [so] look for additional funding. ... older iPads are not compatible with the LEGO we Do kits. We hope to fund raise and use additional individual school funding to purchase new iPads." (Y1_Final_PR)

Use of public online platforms

The project has maintained a website (https://kerrystemcluster.edublogs.org) and a Twitter account (@kerry_stem). The website is consistently organised (though not always easy to navigate). It contains Home and Our Cluster pages which describe the project, and pages allocated to each of the four schools which briefly describes the school and then link to pages for Year 1 and Year 2 where schools provide an account of their project activities. Many of the accounts are simply descriptions of what was done and are heavily dominated by photographs of the pupils, though some accounts do present more details about the organisation of the teaching.

Figure 13: Screenshots from project website.

As of 2 May 2020, the twitter account had 332 followers, with 285 tweets. A plot of the number of tweets per day shows a continuous use of Twitter, increasing over time. The total engagement score (i.e., number of favourites and retweets) was 1.6K, which indicates a high level of engagement.

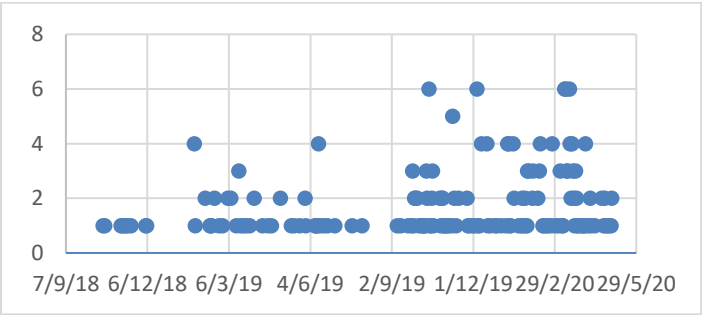

Figure 14: Number of tweets per day by date for Cluster 32.

Observations

Highlights

The project has had significant impact on the teachers in developing leadership capacity and so strengthening middle management, involving teachers in developing CPD for their colleagues, and in developing collaborations within and between schools and with parents, and has established strong links with Third Level partners. The project seems to have been very effective in terms of staff development, the concept of 'community' plays a significant role in the accounts the Cluster provides about its work, and the collaboration between staff, external

partners, Oide TiE and use of the website and social media provides a good example of **connectivism** learning. This in turn leads to a promise of sustainability:

> *"We plan to continue our learning cluster into the future (after the initial three years) and work on other projects together." (CCS2020)*

The strong online presence also seems to reflect an active, well organised and collaborative cluster. The project website provides a positive image of the children's activities with a wide range of digital technologies and STEM and reflects their enjoyment of what they were doing.

Issues

The **IP** was probably over optimistic with respect to what could be done, and underestimated problems over sharing equipment. Feedback from the Department at the proposal stage might have helped the Cluster plan more effectively.

The reports make significant claims for impact which will need to be further evidenced later. The **IP** promises several things in relation to monitoring which have not yet been reported on. Whilst the **Y1_Final_PR** says that the schools "*worked on three step process to become Digital School of Distinction"*, there is no mention of the DLF in any of the project documents.

Planning for wider dissemination, reporting on process as well as outcomes, capturing lessons learned by the project and providing help to other schools that might wish to do something is absent so far.

Lessons learned

The contribution to the development of leadership capacity, and the creation of a collaborative working relationship both across the schools and externally are the most striking aspects of this project. The word 'community' comes up repeatedly in the documents from this Cluster. It would be useful to make comparisons between the effectiveness of this community-oriented approach to the somewhat more top-down approaches adopted by some other Clusters.

It is good that taking part in this project meant that the teachers felt valued, but this raises issues about possible negative impact on schools not selected for participation in such initiatives.

Key factors for enabling innovative practices

- **Students** – The students are active actors in the learning process and are also providing input into the planning for the initiative.
- **Teachers** – The teachers are working collaboratively on the development of teaching approaches and providing formative feedback.
- **School leaders** – School leaders are actively involved in the project, providing support for a school wide change.
- **Shared vision** – The project is based on a shared vision and is part of a wider action plan for change.
- **Wider partnerships** – The project has involved parents, Third Level institutions and industry in collaborative work in developing the project.

Appendix 4: SEF Digital & STEM teacher stories:

Overview

This section contains 12 teacher stories with each following a similar layout, listing the schools involved, an overview of the project, the project aims, how it was implemented, some reflections on how it evolved and links to additional resources. The stories cover a range of topics that include:

- Computational Thinking x 2
- Cross curricular use of digital technologies in primary
- Constructivist use of digital technologies
- Virtual Reality technology in primary school
- STEM x 2
- Creative use of digital [post-primary but no links to curriculum]
- Digital Literacy
- PE at primary
- Robotics

The majority of stories capture practice from predominantly primary school clusters and here the teachers typically adopted a cross curricular approach to the use of digital technologies.

1. Collaborative Approach to Computational & Design Thinking to Improve Primary School Problem-Solving Skills

Schools Involved:

- Talbot Senior National School, Clondalkin, Dublin
- St. Anne's National School, Ardclough, Straffan, Co. Kildare

Project Overview

This project aimed to develop problem-solving skills among primary school students through the integration of computational and design thinking. It involved collaborative work within and between two schools, Talbot Senior National School and St. Anne's National School. Using Lego WeDo, Chromebooks, and iPads, students engaged in hands-on, technology-driven activities that promoted critical thinking and teamwork. The project also fostered self-reflection and peer assessment, encouraging students to evaluate their own learning processes and improve their problem-solving capabilities.

Project Aims

- Improve problem-solving skills among students
- Use technology to design and program robots
- Promote collaboration within and between schools
- Encourage self-reflection and peer assessment

Implementation

The project was focused on 4th class students in both schools and spanned three years. The primary technological tool was Lego WeDo, which allowed students to create and program robots to solve various tasks. Each school was equipped with 30 Chromebooks, and the teachers received training on using Lego WeDo for classroom workshops.

The first phase involved gathering data on the students' attitudes toward problem-solving through discussions and questionnaires. This was supplemented by quantitative assessments to measure the improvement of problem-solving skills, with guidance from the Teaching for Robust Understanding (TRU) framework provided by DCU. Teachers also used a checklist to track students' problem-solving abilities during the workshops, observing aspects like reflection, question posing, and flexible thinking.

Students were given roles such as Builder, Supplier, and Engineer to ensure shared responsibility during the project. A key task assigned to the two schools involved creating a canal boat and a bridge. One school was responsible for designing and programming the boat, while the other developed the bridge. The boat had to move, and the bridge had to open, all controlled by the Lego WeDo 2.0 software. The students then came together to combine their creations, showcasing their work at a Digital STEM event at Athlone Institute of Technology.

In the second year, the project incorporated a digital portfolio system using the Seesaw platform. Students were encouraged to upload videos, photos, and interviews, reflecting on their learning journey. The platform also facilitated peer assessment, with students recording interviews where they discussed what they enjoyed, the challenges they faced, and how they collaborated. Each school purchased additional iPads to enable this more dynamic and interactive approach to learning.

Throughout the project, the teachers held regular meetings with their Oide TiE Advisor to address any challenges and share new ideas. These were crucial in maintaining the project's progress and ensuring a collaborative atmosphere across the schools.

> *"Cluster meetings were essential – either in person (better) or via Zoom (if location is an issue)."*

Teacher Reflections

Teachers noted that the use of digital technology, particularly Lego WeDo, kept students highly engaged throughout the workshops. The division of roles within the student groups helped build fairness and encouraged participation from all members. The teachers found that stepping back and allowing students to explore the technology on their own often led to surprising creativity and innovation.

Cluster meetings among the teachers were viewed as an essential aspect of the project, providing opportunities to refine the approach and generate new ideas. The teachers also reflected on the importance of qualitative data collection, such as student interviews and focus groups, in capturing the nuances of the students' learning experiences. They highlighted the effectiveness of Seesaw in promoting self-reflection and peer assessment, which helped students to become more aware of their own thought processes.

The project faced significant challenges during the COVID-19 pandemic. The restrictions initially halted all in-person activities, and the interactive nature of the Lego sets was particularly difficult to maintain under social distancing rules. While the project resumed with some adjustments, the disruption was noted as a setback to its overall continuity.

> *"Lego WeDo is an excellent resource for all types of learners – it kept the children engaged for the full hour of the workshop."*
>
> *"The children are not afraid of digital technology – sometimes it's best to step back and let the students explore the technology themselves."*

Project Links

https://bawnclough-digitalcluster.weebly.com/

Figure 15: Working with Lego WeDo kits.

2. Developing a Continuum of Computational Thinking and Programming Activities in a Cluster of Schools in Mayo/Sligo

Schools Involved

- Various schools in Mayo and Sligo, including both single-class and multi-class contexts.

Project Overview

The project aimed to develop a coherent and progressive approach to teaching computational thinking and programming across a cluster of schools in Mayo and Sligo. Using the Digital Learning Framework and the European Framework for the Digital Competence of Educators (DigCompEdu) as guidance, the project focused on building a shared understanding of computational thinking, integrating these concepts into the curriculum, and supporting teachers through mentoring and professional development. By pooling resources and collaborating, the schools promoted innovation and creativity in learning, teaching, and assessment, while also providing opportunities for students to engage in real-world problem-solving using digital technologies.

Project Aims

- Develop a shared understanding of computational thinking among teachers and students.
- Integrate computational thinking concepts into various curricular areas in a structured and progressive manner.
- Explore and implement teaching methodologies that enhance digital literacy and problem-based learning.
- Provide coaching and mentoring support to teachers, as well as external resources when needed.

Implementation

The project began with establishing a shared understanding among teachers regarding what computational thinking entails and how it can be applied in primary schools. The schools worked together to define the core skills and concepts, developing a spiral continuum of activities that could be integrated with the primary curriculum. This ranged from using basic tools like BeeBots during early childhood learning to more advanced programming tasks with Lego WeDo and other digital platforms.

Teachers worked collaboratively to create resources, lesson plans, and problem-solving tasks that would help integrate computational thinking across multiple subjects, including mathematics, literacy, and science. These resources were shared among the cluster via a Google Drive repository, which formed the basis of a whole-school plan for each participating institution.

The project also involved extensive professional development for teachers, which included coaching, mentoring, and external training opportunities. Teachers were supported in embedding computational thinking and programming into their teaching practices, and the schools embraced hands-on digital learning activities that allowed students to evaluate and reflect on their experiences.

Throughout the project, there was an emphasis on real-world engagement, with some schools linking with industry partners and the broader community to enrich the learning experience. Students were encouraged to assess their own work and participate in peer assessments, becoming more discerning about the use of digital technologies in the process.

> *"Many schools integrated the equipment with existing curricular areas, such as BeeBots during Aistear and Lego WeDo for Bricks Club."*

Figure 16: Group learning with BeeBots.

Teacher Reflections

Teachers noted the significant impact of the project on both teaching and learning. The collaborative approach to resource development and the shared understanding of computational thinking allowed schools to better integrate digital technologies into the curriculum. Teachers expressed that the project improved their confidence in using digital tools, enabling them to create more interactive and engaging lessons for their students. The professional development sessions were particularly valuable, providing teachers with the skills and knowledge to guide students through computational thinking activities effectively.

Teachers also reflected on the challenges posed by the COVID-19 pandemic, which interrupted the planned face-to-face meetings and training sessions. However, the schools adapted by using digital tools like Google Workspace, email, and WhatsApp to continue collaborating and progressing the project remotely. Despite these challenges, teachers felt that the project helped them develop a stronger understanding of computational thinking and its application in the classroom.

> *"Collaboration around a computer makes learning meaningful and supportive, enhancing creativity and the evaluative power of learners' ideas."*

> *"Computational thinking is emerging as a key competence across all disciplines, combining problem-solving and design to create useful solutions."*

Project Links

https://sites.google.com/spw.ie/mayosligosef/home?authuser=0
https://twitter.com/mayosligosef

3. Digital GMP (Gortskehy-Meelickmore-Patrick)

Schools Involved

- Gortskehy National School
- Meelickmore National School
- St. Patrick's De La Salle BNS

Project Overview

The Digital GMP project was a collaborative effort between three primary schools in County Mayo, aimed at enhancing digital literacy among both teachers and students. By pooling resources and sharing knowledge, the schools integrated digital technologies into various curricular areas, moving from teacher-led instruction to student-driven learning. The project fostered a digital community of practice, where teachers and students worked together on innovative digital projects, learning from each other and enhancing their digital skills in teaching, learning, and assessment.

Project Aims

- Move from teacher-led pedagogy to student-led knowledge creation through the use of digital tools.
- Enhance digital literacy and promote innovative use of technology in teaching, learning, and assessment.
- Collaborate across schools to develop and share best practices in integrating technology into the curriculum.
- Foster creativity and engagement by allowing students to create their own digital content, such as presentations and projects.

Implementation

Each year of the project was divided into two iterations, focusing on different curricular themes. In the first year, the theme was Local History, with students researching their towns (Castlebar and Claremorris) and their schools over the past 100 years. Using tools like Google Slides, Adobe Spark, and Padlet, students created presentations and photo stories, incorporating research, interviews, and historical data. The project allowed students to collaborate in real-time, both at school and at home, using digital platforms to create and share their work.

In the second year, the focus shifted to STEM, with students using Lego WeDo kits and Scratch to create robots and animations. The theme of Climate Change was explored, and students built windmills and other projects while documenting their progress using Adobe Spark and Google Slides. The third year introduced creative writing and literacy, with students creating digital storybooks using Book Creator, focusing on themes related to Mayo and local history.

Collaboration was at the heart of the project, with teachers and students from all three schools working together through shared resources and platforms like Google Workspace. The schools also engaged external partners, including local historians and museums, to enhance learning experiences.

> *"Collaboration was the main ingredient in this project. Working together to plan for the integration of technology into teaching, learning, and assessment was the key to our success."*

Teacher Reflections

Teachers reflected on the positive impact of the project in promoting collaboration and enhancing digital literacy. The shared use of digital platforms allowed schools to plan lessons, create resources, and share content seamlessly. Teachers noted that students became more engaged in their learning when they were given the opportunity to create their own content, leading to a shift from passive consumption to active knowledge creation.

The project also highlighted the importance of flexibility and adaptability, particularly during the COVID-19 pandemic. The digital skills developed during the project allowed both students and teachers to transition to remote learning with greater ease, using platforms like Zoom and Google Workspace to continue collaborating and delivering lessons.

> *"Some students worked in pairs and really enjoyed the fact that they could do this both in school and at home, in real time."*
>
> *"The students were learning a lot. One student shared his presentation with his dad in China, which was a great boost to him at the time."*

Figure 17: Students using Scratch to code their Christmas eCards.

4. Dreams to Reality: Harnessing the Potential of the Digital World

Schools Involved:

- St. Patrick's NS Loch Gowna
- Breifne College, Cavan
- Bruskey NS
- Ballynarry NS
- Lacken NS
- Cavan Institute

Project Overview

The Dreams to Reality project was launched as part of the School Excellence Fund Digital initiative to promote digital literacy and computational thinking across both primary and secondary education. The collaborative project involved schools from primary through secondary levels, focusing on using digital tools like Lego WeDo, Minecraft, and coding to bridge the transition between primary and secondary school learning. Students engaged with technology to develop problem-solving skills, teamwork, and computational thinking, while teachers integrated modern teaching methods to support students' digital proficiency. The project encouraged peer tutoring and self-assessment to foster independent learning.

Project Aims

- Develop digital literacy and computational thinking from early primary education through secondary school.
- Facilitate a smoother transition for students moving from primary to secondary school using technology.
- Incorporate peer tutoring to enhance communication, teamwork, and problem-solving.
- Track the benefits of early digital literacy on students' progress as they advance through the educational system.

Implementation

The project commenced with a cluster planning phase in 2018, during which schools identified their individual strengths and expertise. Over the course of three years, students from junior infants to secondary level participated in various hands-on and digital activities aimed at developing computational thinking and digital skills.

Primary students engaged with BeeBots and BlueBots, starting with simple programming tasks and advancing to more complex coding through tools like Scratch. Unplugged activities such as writing code for everyday tasks helped students understand the importance of precise instructions, which tied into digital activities like Minecraft and Lego WeDo. Senior students in primary schools were involved in creating digital projects using tools like Adobe Spark, which enhanced their presentation skills and allowed them to demonstrate their understanding of STEM topics.

In the post-primary schools, the focus shifted to more advanced computational thinking exercises, including algorithm development and online coding tasks. Students explored various digital platforms like Microsoft Teams, Office 365, and coding software such as Scratch and Blockly. Peer tutoring sessions enabled older students to mentor younger ones, reinforcing the skills learned and fostering a collaborative learning environment.

The project also included several showcase events, where students from different schools gathered to share their work and experiences. One key event was the Showcase Day in 2019, where pupils from all participating schools presented their digital projects and visited Cavan Institute to see advanced digital technology in action. Another notable event was a visit to Combilift, where students observed the practical application of digital technology in industry, further emphasising the real-world relevance of their learning.

> *"Students were excited to work with digital tools, and the collaboration between schools brought a new dynamic to their learning."*

Teacher Reflections

Teachers involved in the project highlighted the positive impact of incorporating digital tools into the curriculum, noting how engaged and motivated the students were during the activities. They observed that the combination of unplugged and digital tasks helped students build essential problem-solving skills while maintaining enthusiasm for learning. The peer tutoring element was particularly effective in fostering leadership skills among older students, and teachers appreciated how the students took ownership of their learning through self-assessment and reflection.

Teachers also mentioned the challenges posed by the COVID-19 pandemic, which temporarily disrupted the project. However, they acknowledged that the groundwork laid by the project in earlier years made the transition to remote learning smoother, as both students and teachers were already familiar with digital platforms such as Teams and Seesaw.

> *"The unplugged activities were a great way to introduce students to computational thinking, and the results were evident when they transitioned to coding tasks."*

> *"The peer tutoring sessions gave older students the chance to solidify their learning, while younger students benefitted from the mentorship and guidance provided."*

Project Links

https://combilift.com/en/combi-ventilate
https://www.nrich.maths.org
https://www.tts-group.co.uk/beebot

Figure 18: Student visit to Combilift factory.

5. Integration of Virtual Reality (VR) in 21st Century Teaching, Learning and Assessment

Schools Involved

- St. Mark's SNS
- Sacred Heart SNS
- St. Thomas' SNS
- St. Anne's Primary

Project Overview

This project involved the use of Virtual Reality (VR) to enhance teaching, learning, and assessment across multiple DEIS (Delivering Equality of Opportunity in Schools) schools. The aim was to provide students, especially those in disadvantaged areas, with access to immersive digital learning experiences that go beyond traditional classroom activities. The project focused on enhancing students' proficiency in SESE (Social, Environmental and Scientific Education), SPHE (Social, Personal, and Health Education), Literacy, and Numeracy by incorporating VR into the curriculum. By exploring virtual environments, students engaged in interactive learning that was otherwise inaccessible. The initiative also promoted mindfulness and well-being activities, using VR to help alleviate stress and anxiety.

Project Aims

- Develop students' proficiency in SESE, SPHE, Literacy, and Numeracy.
- Use VR to provide unique learning experiences that enhance traditional teaching methods.
- Incorporate mindfulness and well-being programmes into the learning process to support mental health.
- Enable students to explore global environments and historical settings virtually, broadening their educational experiences beyond the local context.
- Showcase student learning through digital tools such as Google Expeditions, Google Docs, and Google Slides.
- Establish VR as a cost-effective, scalable tool for improving teaching, learning, and assessment in schools.

Implementation

The project began by integrating VR into everyday teaching activities. Teachers selected virtual expeditions from Google Expeditions (later replaced by Expeditions Pro) and used them to enhance their lessons in various subject areas. Each school was equipped with VR headsets and a teacher tablet to control and guide students through immersive VR lessons. Students explored topics such as virtual archaeological digs, nature trails, and mindfulness activities.

A unique feature of the project was the creation of a collaborative virtual tour of local landmarks in Tallaght. Using 360-degree cameras, teachers and students captured images and recorded information about the landmarks, creating an interactive virtual tour that could be shared across schools. The project was supported by Oide TiE which provided training and guidance to the participating schools. Regular meetings were held between the schools to share ideas, overcome challenges, and assess the project's progress.

Google Workspace for Education was used to facilitate collaboration among teachers. Google Classroom, Google Drive, and shared virtual spaces allowed teachers to co-create lesson plans, share resources, and track the progress of the project.

> *"We were fortunate to have a whole-class set of VR headsets, which made the teacher delivery of lessons more effective and exciting."*

Teacher Reflections

Teachers reflected that VR added an exciting and engaging dimension to the learning experience, making lessons more interactive and accessible for students. The use of VR allowed students to gain insights that were previously unattainable through traditional teaching methods. Teachers appreciated the support provided by the Oide TiE and the collaborative nature of the project, which allowed them to share resources and strategies.

Despite the success of the project, teachers acknowledged that the devices used for VR lessons had some limitations. Older devices tended to overheat, and the battery life and processing speed deteriorated over time. The need for updated technology was noted as a key consideration for future programmes. Teachers also emphasized the need for more VR content specifically designed for primary schools, as much of the content available during the project was limited.

> *"Creating our own virtual tours using 360-degree cameras allowed students to take ownership of their learning, combining technology with creativity to explore their local environment in a new way."*

> *"VR expeditions brought topics to life, providing a way to engage students in immersive learning experiences that sparked curiosity and creativity."*

Project Links

https://x.com/vrdub24

Figure 19: Lesson using VR in action.

6. Machines and Mechanisms

Schools Involved

- Firies National School, Firies
- St. Brendan's National School, Blennerville
- Murhur National School, Moyvane
- Presentation Monastery N.S., Killarney

Project Overview

The Machines and Mechanisms project aimed to incorporate enquiry-based learning in STEM into the school curriculum. The initiative was supported by the School Excellence Fund, promoting collaboration among schools and the use of digital tools like LEGO, K'nex, and iPads. The project provided students with hands-on learning opportunities, encouraging them to work together as developers, designers, and problem-solvers. Through enquiry-based tasks, students were challenged to identify problems, brainstorm solutions, and apply critical and creative thinking. Teachers and students alike benefited from the collaborative approach, sharing best practices and resources across schools to build essential 21st-century skills.

Project Aims

- Actively engage students in enquiry-based learning to enhance problem-solving skills.
- Increase student awareness and appreciation of STEM through practical, hands-on activities.
- Use digital technologies to open up new forums of learning and collaboration.
- Develop students' abilities to work as engineers, designers, and creators using modern tools.
- Empower teachers to collaborate and share effective digital teaching practices, improving instruction and student engagement in STEM.

Implementation

The project began with meetings between lead cluster teachers and school management teams to align the initiative with the Department of Education's Digital Learning Framework. Teachers evaluated their current digital practices and developed a digital learning plan for each school. Digital school committees, consisting of teachers, students, and parents, were formed to guide the project and ensure effective use of resources.

Resources like LEGO Education sets, K'nex, and iPads were procured and organised for use in classrooms. These tools facilitated enquiry-based learning, where students worked together to build and code mechanisms, such as cars, cranes, and cable cars. Students were assigned roles within their teams, such as builder, coder, and reporter. These roles were rotated so that all students gained a understanding of the various aspects of the projects.

The progression of activities was carefully structured across different class levels. Junior and Senior Infants worked with LEGO Duplo, while older students engaged with more advanced sets like LEGO WeDo and LEGO Prime. The activities were linked to the curriculum and provided students with the opportunity to apply scientific, mathematical, and technological concepts to solve real-world problems.

Collaboration was key throughout the project. Schools shared resources, and teams of students worked together on tasks, documenting their learning using iPads and apps like Scratch, Seesaw, and Green Screen. Schools also partnered with external organisations, such as Liebherr and local train stations, to provide students with insights into STEM careers and real-world applications of the skills they were developing.

> *"Children became developers, creators, and change-makers of technology through hands-on enquiry-based learning."*

Teacher Reflections

Teachers expressed that the project had a significant positive impact on student engagement and learning. The combination of digital technology and hands-on enquiry-based activities encouraged students to think critically and work collaboratively. Teachers appreciated the opportunity to share resources and teaching strategies across schools, which enhanced their ability to integrate STEM into the curriculum.

The rotating roles within student teams ensured that every child participated fully and developed a broad range of skills. Teachers also noted the importance of incorporating digital tools like Seesaw, which allowed students to document their progress and reflect on their learning. The project faced challenges due to the COVID-19 pandemic, but the strong foundation of digital literacy and collaboration enabled schools to adapt to remote learning more effectively.

> *"Collaboration among schools and sharing resources enabled us to enhance teaching and learning in STEM."*

> *"Students explored real-world problems, using digital technologies to design solutions, which helped them develop critical thinking and creativity."*

Project Links

https://sites.google.com/view/stemkerry

Figure 20: STEM Kerry project logo.

7. Multimedia Cluster Project

Schools Involved

- Dublin 7 Educate Together (Lead School)
- Presentation Primary School, George's Hill
- St. Mary's Primary School
- St. Paul's CBS, Brunswick Street
- Gardiner Street Convent School

Project Overview

The Multimedia Cluster Project brought together five primary schools to enhance digital literacy through the use of collaborative digital technologies. The project aimed to integrate tools like Microsoft Office 365 and GSuite into the classroom, facilitating communication, creativity, and collaboration among nearly 1,400 students and 100 teachers. Pupils and teachers worked together across schools, creating digital content, sharing resources, and participating in video calls. A core goal was to use digital technologies to transform teaching, learning, and assessment practices across the participating schools, with a focus on inquiry-based learning and cross-curricular activities.

Project Aims

- Enhance the use of digital technologies in teaching, learning, and assessment.
- Foster collaboration between teachers and students across schools.
- Develop pupils' skills in using digital tools to create, share, and communicate.
- Integrate technology into all curriculum areas, promoting a student-centred approach to learning.

Implementation

In the first year, teachers received training in using digital applications such as word processing, email, and content creation tools. Students were introduced to basic computer skills and used video calling to communicate with peers in other schools. The project encouraged an inquiry-based approach to learning, where students worked collaboratively on topics like local, national, and international issues.

Each year group participated in a specific digital project:

- Junior Infants learned basic computer skills.
- Senior Infants created a "Growing" video project for science.
- 1st Class completed a music sound recording project.
- 2nd Class built a Virtual Art Gallery.
- 3rd Class created a photo gallery.
- 4th Class produced a video documentary.
- 5th Class wrote and designed a newspaper/book.
- 6th Class built websites.

Despite the disruptions caused by COVID-19, the project continued into the second and third years, with students and teachers adapting to remote learning. Teachers collaborated via shared online spaces like Google Drive, and students used Seesaw for creating ePortfolios that tracked their learning progress over time. Some classes also used Scratch to create animations and games linked to their projects, which served as assessment tools.

Applications such as Seesaw, Google Drive, and Microsoft Office 365 enabled students to create digital content and share their work with peers, teachers, and parents. Teachers used these tools to assess student progress, give feedback, and plan cross-curricular lessons that integrated digital technologies into all subject areas. Video calls, digital portfolios, and multimedia projects helped build a sense of community among students, despite the physical distance between schools.

> *"Using technology in the classroom enabled our students to connect and work together, even when they couldn't be physically present in the same room."*

Teacher Reflections

Teachers reported that the project had a transformative impact on both teaching and learning. The use of technology encouraged greater collaboration among students and helped them develop skills in problem-solving, communication, and creativity. Teachers also observed improvements in student behaviour, self-confidence, and social skills, particularly when working on group projects. The continuous professional development (CPD) workshops helped teachers develop new digital skills, which they applied to their classroom practices.

Challenges included the limited availability of devices due to COVID-19 restrictions, as well as the need to train new staff on the project's tools and processes. Despite these obstacles, teachers were able to integrate digital technology into their classrooms effectively, making learning more engaging for students.

> *"A reduction in misbehaviour and an increase in pupils' self-confidence were clear outcomes from using digital tools to enhance collaboration and communication."*

> *"Through the project, pupils learned to create digital content, which not only helped them develop IT skills but also fostered a love of learning and exploration."*

Project Links

https://sites.google.com/view/multimediaclusterproject/home

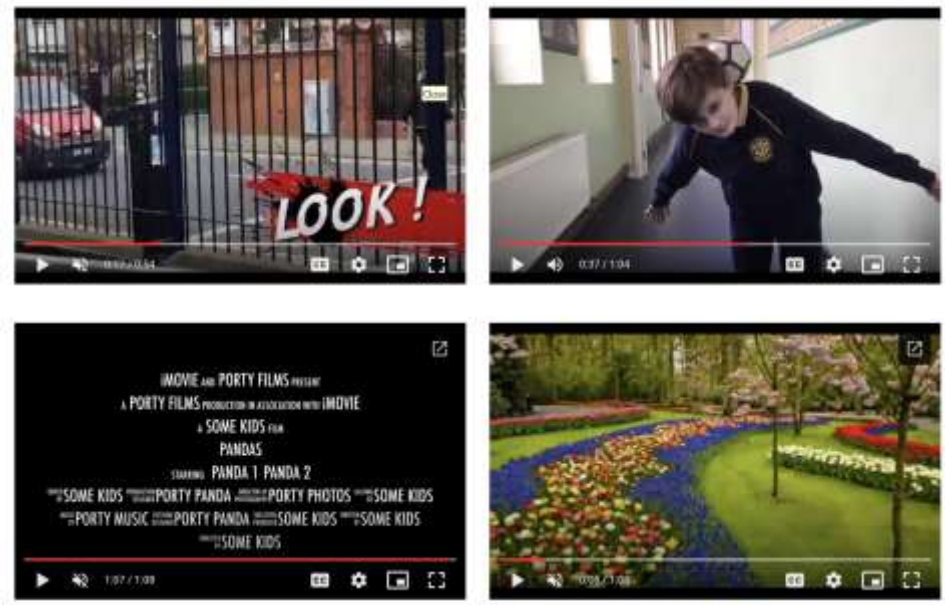

Figure 21: Selection of video documentaries created by 4th classes

8. North-East STEM Champions

Schools Involved

- St. Mary's Parish Primary School, Drogheda
- Aston Village Educate Together National School, Drogheda
- Scoil na gCailíní, Castleblayney

Project Overview

The North-East STEM Champions project brought together three primary schools to promote STEM education through the creation of a professional learning community of teachers and students. The project emphasised active collaboration and knowledge-sharing between educators, encouraging innovative use of digital technologies in teaching, learning, and assessment. By integrating hands-on STEM activities into the curriculum, students engaged in inquiry-based learning, developing key skills in science, technology, engineering, and mathematics. Local businesses and STEM experts were also invited to participate, offering students real-world insights into STEM careers and inspiring them to pursue STEM-related fields.

Project Aims

- Promote the innovative use of digital technologies in teaching, learning, and assessment.
- Build a collaborative community of teachers to share STEM lessons and best practices.
- Engage students in hands-on, inquiry-based STEM lessons to develop skills for the modern workplace.
- Foster connections with local industries and STEM experts to inspire students, particularly girls, to explore STEM careers.

Implementation

The project was coordinated across all three schools through regular meetings to assess progress and plan future activities. Teachers participated in continuous professional development (CPD) to enhance their skills in delivering STEM education. Oide TiE provided training on digital tools such as SeeSaw for recording learning and LEGO WeDo to link coding with science and construction.

The schools engaged students in a variety of STEM activities, including coding with Scratch, using Bee-Bots for literacy and numeracy, and building robots with LEGO WeDo. Students participated in national STEM competitions, such as the Intel Mini Scientist, ESB Science Blast, and VEX Robotics. These activities helped students develop problem-solving and IT skills, while also sparking their interest in STEM subjects.

A key element of the project was the involvement of local businesses and STEM experts, who visited schools to provide role models for students. By showcasing the real-world applications of STEM, the project aimed to inspire students, particularly girls, to see the relevance of STEM in their daily lives and believe in their ability to succeed in these fields. The project culminated in STEM Fairs, where students showcased their work to the wider school community, including parents and local stakeholders. One such fair had over 700 attendees, highlighting the strong community support for the initiative.

Figure 22: Learners collaborating on their projects.

Teacher Reflections

Teachers reflected positively on the collaborative nature of the project, emphasising how pooling knowledge and resources helped to move the initiative forward in creative and innovative ways. They valued the opportunity to work closely with colleagues from other schools, sharing best practices and lesson plans through digital platforms like Google Workspace. The project demonstrated the significant impact that digital technology can have on improving teaching and learning, particularly in science and maths. Teachers noted that the inquiry-based learning approach helped students develop curiosity and critical thinking, skills that will prepare them for the future workplace.

> *"This project helped me with my IT skills, my understanding of STEM, how to build and code a robot, and the amazing opportunity we got when we went to DKIT."*

> *"I learned that science doesn't always go as planned, and I also learned that science is very important in everyday situations."*

Project Links

http://bit.ly/NESTEMChampions

Figure 23: Final project testing and demonstration.

9. STEAM Hub and Centre of Technology Excellence

Schools Involved

- St. Brendan's Community School
- St. Brendan's Boys School
- Mercy Primary School
- Scoil Náisiúnta Osmann
- Crinkhill National School
- Gaelscoil na Laochra

Project Overview

The STEAM Hub and Centre of Technology Excellence project, based at St. Brendan's Community School, aimed to establish a central hub for digital learning and STEAM activities, benefiting both primary and secondary schools in the surrounding community. The project introduced state-of-the-art technology, including Surface Go devices and Micro tools, and developed interactive learning environments such as the Mini DreamSpace 2, located at the school and in Birr Castle Museum. Through professional development provided by Microsoft, educators were equipped to integrate digital tools into their teaching, fostering creativity and critical thinking among students. The project also encouraged collaboration between schools, local community resources, and industry partners to deliver a dynamic and future-oriented learning experience.

Project Aims

- Provide students with access to digital technology to enhance their learning experiences in STEAM subjects.
- Create a hub for STEAM education to serve local primary and secondary schools, fostering collaboration and innovation.
- Offer professional development to teachers, enabling them to integrate digital tools into their teaching practices.
- Promote partnerships with local institutions such as Birr Castle Museum to extend learning opportunities beyond the classroom.

Implementation

The project began by providing participating schools with Surface Go devices and Micro technology, enabling students to engage in hands-on STEAM activities. Teachers received professional development through Microsoft to ensure the effective use of these digital tools in the classroom. A central feature of the project was the creation of Mini DreamSpace 2, a dedicated space for interactive learning, where students could explore subjects like robotics, Minecraft, and digital design. This space was used both at St. Brendan's Community School and in Birr Castle Museum, expanding the reach of the project to a wider community.

The project also included outreach to local primary schools, where students were introduced to STEAM through workshops and activities. By linking with local partners, such as Birr Castle and other community resources, students gained insights into real-world applications of the skills they were developing. Ongoing collaboration between the schools and industry partners ensured the project remained relevant and impactful.

Teachers were a key focus of the project, with continuous professional development enabling them to integrate digital tools into their lessons effectively. Microsoft provided CPD opportunities, while schools worked collaboratively to share strategies and resources, building capacity for digital learning across the region.

> *"Collaboration with local partners, such as Birr Castle Museum, provided valuable real-world learning experiences for students."*

Figure 24: Mini Dream Space 2 in St Brendans Community School, Birr.

Teacher Reflections

Teachers reflected on the importance of technology in enhancing student learning and engagement. The introduction of Surface Go devices and Microtechnology allowed students to interact with STEAM subjects in new and creative ways, developing problem-solving and critical-thinking skills. The Mini DreamSpace 2 provided an innovative environment for students to collaborate on digital projects, which further engaged them in learning.

Professional development was another crucial element of the project. Teachers noted that the training provided by Microsoft helped them to better integrate digital tools into their teaching, making lessons more interactive and student-centred. They emphasized that technology should be viewed as a tool to facilitate learning rather than a solution on its own, requiring careful planning and integration into the curriculum.

The project faced challenges, particularly during the COVID-19 pandemic, but the foundations laid in digital literacy helped schools transition to remote and blended learning more smoothly. Teachers expressed that the skills gained through the project were invaluable during this period, as they adapted to new ways of teaching and engaging students.

> *"Technology played a significant role in enabling students to engage in interactive learning experiences, fostering creativity and innovation."*

> *"The professional development opportunities provided by Microsoft gave teachers the skills to effectively incorporate digital technology into their teaching."*

Project Links

https://stbrendanscsbirr.ie/Birr-STEM-Cluster/Index.html
https://www.birrcastle.com

10. Technology to Collaborate, Create, and Innovate

Schools Involved

- Cork Educate Together Secondary School
- Firhouse Educate Together Secondary School
- Malahide Portmarnock Educate Together Secondary School
- Clonturk Community College

Project Overview

The "Technology to Collaborate, Create, and Innovate" project focused on establishing a Maker Space in each participating Educate Together secondary school. The project, developed in partnership with Make Create Innovate and Oide TiE, aimed to support student learning in physical computing through the use of digital and creative technologies. By fostering a constructivist and inquiry-based learning environment, the project encouraged students to take ownership of their learning, work collaboratively, and explore innovative solutions using technologies such as Raspberry Pi, Arduino, and 3D printers.

The Maker Spaces provided an interactive platform for cross-curricular activities, encouraging students to integrate various subjects like science, history, geography, and digital literacy into their projects. This hands-on approach facilitated the development of problem-solving, critical thinking, and digital literacy skills, with students creating projects that spanned multiple disciplines.

Project Aims

- Establish sustainable Maker Spaces in each participating school.
- Train staff to use Maker Space technologies and develop cross-curricular activities.
- Empower students to deliver workshops and mentor younger peers in using the technologies.
- Foster collaboration between schools, enabling the exchange of skills and ideas among teachers and students.

Implementation

Junior Cycle students were introduced to the Maker Space through dedicated workshops held throughout the school year. Each class participated in three two-hour workshops, during which they worked in groups to explore themes such as mapping Medieval Cork, soil moisture sensors, and 3D printing. Students researched the use of various technologies, such as Raspberry Pi, micro:bits, and 3D printers, and applied this knowledge to design and build projects.

Collaboration was a key feature of the project, with students working together in groups, reflecting on their work, and discussing how the projects integrated multiple subjects. Teachers from different subject areas collaborated to develop interdisciplinary lessons that made use of the Maker Spaces, enriching the learning experience. Students were encouraged to continue using the Maker Spaces during lunch breaks, fostering independent exploration and innovation.

Role of Technology

Technology played a central role in this project, with the Maker Spaces equipped with tools like microcontrollers, 3D printers, and programming languages such as Scratch, Python, and Arduino. Students used these technologies to prototype designs, program interactive projects,

and create solutions to real-world problems. Projects ranged from creating soil moisture sensors to designing renewable energy systems. The Internet of Things (IoT) was also explored, with students developing projects that connected sensors and actuators to the internet.

The Maker Spaces democratized access to advanced technologies, providing students with the opportunity to experiment with tools that were previously out of reach due to cost. This hands-on learning environment encouraged students to become creators, not just consumers, of technology.

> *"Maker Spaces provide an opportunity for students to experiment with technology, taking ownership of their learning while developing practical and creative skills."*

Figure 25: Students create an interactive map of Cork.

Teacher Reflections

Teachers noted that the project had a significant impact on student engagement and learning. The hands-on nature of the Maker Spaces allowed students to explore and experiment with technologies in a way that deepened their understanding of various subjects. The collaborative aspect of the project fostered a sense of ownership among students, as they were able to see the tangible results of their work.

Teachers also reflected on the importance of having dedicated time for staff collaboration. Rather than simply training each other, teachers found that the most effective approach was to work together to develop ideas and activities, building on each other's contributions. This collaborative process led to more creative and innovative projects being developed in the Maker Spaces.

> *"Students were encouraged to reflect on what they learned and how the project connected different subjects, reinforcing the cross-curricular approach."*

> *"True collaboration is not simply to work with someone but to 'think' with someone; to build ideas on top of ideas with others so that everyone reaches a point they couldn't have reached alone."*

Project Links

https://x.com/makersinschools
https://www.makersinschools.ie

11. Using Digital Mobile Technologies for Teaching, Learning, and Assessment of Fundamental Movement Skills in Physical Education

Schools Involved

- St. Colman's NS, Derrinabroc
- Scoil Íosa, Carracastle NS
- St. Nathy's College, Ballaghaderreen
- Scoil Mhuire, Brusna NS

Project Overview

This project aimed to enhance the teaching and learning of Physical Education (PE) across four primary schools and one secondary school by integrating digital mobile technologies. The initiative focused on using constructivist pedagogies, allowing students to engage in peer assessment and feedback while developing fundamental movement skills (FMS). The use of digital portfolios and mobile devices enabled students to record, assess, and reflect on their progress, fostering a sense of ownership over their learning. Teachers also collaborated to improve their own digital competencies and share best practices within the cluster.

Project Aims

- Improve the teaching and learning of PE through the use of digital technologies.
- Develop students' fundamental movement skills using video recording and peer assessment.
- Create digital portfolios to track students' progress over time.
- Facilitate professional development for teachers in using digital tools in PE and beyond.
- Strengthen the connection between primary and secondary schools by promoting collaboration.

Implementation

The project was organised with support from Oide TiE, which provided training in the use of iPads, Google Workspace, and Seesaw for creating digital portfolios. Teachers used the "Move Well, Move Often" booklet and website to introduce students to fundamental movement skills. During PE lessons, students recorded their performances using mobile devices and assessed their peers based on key skill objectives highlighted in videos.

Digital resources were shared through apps like Seesaw at the primary level and Showbie at the secondary level. Students compiled videos of their progress into ePortfolios, which allowed both students and teachers to monitor development from the exploration to the mastering stage. This approach aligned with the proposed Physical Education Leaving Certificate subject, helping secondary students develop assessment skills that could be applied at higher levels.

Cluster meetings were held to share lesson videos, ideas, and progress, enabling ongoing reflection and evaluation of the project. Schools collaborated closely, sharing resources and digital content to create a cohesive learning experience across different age groups.

> *"Integrating technology into PE has allowed us to marry activity with modern digital tools, producing really worthwhile outcomes."*

Teacher Reflections

Teachers noted that the use of digital technologies significantly enhanced the PE learning experience. Recording and reviewing videos of FMS allowed students to observe their own performance and provide constructive feedback to peers. This visual approach to learning helped students understand their strengths and areas for improvement more clearly, making the learning process more interactive and engaging.

The project also improved teachers' confidence in using digital tools, thanks to the ongoing professional development and collaboration with colleagues. Teachers felt that digital portfolios were an invaluable resource, particularly during the COVID-19 pandemic, when remote learning required new ways of assessing students' progress.

However, challenges included the initial learning curve for teachers unfamiliar with digital technologies and the logistical difficulties of recording and editing videos, particularly in large classes. Despite these challenges, the benefits of integrating digital tools into PE were clear, with both students and teachers expressing a greater enjoyment and engagement in the subject.

> *"By digitally recording themselves performing fundamental movements, our project allowed pupils to utilise technology as an extremely valuable assessment tool to monitor their own and their peers' progress."*

> *"Students enjoyed using video technology in PE class. They loved assessing their peers and giving constructive feedback."*

Figure 26: Project feature on RTE news2day.

Project Links

https://sites.google.com/derrinabrocns.ie/clusterproject/home
https://x.com/SEFDigtialPE

12. West Cork Robotics

Schools Involved

- Togher National School, Dunmanway, Co. Cork
- Drinagh National School, Co. Cork
- Cappabue National School, Kealkil, Co. Cork

Project Overview

West Cork Robotics was a collaborative initiative between three primary schools in West Cork, in partnership with the School of Computer Science and Information Technology at University College Cork (UCC). The project aimed to explore and demonstrate the innovative use of robotics and coding in enhancing teaching, learning, and assessment. The project aimed to improve digital skills for both teachers and students by integrating digital technologies into the Irish Primary Curriculum. Students engaged in hands-on activities focused on creativity, collaboration, and computational thinking.

Project Aims

- Provide opportunities for active learning, allowing students to explore mathematical and spatial concepts using digital technologies.
- Foster collaboration and teamwork through group-based learning activities involving coding and robotics.
- Develop computational thinking and problem-solving skills in students through coding languages and robotics.
- Enhance digital literacy and encourage creativity and innovation by allowing students to design, build, and test solutions to various challenges.
- Demonstrate effective use of digital technologies in teaching and learning, sharing best practices with other schools.

Implementation

In the first year, the project introduced high-level block-based programming languages such as Scratch and Scratch Junior to students across all participating schools. These tools allowed students to create interactive stories, games, and animations, enhancing their computational thinking and problem-solving skills. Younger students also worked with BeeBots, simple programmable robots designed for early learners, to explore sequencing, directional skills, and teamwork.

The second year of the project expanded to more advanced robotic kits such as LEGO WeDo 2.0, Photon Robots, and Spike Prime, providing students aged 7-11 with the opportunity to build and program robots. These activities helped students develop a deeper understanding of programming concepts like loops, sensors, and event sequencing, while encouraging creativity through the design and testing of their robots. Collaboration was a key element, with students working in small groups to complete tasks.

As the project progressed, students also engaged with more complex technologies, including the LEGO EV3, VEX IQ robotics kits, and 3D printing with Tinkercad. These tools were introduced to help students develop critical thinking, design thinking, and technical skills. Cross-curricular integration was encouraged, connecting robotics and coding activities with subjects like mathematics, science, and engineering. The project concluded with a showcase event where

students presented their projects and shared their learning experiences with the school community.

> *"Block-based programming allows young students to learn the fundamentals of coding and computational thinking without worrying about syntax errors or complex code."*

Figure 27: Programming a Mindstorms Robot.

Teacher Reflections

Teachers reflected that the project significantly enhanced student engagement, fostering creativity, innovation, and collaboration. The introduction of block-based programming languages like Scratch allowed younger students to develop a strong foundation in coding, while older students thrived in using more advanced robotic kits to solve complex challenges.

Teachers also highlighted the importance of professional development, as many of the technologies were new to them. Training and support from UCC and external facilitators were essential for building confidence in integrating robotics and coding into the curriculum. Teachers also noted the long-term impact of the project, with students demonstrating greater interest in STEM subjects and continuing to apply computational thinking in their daily learning.

The COVID-19 pandemic presented challenges, limiting in-person workshops and group activities. However, the schools adapted by focusing on remote collaboration and virtual learning. Teachers expressed hope that future face-to-face events and showcase opportunities could resume.

> *"These tools help students develop computational thinking, critical thinking, design thinking, and problem-solving skills."*

> *"Robotics and coding enhanced student engagement, encouraged creativity and innovation, and fostered collaboration and teamwork."*

Project Links

https://westcorkrobotics.edublogs.org/

Appendix 5: Post-project survey

Schools Excellence Fund Digital and STEM Programme (2018-2022)

Introduction

Thank you for your involvement in the SEF DIGITAL and STEM Programme. Now that the programme has come to an end, we would like to invite you to look back on your experience of the programme as a whole, and of the proect(s) you were involved in and to share your thoughts with us.

This will probably take about 20 minutes minutes of your time (but it depends on how much you wish to say).

Please provide your feedback by **15 April.**

ALL

1. Was your involvement with the SEF DIGITAL and STEM Programme as a:
   a) Cluster Co-ordinator (and teacher)
   b) Teacher (but not Cluster Co-ordinator)
   c) Member of PDST providing support to one or more clusters.
2. Which projects(s) were you involved with (select one or more from the drop down list. If you do not recognize your project(s) in the list then select 'Other' and enter the title you knew the project by in reponse to the next question:
3. Name of project(s) if not identified in the list provided:

Section 1- Project Experience

TEACHER or CLUSTER COORDINATOR

4. To what extent do you agree with the statement: The overall experience of the project was positive?
5. To what extent do you agree with the statement: The project successfully achieved its objectives?
6. To what extent do you agree with the statement: The students responded positively to the project?
7. To what extent do you agree with the statement: The project had a positive impact on students' learning?
8. To what extent do you agree with the statement: The project improved my teaching experience?
9. To what extent do you agree with the statement: The project had a positive impact on the school as a whole?
10. To what extent do you agree with the statement: The support from other schools in my cluster during the project was helpful?
11. To what extent do you agree with the statement: The finance provided from the Programme for the project was sufficient for the needs of the project.

12. To what extent do you agree with the statement: The support privided by the Programme, PDST and other support brought in was sufficient?

Responses:

Strongly Agree
Agree
Neither Agree nor Disagree
Disagree
Strongly Disagree
N/A

CLUSTER COORDINATOR

13. To what extent do you agree with the statement: The support from the Programme and PDST in supporting me as a Cluster Coordinator met my needs

PDST ADVISOR

14. To what extent do you agree with the statement: The support PDST was able to provide to the Clusters was sufficient for their needs.

ALL

15. If you wish to do so, then please add any additional comments relevant to your experience of the project(s) you were involved in and the Programme as a whole.

Section 2- Sustainable Practice and Long-Term Impact

TEACHER or CLUSTER COORDINATOR

16. To what extent do you agree with the statement: I have continued to use the technology and teaching methods learned from the project after the funding period ended.
17. To what extent do you agree with the statement: The project has resulted in sustainable practices that will continue to benefit my school in the long term?
18. To what extent do you agree with the statement The project has had a significant long-term impact on our school's ability to integrate technology into teaching and learning?
19. To what extent do you agree with the statement The project has had a significant long-term impact on our school's ability to integrate technology into teaching and learning?

Responses:

Strongly Agree
Agree
Neither Agree nor Disagree
Disagree
Strongly Disagree
N/A

20. Has funding (for replacement/updating of equipment, training and support) been an issue for sustainability and long term impact of the project Yes/No
21. If answer to previous question is Yes, then please explain the issues that have arisen in relation to funding.
22. What barriers have you faced in sustaining the practices from your school-based project?

PDST

23. To what extent do you agree with the statement The project has had a significant long-term impact on our school's ability to integrate technology into teaching and learning?

ALL

24. If you wish to do so, then please add any additional comments relevant to sustainable practice and long-term impact, both of the project(s) you were involved in and the Programme as a whole.

Section 3- Future initiatives

ALL

25. If DfE was to look to fund other digital initiatives similar to this, what changes would you like to see in the process of application, funding, support and monitoring?
26. How can we ensure that any such future initiatives have a sustained impact beyond the project's funding period?
27. Do you have any suggestions for similar initiatives in the future?

Section 4: Lessons learned

ALL

Please provide brief accounts in reponse to the follwing four (final) questions outlining what you felt you learned about these issues as a result of your involvement in your project in the Programme as a whole. If you wish, you can reference existing project web sites, online materials etc where evidence or your thinking on these issues is developed further:

28. What is good classroom practice in relation to the embedding of digital technologies and constructivist approaches to teaching, learning and assessment at primary and post primary levels?

29. Which infrastructural models can best support innovation in relation to teaching, learning and assessment, communication, and collaborative planning?

30. What are the most appropriate support models that can be put in place to facilitate innovation in schools?

31. How the embedding of digital technologies in teaching, learning and assessment within and across all schools can be achieved?

Sign off

Thank you for taking the time to complete this survey.

List of abbreviations

General

CBA	Classroom Based Assessment
CPD	Continuing Professional Development
DE	Department of Education
DES	Department of Education and Skills (now Department of Education and Youth)
ICT	Information Communications Technology
PDST	Professional Development Service for Teachers (now Oide)
Oide TiE	Oide Technology in Education
SESE	Social, Environmental and Scientific Education
SSE	School Self-Evaluation
STEAM	Science, Technology, Engineering, Arts, and Mathematics
STEM	Science, Technology, Engineering and Mathematics
VLE	Virtual Learning Environment

Source documents

IP	Initial Cluster Proposal
3YP	3 Year Plan
B	Budget
Y1_Mid_PR/ER	Year 1Term 1 Progress/Expenditure Report
Y1_Final_PR/ER	Year 1 End of Year Progress/Expenditure Report
CCS2020	Cluster Coordinators Survey April 2020